AUTOMATED PENGUIN MONITORING SYSTEM

Designed by Pierre Retief for BirdLifeSA

Revision: DRAFT

INTRODUCTION

The Automated Penguin Monitoring System is used to weigh African Penguins in the wild, without the need to physically handle them.

The system consists of both hardware and software components. This manual is written as a step-by-step guide to building a complete system. The hardware components are covered first, followed by the software setup. Finally, the software needed to process the data and display it on a custom dashboard is discussed. A trouble shooting guide is also included at the end of the document.

OVERVIEW

To build the APMS, some basic tools are required. A full list can be found at:

s3://apmsbirdlife /APMS /Documents/BirdLife_toolkit.xlsx

See the section AWS S3 CLOUD STORAGE for more information regarding accessing the cloud. Please add to the tool list as/when needed. In this manual, commands to be executed in the command line will start with ***$*** and be in bold and italics. Example: ***$ sudo apt update***

The APMS consists of the following key components:

- 24V Solar Power Supply
- 4G Router
- Two scales, mounted on a platform.
- APMS Main Unit
- BioMark RFID Reader.

- The 24V Solar Power Supply provides power for the APMS Main Unit, scales, 4G Router and BioMark RFID Reader. This is site dependent.

Figure 1 - Solar Panel on Bird Island

- The 4G router is used for the main internet connection and depending on the strength of the cell signal at the site, either an outdoor router, or indoor router is used.

Figure 2 - Outdoor Router

Figure 3 - Indoor Router

- Two scales are used to increase accuracy and to indicate whether the penguin is heading out to- or coming back from the ocean. These scales are mounted on a platform.

Figure 4 - Scales Platform

- All the electronics are housed inside an IP67 rated enclosure, referred to as the APMS Main Unit. The section "APMS Main Unit" described how to build this box.

Figure 5 - APMS Main Unit

- The BioMark reader is an external device, used for picking up RFID pit tags. The APMS connects via a USB-Serial interface, internal to the reader. See the "BioMark" section for more information.

Figure 6 - BioMark Reader

- The data is downloaded daily, processed, and displayed on a Grafana Dashboard. This is handled by the APMS Server off-site. See *** for more information regarding the Dashboard

Figure 7 - APMS Dashboard

SOLAR POWER SUPPLY

Each site is different and requires planning to ensure components are correctly sized. A basic system and the larger system at Bird Island are discussed below.

BASIC SYSTEM

A basic solar system is shown below and can be used for a site that is running 1 BioMark RFID reader, an APMS Main Unit and a 4G Router. This is the setup for Stony Point, Dyer Island and Robben Island. Face the solar panels North and tilt at an angle of approximately 45 degrees. An up-to date parts list can be found at s3://apmsbirdlife /APMS/Documents/solar_parts_list.xlsx

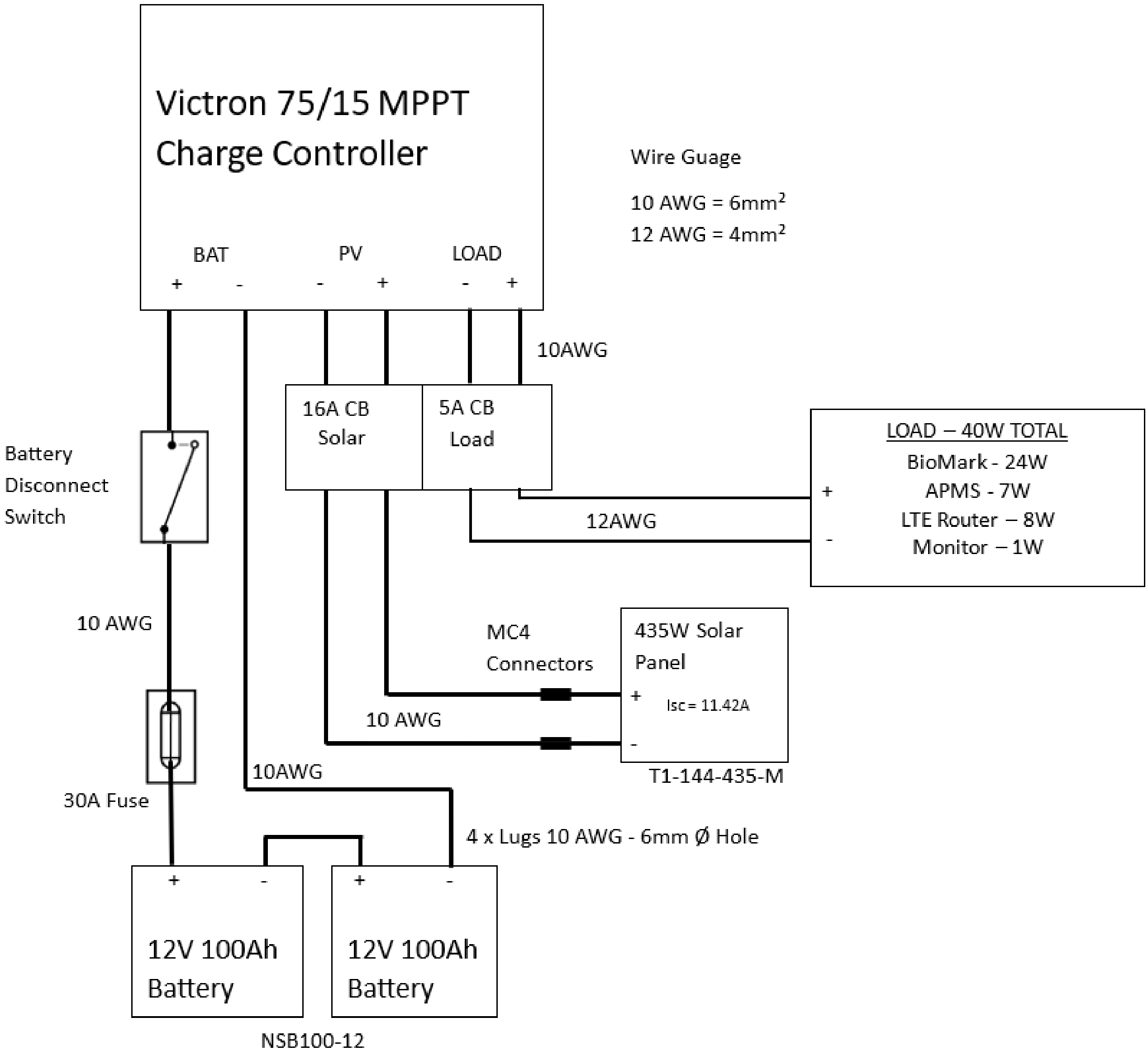


Figure 8 - Basic Solar Power Supply

BIRD ISLAND SOLAR SYSTEM

The system on Bird Island is slightly larger as it is powering two BioMark readers and has provision for a future live feed camera. See FIGURE 9 - BIRD ISLAND SOLAR POWER SUPPLY for a wiring diagram

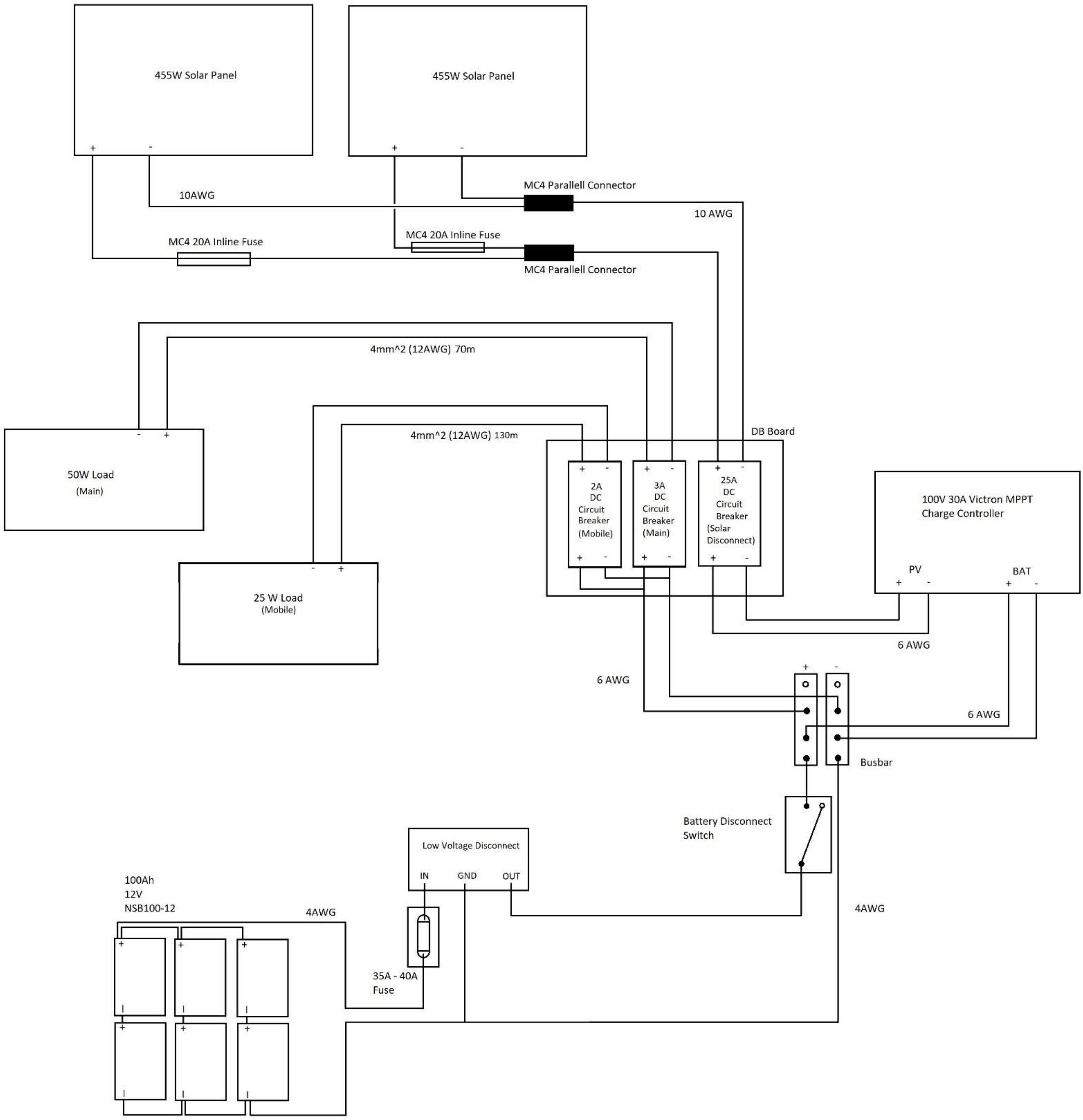


Figure 9 - Bird Island Solar Power Supply

INTERNET AND WI-FI

To monitor the site and upload data daily, a reliable internet connection is needed. This is accomplished by using a 4G Router. Wi-Fi has been used and has proven very reliable. If ever needed in future, a LAN cable can be used instead.

Currently, Vodacom seems to be the strongest and most stable provider at the different sites. The Vodacom app also makes it easy to load and keep track of data and SMS bundles. Remember to copy down the IMEI number from the sim card before installing it into the router, as this number is needed for the app.

When installing the APMS on site, take both a Vodacom and MTN sim card, and check which one has a better signal. Be sure to test the sim cards in a phone, and make sure they can access the internet. https://www.speedtest.net/ can be used to test internet connectivity.

Depending on the site, an indoor router or outdoor router is used.

To prepare either router for use with the APMS:

- Get a suitable SIM card, activate it, and load data.
- Test the sim in a phone to ensure it works.
- Insert the SIM card into the 4G router and set up according to router manual.
    - Use “apms” as the SSID and “africanpenguins” as the password
- Connect to the router from a phone or PC and test the internet is working.

More specifics regarding the Outdoor and Indoor router can be found on the next page.

OUTDOOR ROUTER

For more remote sites, with weaker cell reception, such as Bird- and Robben Island, an outdoor router is used. The router is purchased from https://www.outdoorrouter.com/.

Currently used model is: UK CAT6 Outdoor 4G Router (EZR33L-U4).

When ordering, ask to have a 5m DC Cable included, as we do not use 48V POE. The cable comes with a DC connector that plugs into the circuit board. (It needs to be bent in a slightly awkward way to fit)

The router runs directly on the 24V Solar Power supply. A circuit breaker for the router should be installed and will also act as a power switch, as there is no easy way to power cycle the router.

The contact person is Tammy He (tammy@outdoorrouter.com)

Mount the router on a pole or to a wall, using the hardware supplied with the router

After installation, ensure the unit is closed properly and all 4 antennas are screwed on tightly.

> NOTE: Ensure gasket seats properly and is not folded over when closing up. Tighten the four corners in a "criss-cross" pattern to ensure even closing.

Figure 10 - Outdoor Router Before Closing

INDOOR ROUTER

If the cell signal is strong enough, a cheaper indoor router can be used. The router currently used is the DLINK DWR-M920, purchased from Takealot. It runs on 12V DC and has external antennas for the Wi-Fi, as well as for the cell signal. (If a different router is to be used, ensure it runs on 12V and has decent antennas).

- Cut off the 12V power supply and replace it with the 12V DC plug (RS: 138-9405).
    - This connector will plug into the APMS Main Unit (12V DC IN) when the router is to be used.

Figure 11 - Indoor Router

SCALES PLATFORM

The Scales Platform consists of two scales, mounted on a wooden base, with a RFID Antenna running through the middle. There are two “cable guides” to guide the scale cables and keep it in place.

OVERVIEW

The scales are provided by R&D WEIGHING (PTY) LTD, and the contact person is Peter Senekal (066 271 0540). Depending on the distance between the scales and the Main Unit, extra scale cable might be needed. Please specify this in the order. The company can also solder the extended cable if needed. 5kg and 1kg calibrated weights are also purchased from R&D Weighing and are used for scale calibration. Please find a suitable enclosure for the weights and keep them indoors to prevent corrosion. A parts list can be seen in TABLE 1 - SCALES PLATFORM PARTS LIST and the drawing can be seen in FIGURE 12 - DRAWING OF SCALES PLATFORM.

Table 1 - Scales Platform Parts List

Name	Quantity	Description	Where to buy / Part Number
Scale	2	Stainless Steel scale with B6N load cell	RnD Weighing
Indicator	2	Laumas Indicator - TLB485	RnD Weighing
1 kg weight	1	Used for calibration	RnD Weighing
5 kg weight	1	Used for calibration	RnD Weighing
Scale Cable	as needed	Extra scale cable (optional)	RnD Weighing
Base board	1	Base of platform. Marine Ply 1200x500x15	https://timbuildwoodstock.co.za/marine-plywood/
Cable Guide	2	To guide scale cable, wood	Hardware store – Builders Warehouse
RFID block	1	RFID mounting block to hold antenna, wood	Hardware store – Builders Warehouse
6 pin Plug	2	Plug to connect scale to Main Unit	RS: 111-5758
Canvas	±4m	To cover everything. Olive/Dark Green	Ludal Interior in Betty’s Bay s Bay https://www.ludalinterior.co.za/

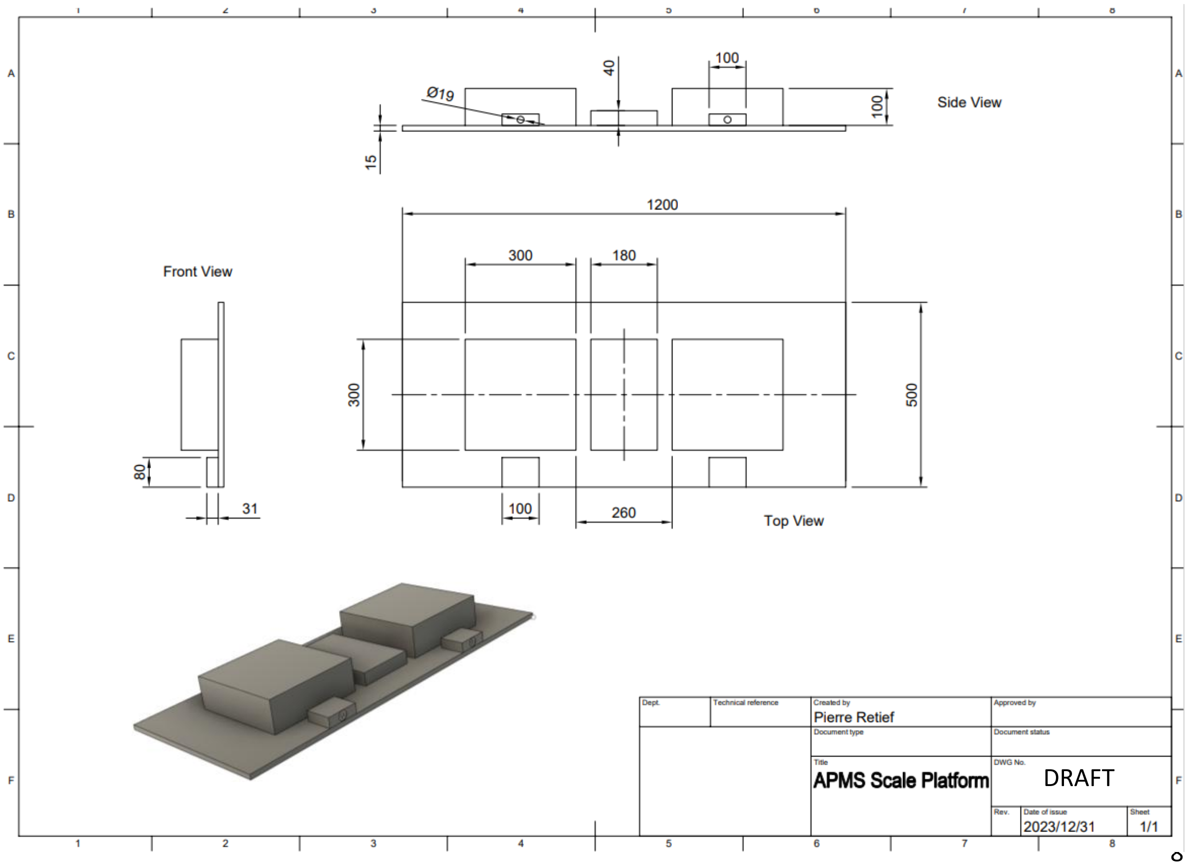


Figure 12 - Drawing of Scales Platform

SCALES

The scales consist of a stainless-steel frame with a type B6N loadcell mounted underneath. A pan rests on top of the frame to provide a flat surface. Note that the scales at Stony Point and Bird Island have older frames that are 90 degrees with respect to the new scales. If the scales need to be replaced in future, new holes can be drilled in the platform. Usually, however, only the load cell needs to be replaced and thus the existing frame can be used.

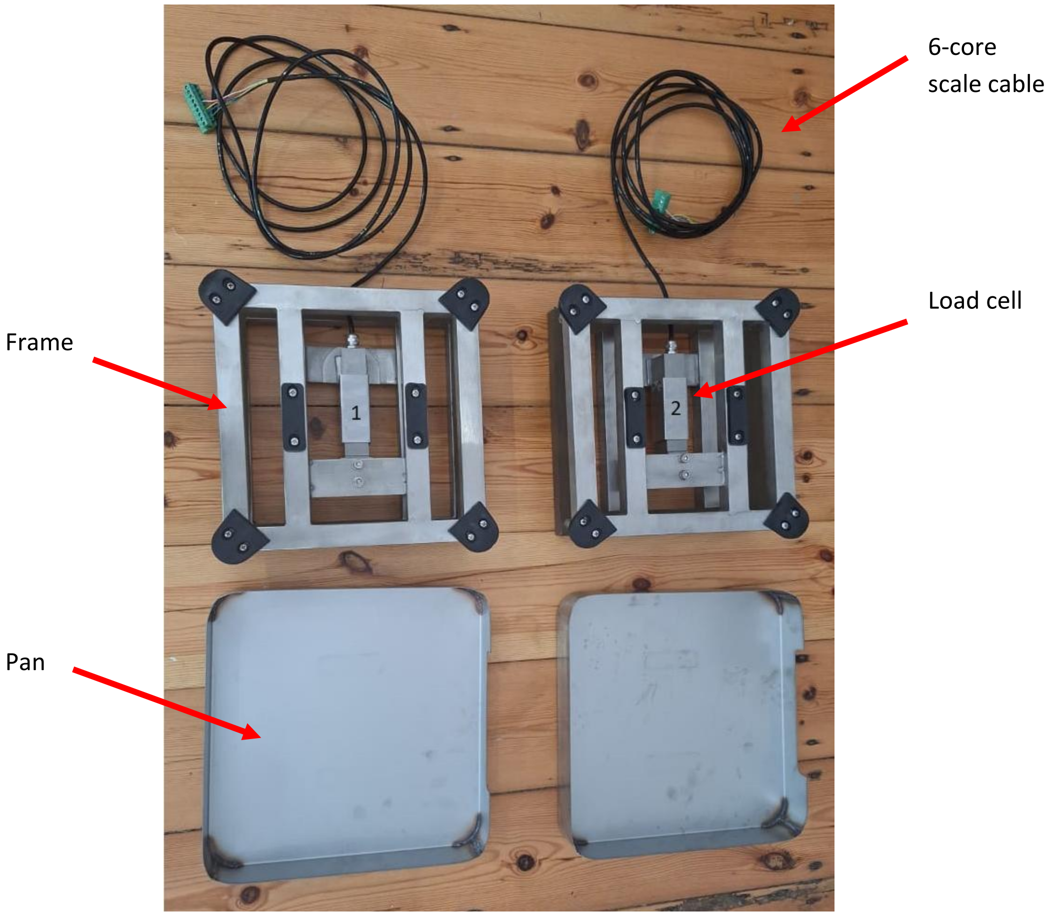


Figure 13 - Scales with Pans

The initial systems (Stony Point and Bird Island) used 50kg load cells. For the two newer systems (Dyer Island and Robben Island), a more sensitive 30kg loadcell is used. Use the 30kg load cell going forwards.

Note: The maximum safe loading weight of the loadcell is 150%. Thus, for a 50kg load cell, the max load is 75kg, and for the 30kg load cell, the max load is 45kg.

The part numbers for the load cells are as follows:

Stony Point and Bird Island: B6N-C3-50kg-3B6-S1

Dyer Island and Robben Island: B6N-C3-30kg-3B6-S1

Figure 14 - 50kg Load Cell

BUILDING THE PLATFORM

Start by gathering all the required parts for the platform.

Take note of the following while building the platform:

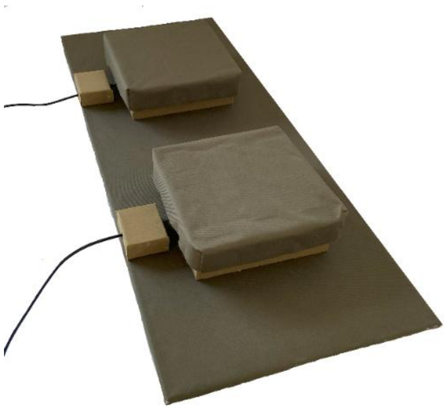


Figure 15 - Scales Platform without RFID block

- Label the scales and indicators immediately upon removing them from their packaging, as the scales and indicators are tied to each other.
- Cover the wooden base and scales with canvas. The canvas can be glued to the wood and scales using Contact adhesive. Silicon has also been used. The cable guides and RFID block can be screwed down from underneath.
- The scales are mounted using the threaded holes where the feet used to screw in.
- Mount the scales to the base using 4 ***mm bolts per scale.
- Scale 1 is colony side and Scale 2 is ocean side.
- The distance between the pans of the scales is set to 26 cm
- It is important to ensure the frame of the scale is flush with the base.
- Cover the cable guides with canvas and secure to the base. Feed the scale cables through the cable guide. See FIGURE 15 - SCALES PLATFORM WITHOUT RFID BLOCK
- Cover the RFID Mounting Block with canvas and secure it between the scales. Drill holes to allow zip ties to be used to clamp the RFID antenna to the block.
- Solder a 6-pin socket (RS 111-5764) to the end of each scale cable. This will plug into the 6-pin connector on the Main Unit.
- After the Scale Indicators have been installed and configured, test and calibrate the platform on a solid, level floor. This will confirm the scales are 100% functional before taking it to the field. See INSTALLING THE SCALE INDICATORS and ROUTINE CALIBRATION OF SCALES for more information.
- See FIGURE 16 - ROBBEN ISLAND SCALES PLATFORM for a completed platform in the field

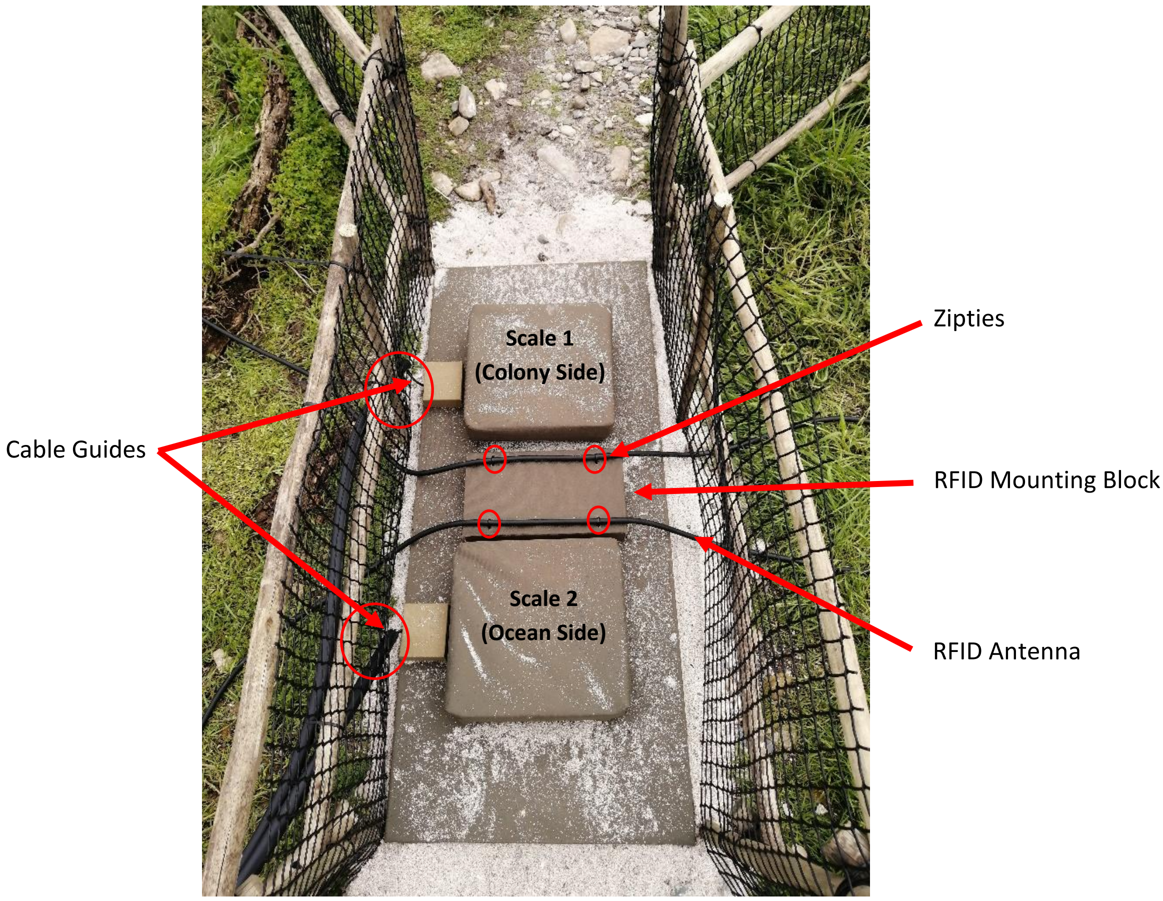


Figure 16 - Robben Island Scales Platform

APMS MONITOR

A separate circuit, called the APMS Arduino Monitor, or just "Monitor" for short, is used to constantly check different system parameters, and gives the administrator the ability to diagnose system faults remotely.
All the components needed to build the Monitor are listed in Table 2 - APMS Monitor Parts List. A website or RS part number is given, accurate at time of writing this manual.
The circuit diagram can be seen in FIGURE 17 - APMS ARDUINO MONITOR CIRCUIT DIAGRAM

Table 2 - APMS Monitor Parts List

Name	Designation	Quantity per board	RS Part Number	Notes
Strip board - 100 x 160		1	206-5879	RS
Fuse holder	FH1	1	765-2973	Min Order Quantity: 5
3A fuse	F1	1	668-5963	Min Order Quantity: 10
Monitor Mounting Plate		1		3D Printed. Design files on AWS S3
PCB Screw Terminal	J1, J2	2	193-0564	Min Order Quantity: 5
SIM800L	U2	1		Prototype DIY
SIM800L Antenna Cable		1		Prototype DIY
SIM800L Antenna		1		Prototype DIY or RS: 202-4179
1uF Capacitor	C1	1	538-1578	Min Order Quantity: 5
33pF Capacitor	C2	1	194-0457	Min Order Quantity: 10
10pF Capacitor	C3	1	194-0443	Min Order Quantity: 10
1000uF Capacitor	C4	1	711-1150	Min Order Quantity: 10
5.1V Zener	D1	1	544-3597	Min Order Quantity: 20
10uF 50V Capacitor	C5	1	811-8373	Min Order Quantity: 5
K7805 Voltage Regulator 500mA	U3	1	193-3976	RS
22uF Capacitor	C6	1	190-2953	Min Order Quantity: 10
Nano Every	U4	1	192-7586	RS
Push Button	SW1	1	135-9445	Min Order Quantity: 20
10k Trimpot	RV1	1	769-2202	RS
Temperature Sensor	U5	1	811-5595	Min Order Quantity: 10
NPN Transistor KSP2222ABU	Q1, Q2	2	739-0385	Min Order Quantity: 10
Resistor 22k	R9	1	135-954	Min Order Quantity: 10
Resistor 10k	R4, R10	2		https://www.prototypediy.co.za/
Resistor 4k7	R8	1		https://www.prototypediy.co.za/
Resistor 27k	R3, R6	2		https://www.prototypediy.co.za/
Resistor 5k6	R5, R7	2		https://www.prototypediy.co.za/
Resistor 6k8	R2	1		https://www.prototypediy.co.za/
Resistor 1k2	R1, R11	2		https://www.prototypediy.co.za/
LM2596 DC-DC Converter	U1	1		www.netram.co.za
Logic Level Converter	U6	1		https://www.netram.co.za
40 Pin Header Socket for Pi	J3	1	674-1252	RS
36 Pin Male Headers		5	547-3166	RS
30 Pin Female Header		2	767-9842	RS
Schottky Diode	D2	1	168-7175	comes in pack of 10
40 Pin Ribbon Cable		1		www.pishop.co.za
Buzzer	BZ1	1	771-6957	comes in pack of 5
M2.5 x 8mm machine screws		4	914-1567	RS

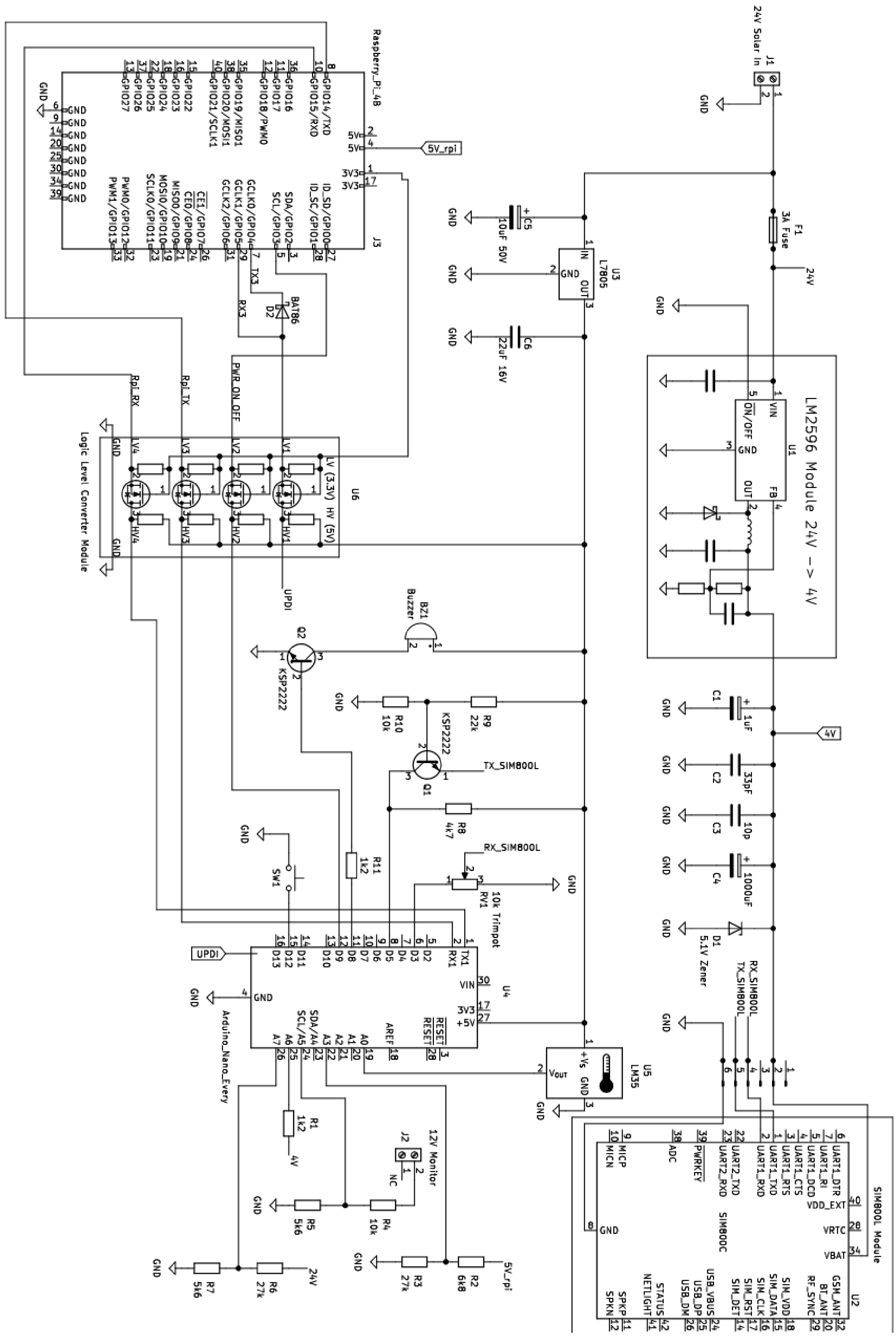


Figure 17 - APMS Arduino Monitor Circuit Diagram

OVERVIEW

The Monitor uses a separate sim card and can thus be used to fault find the system even when the main internet is down. The Arduino can be programmed through the Raspberry Pi once configured. Currently, the main features are:

- Monitors the 4V internal power supply, used to power the SIM card module.
- Monitors the 5V power supply, used to power the Raspberry Pi.
- Monitors the 12V power supply, used for the Scales and the Indoor Router.
- Monitors the 24V solar power supply.
- Monitors the ambient temperature inside the APMS Main Unit
- Alerts the administrator when the Monitor or Raspberry Pi boots up via SMS.
- Able to shut down and power up the Raspberry Pi via SMS.
- Able to send an APMS status report on request via SMS.
- Exchanges status reports with the Raspberry Pi every hour.

See USING THE APMS MONITOR for more information about the features mentioned above.

As the Monitor circuit is still a prototype, with relatively few components, the board is built on a piece of stripboard directly from the circuit diagram. Four boards have been built thus far, one for Stony Point, Bird Island, Dyer Island and Robben Island. If many more boards will be built in future, a PCB can be designed.

As most components are quite standard, it should not be hard to source parts or find suitable replacements.

The circuit board is mounted on a custom designed, 3D printed Monitor Mounting Plate. The mount has been designed to fit the existing DIN rail, for easy insertion/extraction. If no spare mounting plate is available, one must first be printed (print with ABS plastic to prevent warpage in the summer heat). The design file is located at: S3://apmsbirdlife/APMS/3D_prints/Monitor_Mount/Monitor_V_0_5.3mf

PROGRAMMING VIA THE ARDUINO SOFTWARE

The heart of the circuit is the Arduino Nano Every microcontroller. This is the successor to the popular Arduino Nano and provides slightly better capabilities for a lower price. The Arduino can be programmed in numerous ways. Two ways are discussed in this manual. The first will be using an unaltered Arduino and the standard Arduino Software, while the second method involves using the Raspberry Pi, together with some hardware modifications to the Arduino.

- Before building the circuit, it is a good idea to test the Arduino. As the Arduino is straight out of the box, the standard Arduino software will be used for the initial setup.
  - Install the Arduino Software (https://www.arduino.cc/en/software)
  - Install the Arduino megaAVR Core.
    - From the Tools menu, select “Boards”
    - Select “Boards Manager”
    - Search for and install “megaAVR”
  - Download the test sketch from:
    S3://apmsbirdlife/APMS/software/monitor_test_sketch.ino
  - Use a Micro USB cable to connect to the Arduino and select the appropriate com port
  - Click “Upload”
  - If the upload was successful, the LED will light up and the buzzer will sound when the button is pressed. [Untested]

> NOTE: If the Arduino has already been modified for UPDI, programming via the USB port will not be possible. First remove the soldered link and disconnect the UPDI wire. See UPDI PROGRAMMING

PREPARING THE STRIPBOARD

The 100x160 stripboard needs to be reduced to fit inside the Main Unit.

- Mark the four corners for drilling with a permanent marker. The locations of the holes are important as they correspond to the holes in the mounting plate. See FIGURE 18 – PREPARING THE STRIPBOARD
- The board does not have to be broken at the line, as seen in the picture, and a larger PCB can be obtained by breaking off a smaller piece. Just ensure the spacing between the holes are correct, as in the picture.

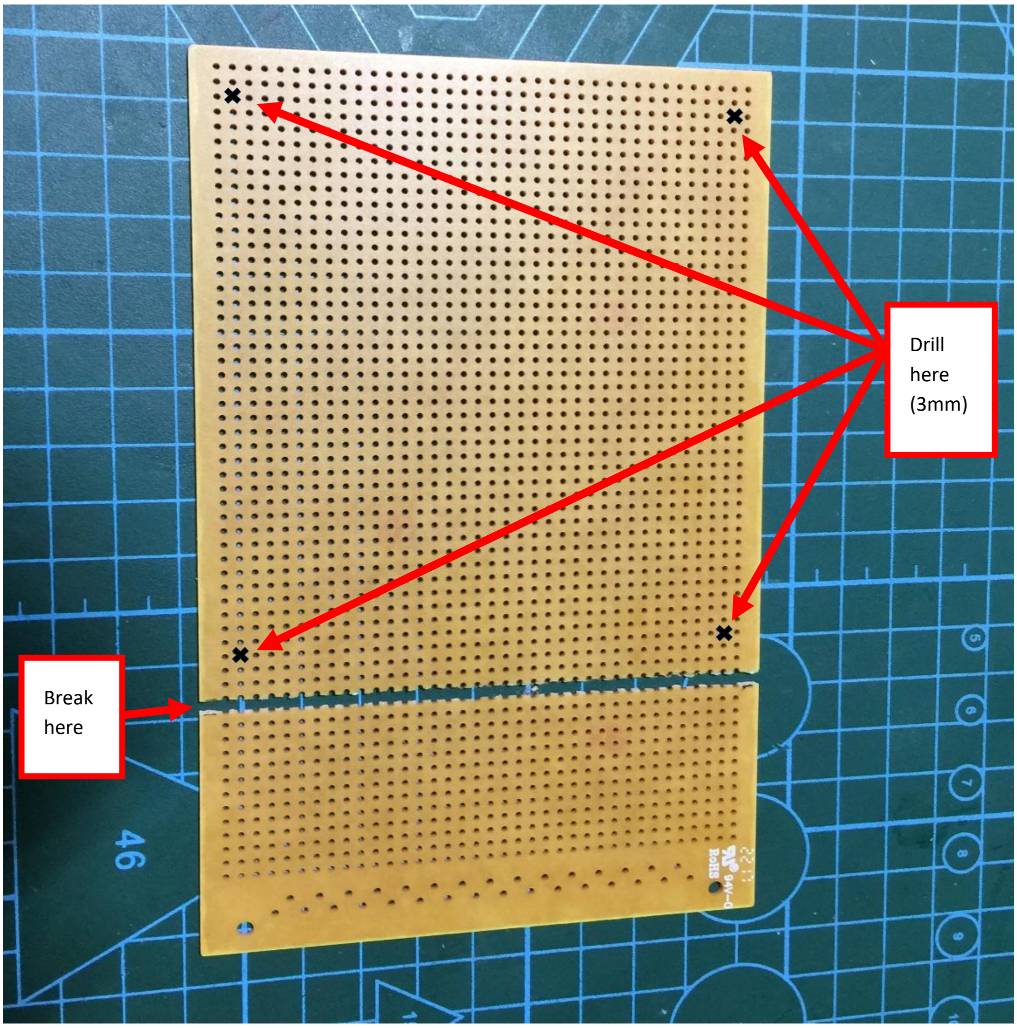


Figure 18 – Preparing the Stripboard

- Use the 3mm drill bit to drill the four corner holes in the circuit board.
  - Do a test fit as shown in FIGURE 19 – TEST FIT OF BLANK BOARD ON MOUNTING PLATE. Use the four 2.5mm machine screws to secure the blank circuit board to the mounting plate. <u>Note, do not exert too much force as the screws are going into plastic. (ABS)</u>

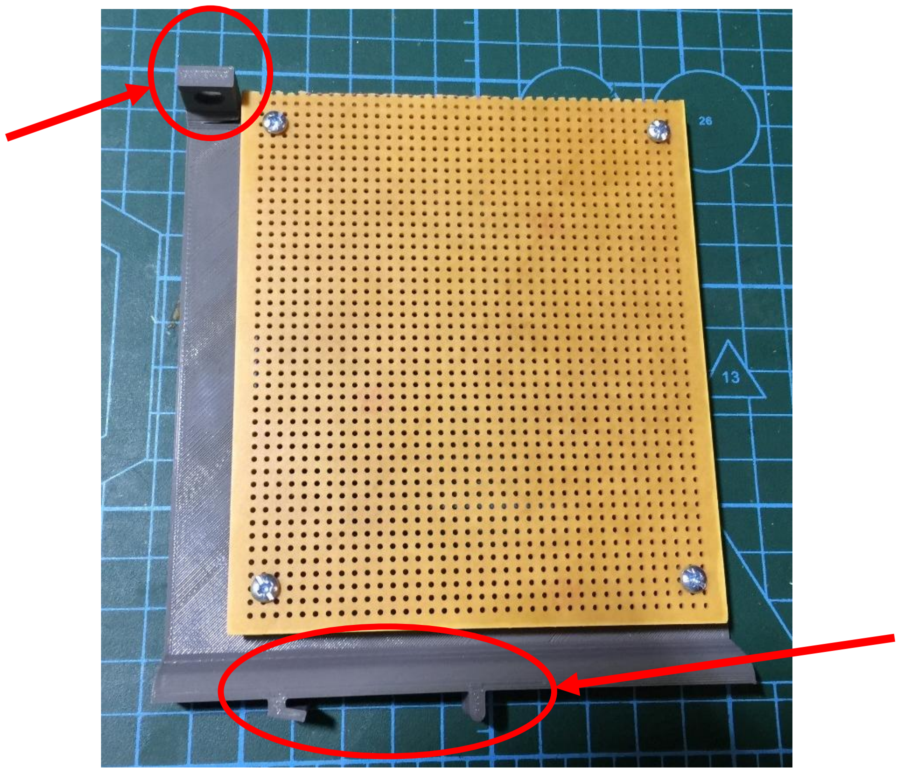


Figure 19 – Test fit of blank board on mounting plate

- The SIM800L antenna can be installed on the mounting bracket provided.
- If the signal at the site is particularly weak, an antenna with higher gain can be used: Current experimentation is with an omnidirectional antenna (RS Part Number: 202-4179).
- Remove the blank circuit board from the mounting plate and start building the circuit according to the circuit diagram.

4V AND 5V POWER SUPPLIES

- Start with the main 24V DC connector, fuse holder and the LM2596 DC-DC converter.
- Insert a 3A fuse into the fuse holder and provide 24V to the main DC connector. Adjust the output voltage of the LM2596 DC-DC converter to 4.0 V.
- Install the K7805 voltage regulator, capacitors C5 and C6 and check for 5V on the output.
- Break one of the 30 pin female header strips in half, to obtain two 15 pin strips. Use these two strips as a socket for the Arduino Nano. See FIGURE 20 - PARTIALLY BUILT MONITOR FOR STONY POINT.
- Test for 5V between the GND and VCC pins on the Arduino header. Do not install the Arduino yet.

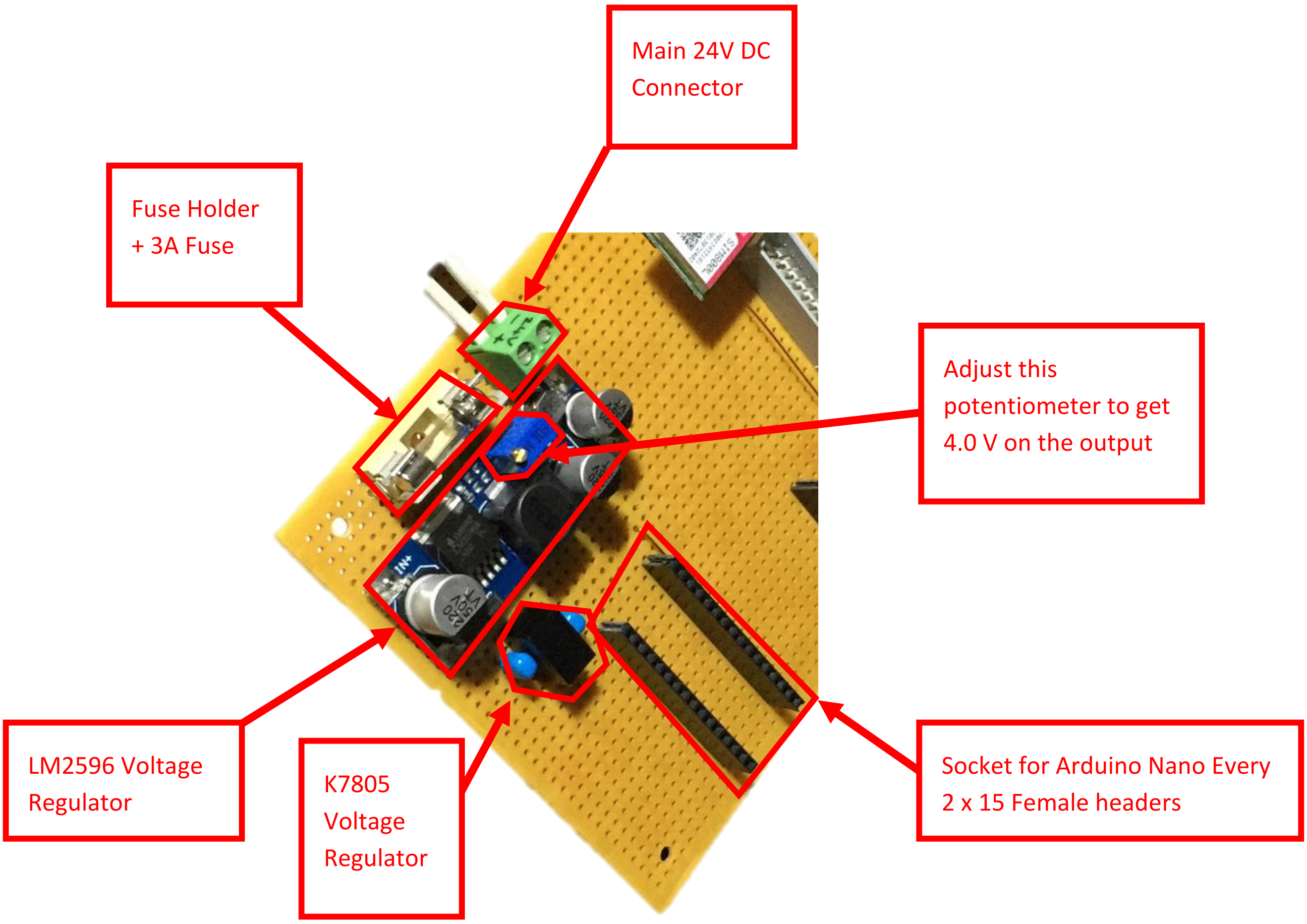


Figure 20 - Partially built Monitor for Stony Point

SIM800L POWER SUPPLY

- The Monitor uses a SIM800L module to send/receive SMS messages. Four pins are used: VCC, GND, RXD and TXD. VCC requires a supply voltage of 3.4V – 4.4V. The monitor uses the LM2596 regulator to provide a stable 4.0V.
- The first three monitors (Stony Point, Bird Island and Dyer Island) all use a smaller green PCB. As this module has been difficult to procure, a larger module with a red PCB was used at Robben Island and will be used for future boards. As we only use 4 pins, the extra features of the red board are not used.
- Unfortunately, the modules are not directly interchangeable as the RX and TX pins are reversed (see pictures below).
- An adapter board can be made, or the Monitor PCB modified, if a green module at the first three sites ever needs to be replaced with a red one.

Figure 21 - Green SIM800L Module (used at Stony, Bird and Dyer)

Figure 22 - Red SIM800L Module used at Robben

Install the SIM800L Module header in the top left section of the board, as the module needs to be located close to the antenna mounting bracket. See FIGURE 23 - ROBBEN ISLAND MONITOR

Use female header pins to create a socket for the module. This enables the module to be easily removed for testing or to be replaced if faulty.

- Install capacitors C1 – C4 and Zener diode D1.
- With the SIM800L not inserted, confirm on the header that there is 4.0 V between VCC and GND.

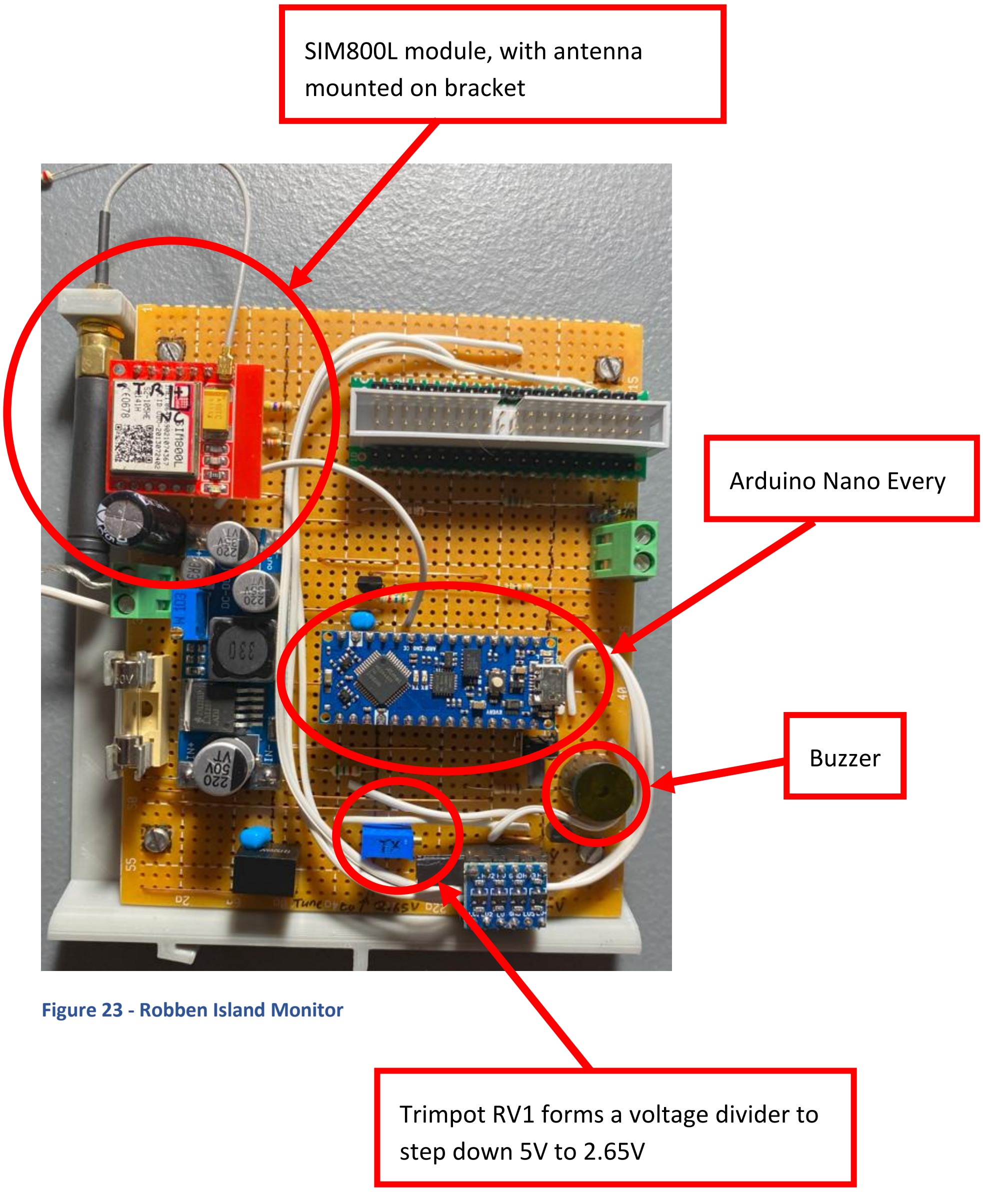


Figure 23 - Robben Island Monitor

- <u>SIM800L RX and Arduino TX</u>
  Since the TX of the Arduino is between 0V and 5V, a simple voltage divider is used to step the voltage down to a level suitable for the SIM800L. Install Trimpot RV1 as per the schematic.
  With the SIM800L module and Arduino <u>not</u> inserted, apply 5V to the voltage divider (pin D3 – See FIGURE 24 - ARDUINO NANO PINOUT). With 5V applied to D3, a voltage divider is created by trimpot RV1. Adjust the trimpot until the voltage on the centre pin, is 2.65 V. This pin will be connected to the receive line of the SIM800L.

- <u>SIM800L TX and Arduino RX</u>
  The voltage level of the SIM800L needs to be stepped up for use with the Arduino. To achieve this, a circuit consisting of three resistors (R8, R9 and R10) and one NPN transistor (Q1) is used, as recommended by the SIM800L manual. Install the above-mentioned resistors and transistors and ensure the input of the circuit is connected to the transmit line of the SIM800L.

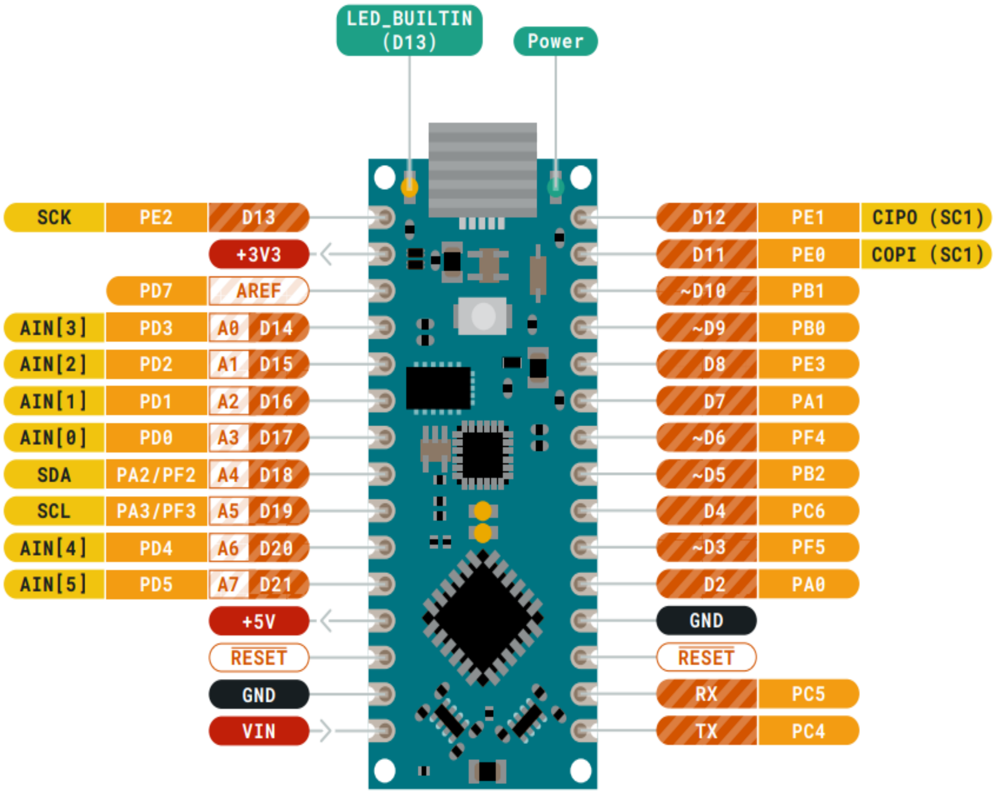


Figure 24 - Arduino Nano Pinout

The connector used to connect the Monitor to the Raspberry Pi can be seen in **Error! Reference s ource not found.**. To use the connector with stripboard, the pins must either be bent outwards, or a “breakout” board made from a piece of scrap protoboard (See **Error! Reference source not found.**). B oth methods have been used and are effective. A pre-made break-out board can be purchased if available. If a PCB is designed in future, it will be possible to solder the connector directly on the board.

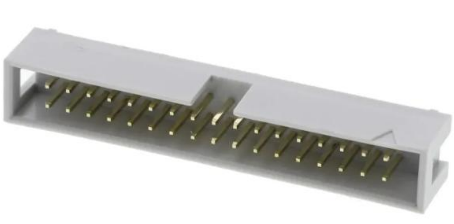


Figure 25 - Raspberry Pi Connector

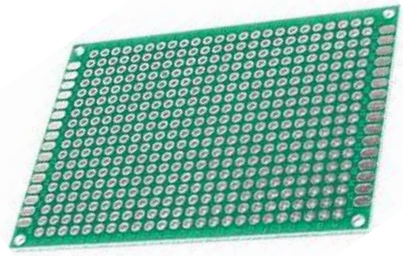


Figure 26 - Proto Board

The Raspberry Pi 40-pin Connector pinout can be seen in FIGURE 27 - RASPBERRY PI GPIO PINOUT.
A simple voltage check of 3.3V on pin 1 and 5V pin 2 should confirm orientation if unsure.
The Raspberry Pi GPIO pins and their functions can be seen in TABLE 3 - RASPBERRY PI GPIO PINS AND FUNCTIONS

Pin	Function
GPIO3	Power ON/OFF
GPIO4	Raspberry Pi UART3 TX
GPIO5	Raspberry Pi UART3 RX
GPIO14	Raspberry Pi TX
GPIO15	Raspberry Pi RX

Table 3 - Raspberry Pi GPIO Pins and Functions

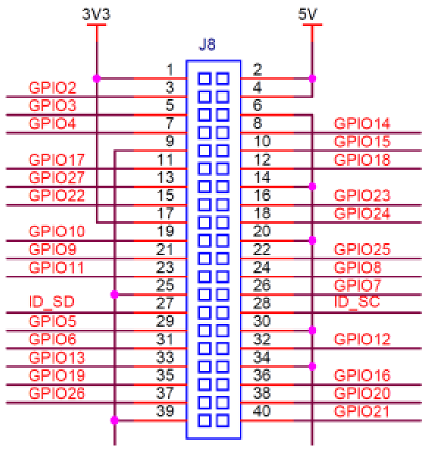


Figure 27 - Raspberry Pi GPIO Pinout

A 40-pin ribbon cable is used to connect the Monitor to the Raspberry Pi. See FIGURE 28 - 40 PIN RIBBON CABLE. After building the Monitor, check the voltages on the other end of the ribbon cable, to ensure it is not above 3.3V, before plugging it into the Pi.

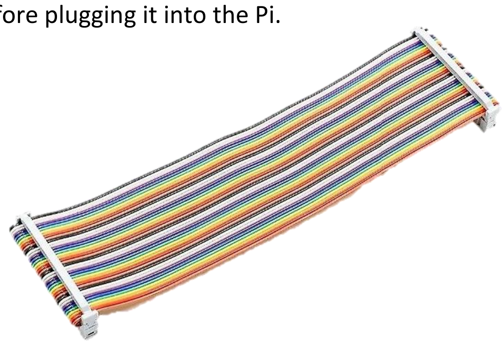


Figure 28 - 40 Pin Ribbon Cable

LOGIC LEVEL SHIFTER

- Since the Pi uses 3.3 V for communication, and the Arduino 5 V, a logic level shifter is needed. These modules are available cheaply and two 6-pin female headers should be used as a socket for easy install/removal. See FIGURE 29 - DYER ISLAND MONITOR.
- The level shifter uses the 3.3V on pin 1 of the Raspberry Pi connector for the LV connection and the 5V Arduino supply voltage for the HV connection.

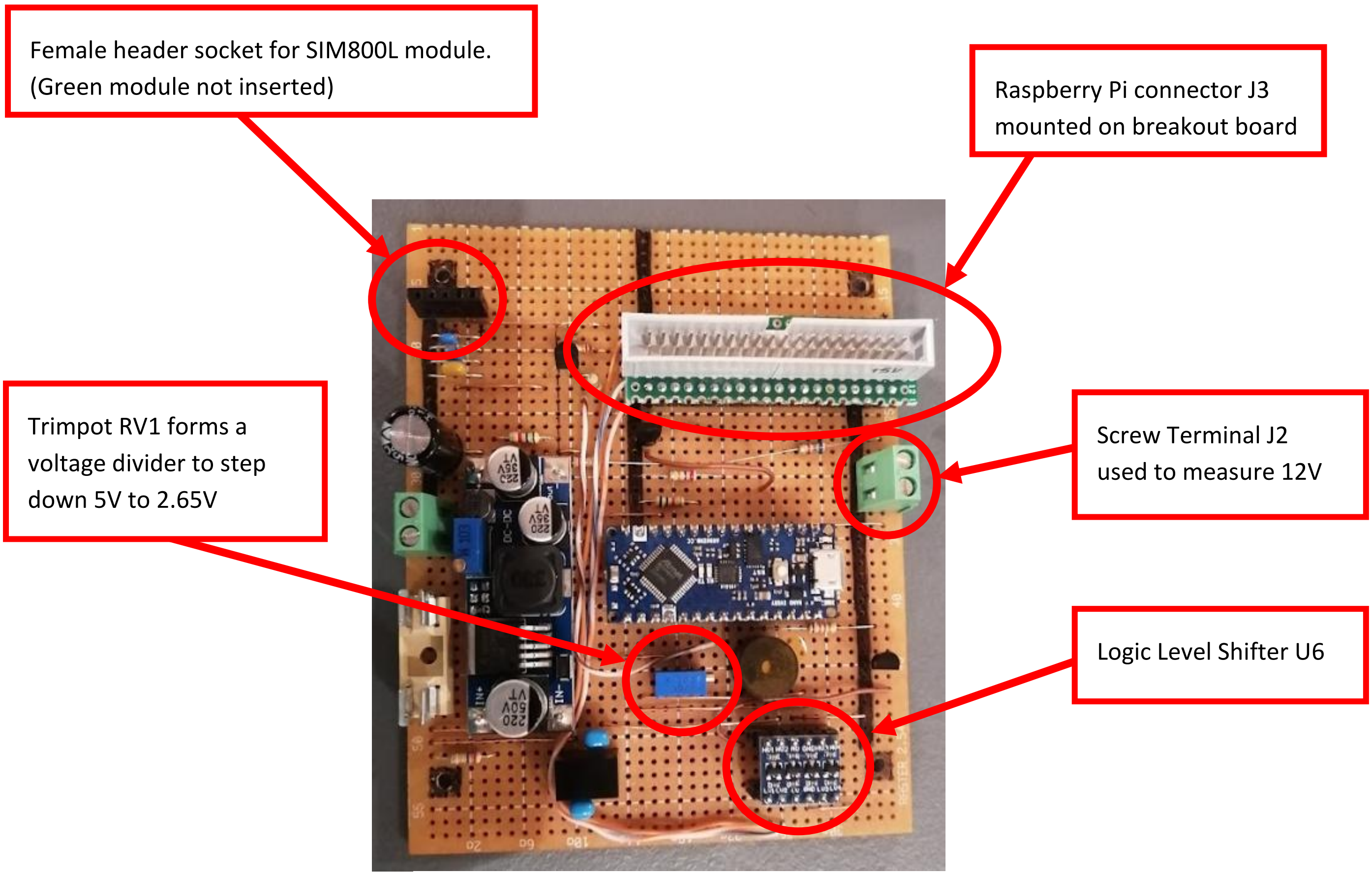


Figure 29 - Dyer Island Monitor

UPDI PROGRAMMING

The Arduino Nano Every uses UPDI (Unified Program and Debug Interface) for programming. This uses only 1 wire for programming, but a more complex communication protocol is needed. To use UPDI a wire needs to be soldered to a pad located on the underside of the Arduino. A link must also be soldered. See FIGURE 30 - ARDUINO NANO EVERY UPDI WIRE AND LINK

NOTE: After the link has been made, the board can no longer be programmed via the Arduino software and USB cable. It will be programmed via the Raspberry Pi using the UPDI wire. If the Arduino software needs to be used during fault finding, unsolder the link. The pads can break off with successive heating sessions, so be cautious.

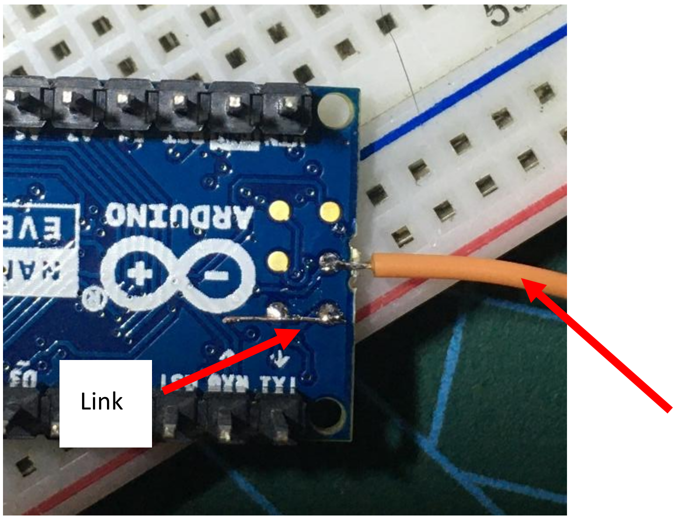


Figure 30 - Arduino Nano Every UPDI Wire and Link

- To program the Arduino remotely via the Raspberry Pi, a utility called “pymcuprog” is used. This utility allows the Raspberry Pi to program the Arduino via a serial connection. This serial connection is established using UART3 (RX3 and TX3) on the Raspberry Pi as a serial port. See TABLE 3 - RASPBERRY PI GPIO PINS AND FUNCTIONS.

- To program the Arduino via a serial port, a hardware modification is required. The pymcuprog documentation (https://pypi.org/project/pymcuprog/) describes various combinations of diodes and resistors to achieve this, for various setups. After experimentation, using only a Schottky diode (D2) delivered the best results. Note the cathode of D2 is connected to TX3.

- The UPDI wire is connected to HV1 on the level shifter. See FIGURE 17 - APMS ARDUINO MONITOR CIRCUIT DIAGRAM

- A script has been written to compile and upload the firmware to the Arduino, from the Raspberry Pi, thus the user does not need to use pymcuprog directly. See PROGRAMMING THE ARDUINO MONITOR VIA THE PI.

VOLTAGE MONITORING

If any of the main supply voltages are outside allowable limits, an SMS will be sent to the operator. Currently, the following important voltages are monitored:

- 4V: This is the SIM800L supply voltage. This voltage is generated by the LM2596 voltage regulator and is fed to analog input A6 of the Arduino, via resistor R1.

- 5V: This is the 5V supply for the Raspberry Pi. The 5V is picked up from pin 2 or 4 on the Raspberry Pi connector. See FIGURE 27 - RASPBERRY PI GPIO PINOUT.
  As this voltage is close to, and might exceed, the Arduino maximum safe voltage, a voltage divider consisting of R2 and R3 is used to step the voltage down. The Arduino uses A3 to measure this voltage. Note that this 5V is separate and should be isolated from the Arduino 5V supply. This is in essence measuring the output of the 5V Meanwell Power Supply.

- 12V: This is the 12V supply for the scales and the router (if an indoor router is used). This voltage is fed via a wire from the Meanwell 12V Power Supply to screw terminal J2. Install the screw terminal in the top right section of the board to allow for easy connecting when installing the board. See FIGURE 29 - DYER ISLAND MONITOR. Resistor R4 and R5 is used to step the 12V before connecting it to A5 of the Arduino

- 24V: This is the main solar supply voltage. This supplies power to the BioMark reader, the APMS Main Unit and the Outdoor Router (if one is installed). The voltage is measured at the input of the Monitor and is stepped down via resistor divider R6 and R7 before being connected to A7 of the Arduino.

TEMPERATURE MONITORING

The ambient temperature inside the Main Unit is monitored using a LM35 temperature sensor (U5). The sensor operates on 5V and needs no other components to operate. The output of the sensor is read directly by analog pin A0 on the Arduino.

BUZZER AND BUTTON

A buzzer and button are included for debugging and future options. Currently the Monitor will beep once initialisation is complete, and the button will power the Raspberry Pi down/up.

- The buzzer is switched via transistor Q2 with base current being limited by resistor R11.
- The button is connected directly to pin D12 on the Arduino and makes use of the built-in pull-up resistor, thus requiring no other components. Place the button on the board in a location where it will be easy to press once the board is installed.

TESTING THE MONITOR CIRCUIT BOARD

Before testing communication between the Raspberry Pi and the Arduino Monitor, some basic tests must be done to ensure no components will be damaged. A dual variable power supply will be needed to test the various voltages.

Ensure the firmware was uploaded successfully as explained in the section PROGRAMMING VIA THE ARDUINO SOFTWARE

Plug the SIM800L into its socket and connect the antenna.

- Plug the Arduino into its socket.
- Plug the Level Shifter into its socket
- Plug the 40-pin ribbon cable into the socket on the Monitor but do not plug the other end into the Pi.
- Power the Monitor via the main 24V DC connector
- Observe lights flashing on the SIM800L
    - The green light on the SIM800L should flash rapidly at first, and after about 20 seconds, it should start flashing slowly. This indicates that the module was able to connect to the cell network successfully. Try different antennas and orientations for best results.
- Check that there is 5.0V on the Arduino and Level Shifter supply pins.
- Check that there is 4.0V on the SIM800L supply pin.
    - Check that the same 4.0V is measured on pin A6 of the Arduino.
- Check that none of the pins on the Raspberry Pi connector has 5V on it.
    - Note pin 2 and 4 will have 5V coming from the Pi when it is connected at a later stage
- Supply 12V to connector J2 and ensure the voltage has dropped to below 5V on pin A5 of the Arduino
- Measure the stepped-down solar supply voltage on pin A7 of the Arduino. The 24V should have been stepped down to a value less than 5V.
- Measure the output voltage of the temperature sensor on pin A0 of the Arduino. The sensor outputs 10mV per °C, so at 25°C, the voltage should be about 0.25V.

APMS MAIN UNIT

The APMS Main Unit consists of all the electronics needed for the system.

For a complete parts list, see TABLE 4 - APMS MAIN UNIT PARTS LIST

For a system diagram, see FIGURE 31 - APMS SYSTEM DIAGRAM

Table 4 - APMS Main Unit Parts List

Name	Quantity	Description	RS Part Number
Enclosure	1	Housing for the APMS system	508-2798
DIN Rail	1	DIN rail for mounting electronics	188-2538
DC 30V 5A Socket	1	DC panel socket for main 24V solar input voltage	705-1525
DC 30V 5A Plug	1	For cable coming from 24V Solar Supply	705-1519
USB panel mount socket	1	To connect to BioMark Reader	111-6761
DC panel socket (12V 1A)	1	12V output for LTE Router	111-3725
DC 12V 1A Plug	1	Plug for Indoor Router	138-9405
6-pin socket	2	Sockets for scales	111-5764
6-pin Plug	2	Plugs for scales	111-5758
USB Cable	1	Mini USB cable to connect to BioMark Reader	822-3226
12V 30W PSU	1	12V Power Supply for Router and scales	175-7778
5V 15W PSU	1	5V Power Supply for Raspberry Pi	175-7775
Fuse Holder	1	Fuse holder	501-871
3A Fuse	1	Main Fuse	668-5963
USB-RS485 Converter	2	USB to RS485 converter	687-7834
64GB SD Card	1	Main storage for APMS operating system.	Takealot
64GB USB Flash drive	1	USB Flash drive for data backup	Takealot
Raspberry Pi 4	1	Main computer for APMS system	182-2096
Heatsinks	3	For cooling Raspberry Pi processors	201-2169
Fan	1	For cooling Raspberry Pi processors	789-7858
2-pin Female Header for Fan	1	Same as used for APMS Monitor	767-9842
USB-C cable for power	1	Cabe for powering Raspberry Pi	192-4730
Mount for Raspberry Pi	1	3D Printed DIN Rail Mount for Raspberry Pi	Printed in-house
Mount for Arduino Monitor	1	3D Printed DIN Rail Mount for APMS Monitor	Printed in-house
3D Printed Fan Mount	1	To mount the cooling fan above the CPU	Printed in-house
Monitor Board	1	Used to monitor health of main system	Built in-house
SIM card	2	Sim card for main system and monitor board	Vodacom shop
Internal wiring		Used for various connections inside box	Hardware store
Ferrules		For better contact with screw terminals	Takealot
Heat-shrink		Insulate connections	Takealot
M2.5x8 machine screws	4	Screws to secure the Raspberry Pi	914-1567
M3x16 machine screws	4	Screws to secure the fan	190-462

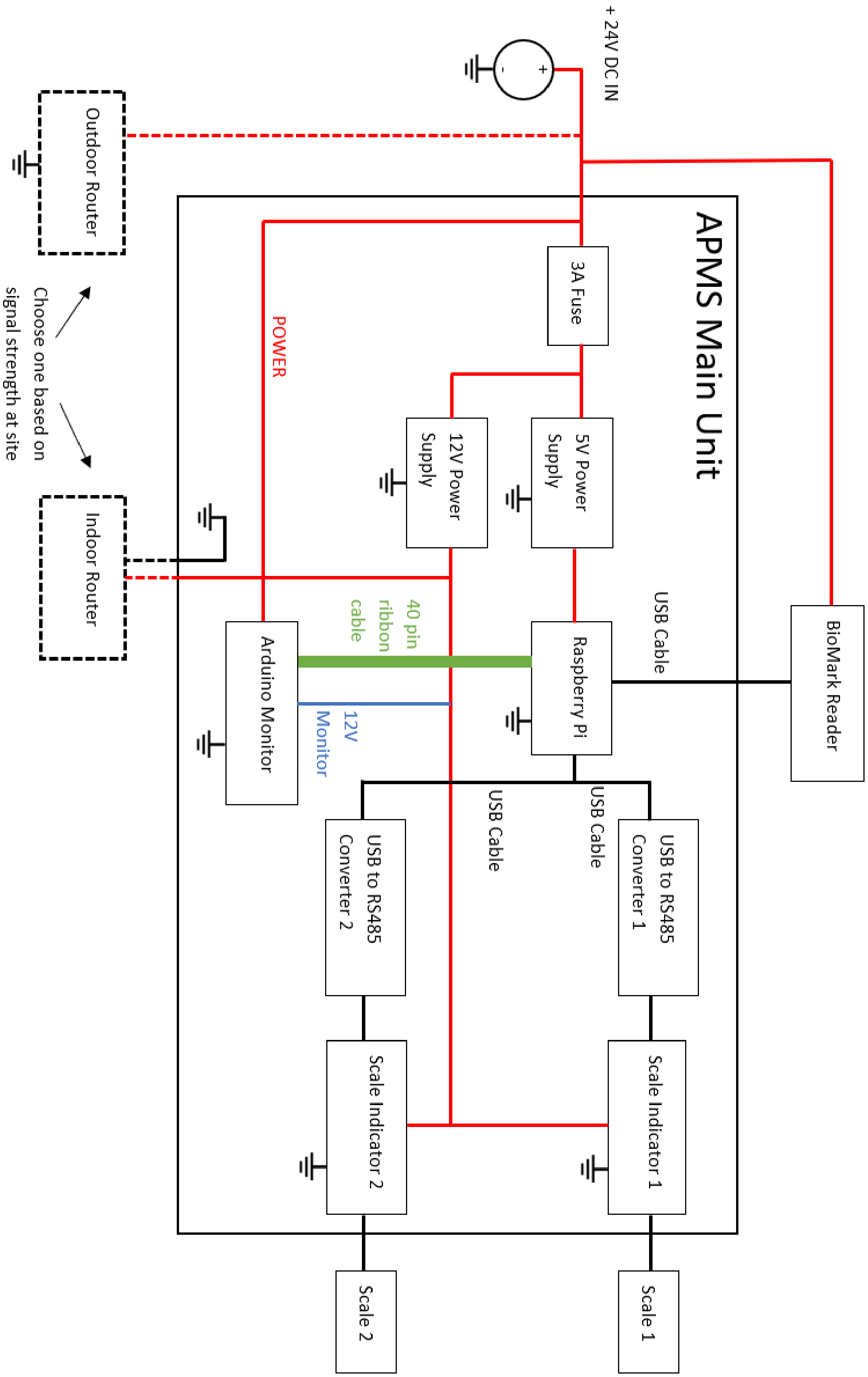


Figure 31 - APMS System Diagram

PREPARING THE USB PANEL MOUNT SOCKET

The USB Panel Mount Socket does not have adequate strain relief. Use Pratley Putty to reinforce the cables. Take note to avoid getting the putty on the threads. Please see FIGURE 32 - PRATLEY PUTTY AND USB SOCKET WITH STRAIN RELIEF and FIGURE 33 - CLOSE-UP OF AREA STRENGTHENED.

Figure 32 - Pratley Putty and USB Socket with strain relief

Figure 33 - Close-up of area strengthened

PREPARING THE MAIN UNIT ENCLOSURE

Prepare the enclosure before starting the build. RS Part numbers included in brackets.

- Open the enclosure (508-2798) and screw down the DIN rail (188-2538) using the screws provided.
    - Note, it takes a fair bit of force.
- Print the template seen in FIGURE 35 - FRONT TEMPLATE. Check the dimensions with a ruler after printing.
- Glue/stick the template on to the front of the enclosure and drill the holes as indicated on the template.
    - The 11mm and 21mm diameter holes are an odd size. Either go one size smaller and enlarge, or use one size larger, with it being a slightly loose fit.
- Do a test fit of all the sockets: Main DC socket (705-1525), USB Panel Mount Socket (111-6761), 12V DC socket (111-3725) and the two scale sockets (111-5764)
- The sockets can be removed to ease in the rest of the build.
- Use a label maker, or a permanent marker, and label each socket. See Figure 34 - APMS Main Unit Front

Figure 34 - APMS Main Unit Front

Note This is Version 1.1 of the APMS. More information can be found in the APMS Report. The main changes were the following: • Improved Fan with newly designed fan bracket. • Use of RS485 converters, instead of RS232. • Scales were mounted on a platform instead of inside boxes

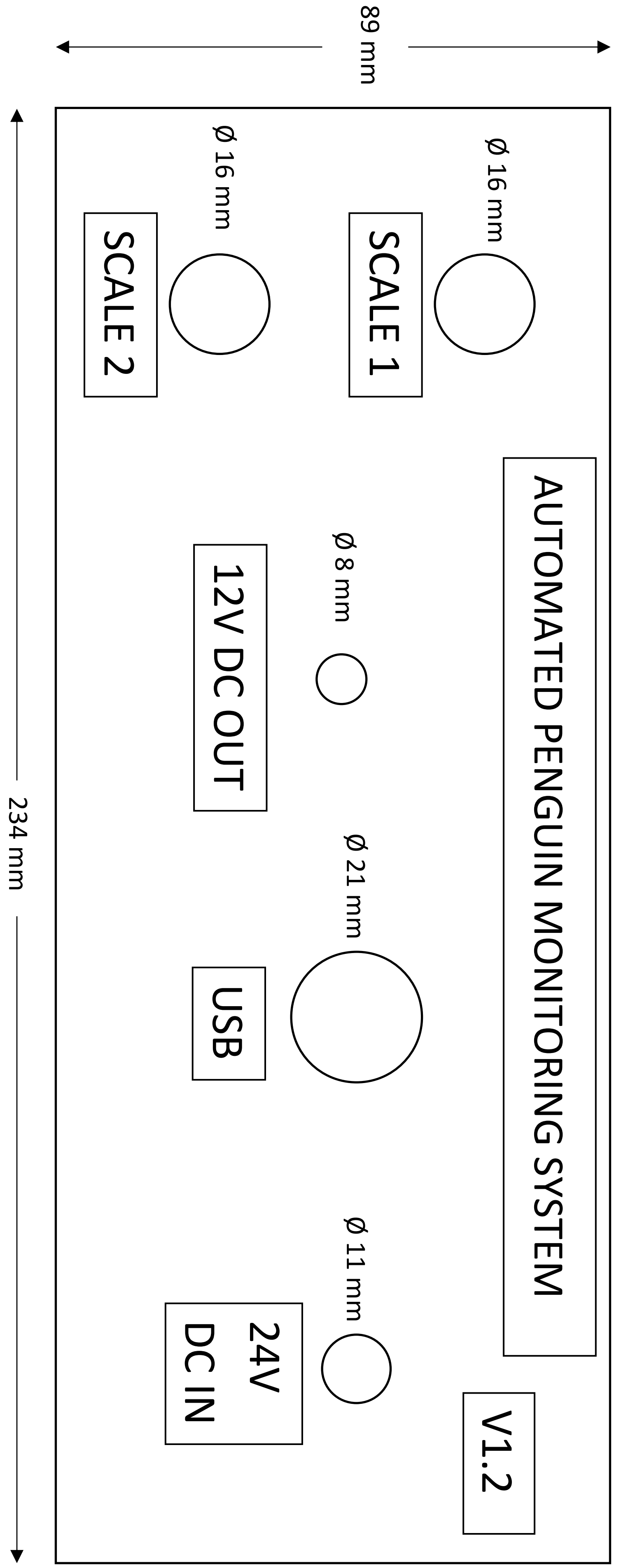


Figure 35 - Front Template

MOUNTING THE RASPBERRY PI

Use four 2.5mm machine screws to mount the Raspberry Pi on the mounting plate. See FIGURE 36 - RASPBERRY PI 4 WITH MOUNTING PLATE

Add heatsinks (RS: 201-2169). See FIGURE 37 - RASPBERRY PI MOUNTED WITH HEATSINKS

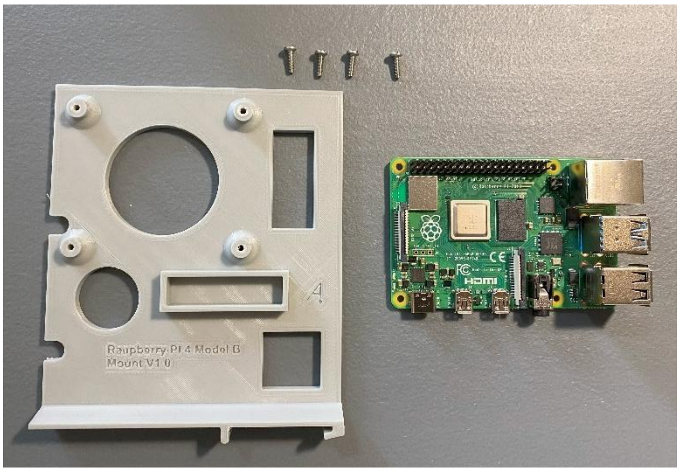


Figure 36 - Raspberry Pi 4 with Mounting Plate

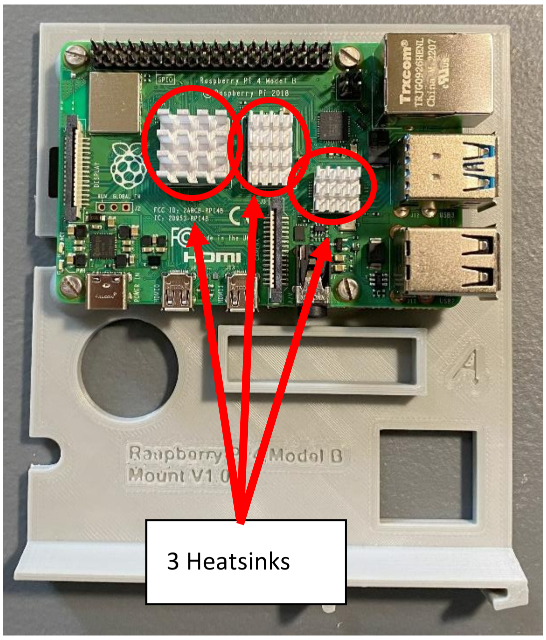


Figure 37 - Raspberry Pi Mounted with Heatsinks

PREPARING THE FAN

Solder a 2-pin female header connector the ends of fan cable. Use the four M3 screws and mount the fan to the fan bracket. See FIGURE 38 - CPU FAN MOUNTED ON BRACKET

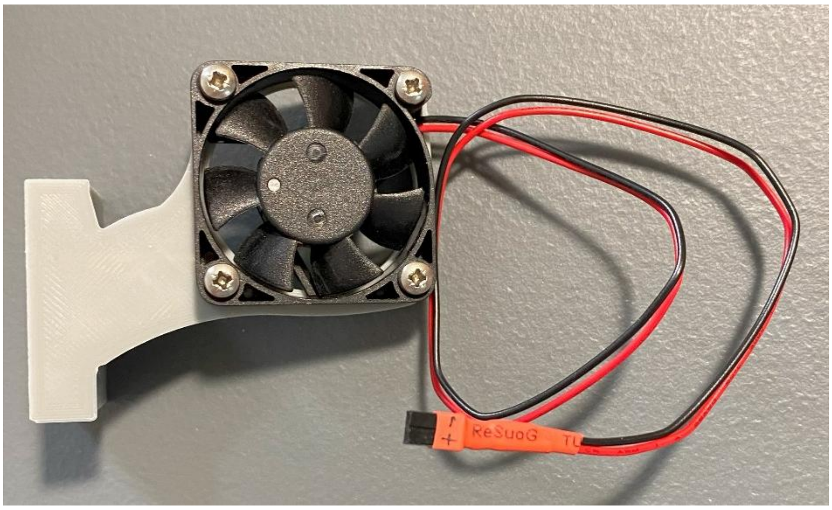


Figure 38 - CPU Fan Mounted on Bracket

The fan + bracket fits into the slot provided on the Raspberry Pi mounting plate. The fan bracket slides in with a friction fit to allow easy installation/removal. Even though 5V power is provided to the fan from the Monitor, the 5V is retrieved from the Raspberry Pi connector, and not from the 5V supply on the Monitor itself. This is done to prevent the fan stopping should the Monitor fail.

WIRING THE APMS MAIN UNIT

- The location of the DIN mounted components, as well as the power wires, can be seen in FIGURE 39 - LOCATION OF DIN MOUNTED COMPONENTS AND POWER WIRES
- The main power wires consist of a ground wire, a 5V and a 12V wire.
- The APMS uses a DIN mounted Fuse Holder (501-871) with a 3A Fuse (668-5963) to provide protection.
- Two Meanwell Power Supplies are used to step the 24V DC down to the required voltages:
    - 5V 15W Power Supply (175-7775) for the Raspberry Pi
    - 12V 30W Power Supply (175-7778) for the scales (and indoor router if used).
    - Note the polarity of the two power supplies, as they are opposite to each other.

- Install the 24V DC and 12V DC connectors.
- Mount the Arduino Monitor, Raspberry Pi, Fuse Holder, 5V Power Supply, 12V Power Supply and the two Scale Indicators on the DIN Rail.
    - If the Arduino Monitor has not been built yet, use the 3D printed mounting plate is this step.

- Run a ground wire from the Main 24V DC Connector to the following components:
    - Arduino Monitor
    - Input and Output of 5V Power Supply
    - Input and Output of 12V Power Supply
    - Raspberry Pi
    - Two Scale Indicators
    - 12V DC connector

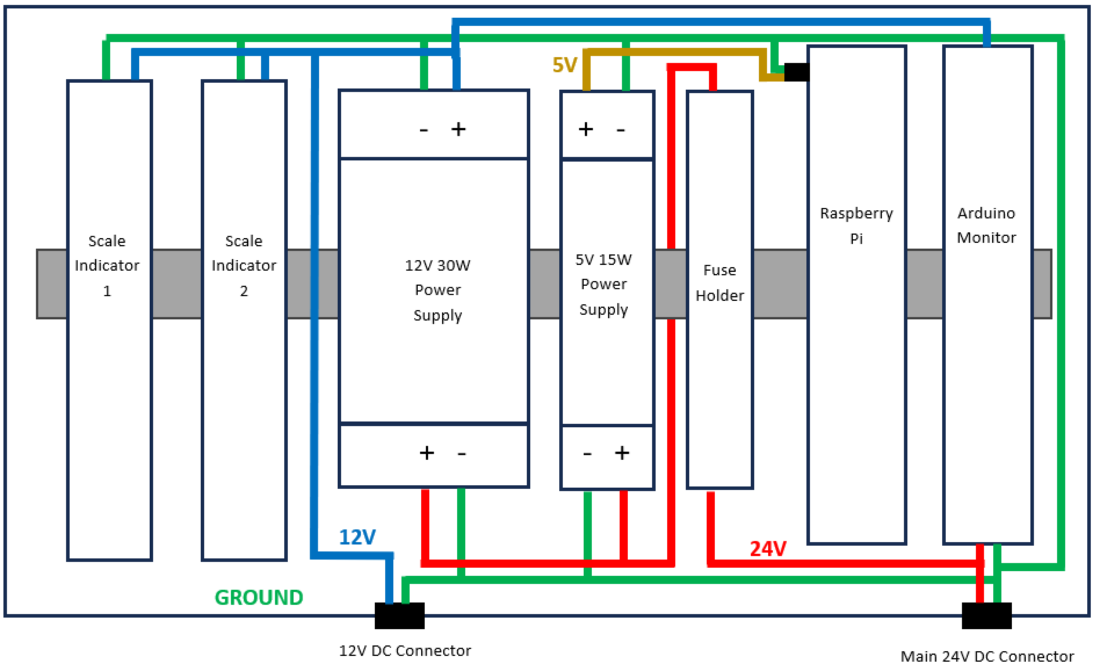


Figure 39 - Location of DIN Mounted Components and Power Wires

- With the ground wire laid out, the locations of the different components are established. The Arduino Monitor, Raspberry Pi and the two Scale Indicators can now be removed to ease the rest of the wiring.

- Run two positive wires from the Main 24 DC Connector:
  - One through the fuse holder and to the input of the 5V and 12V Power Supplies,
  - One to the power connector on the Arduino Monitor.
    - The Arduino Monitor has its own 3A fuse. This allows the Monitor to continue operating even if a fault caused the main fuse to blow.

- On the output of the 5V power supply, run a wire for the Raspberry Pi. This positive 5V wire, together with the ground wire created earlier, will be connected to a USB-C cable, to power the PI. Currently, the most cost-effective method to obtain a quality cable with a USB-C connector, is to use a USB-B to USB-C cable from RS Components (192-4730) and to cut off the USB-B connector. Please update this manual when a better alternative is sourced. Keep the cable running to the Pi short to avoid extra cabling in the Main Unit.

- Run a positive 12V wire from the output of the 12V Power supply to the following locations:
  - Arduino Monitor 12V monitoring screw terminal
  - Both Scale Indicators
  - 12V DC Socket
- Crimp ferrules to the ends of all the power wires
- See FIGURE 40 - POWER SUPPLIES AND WIRING

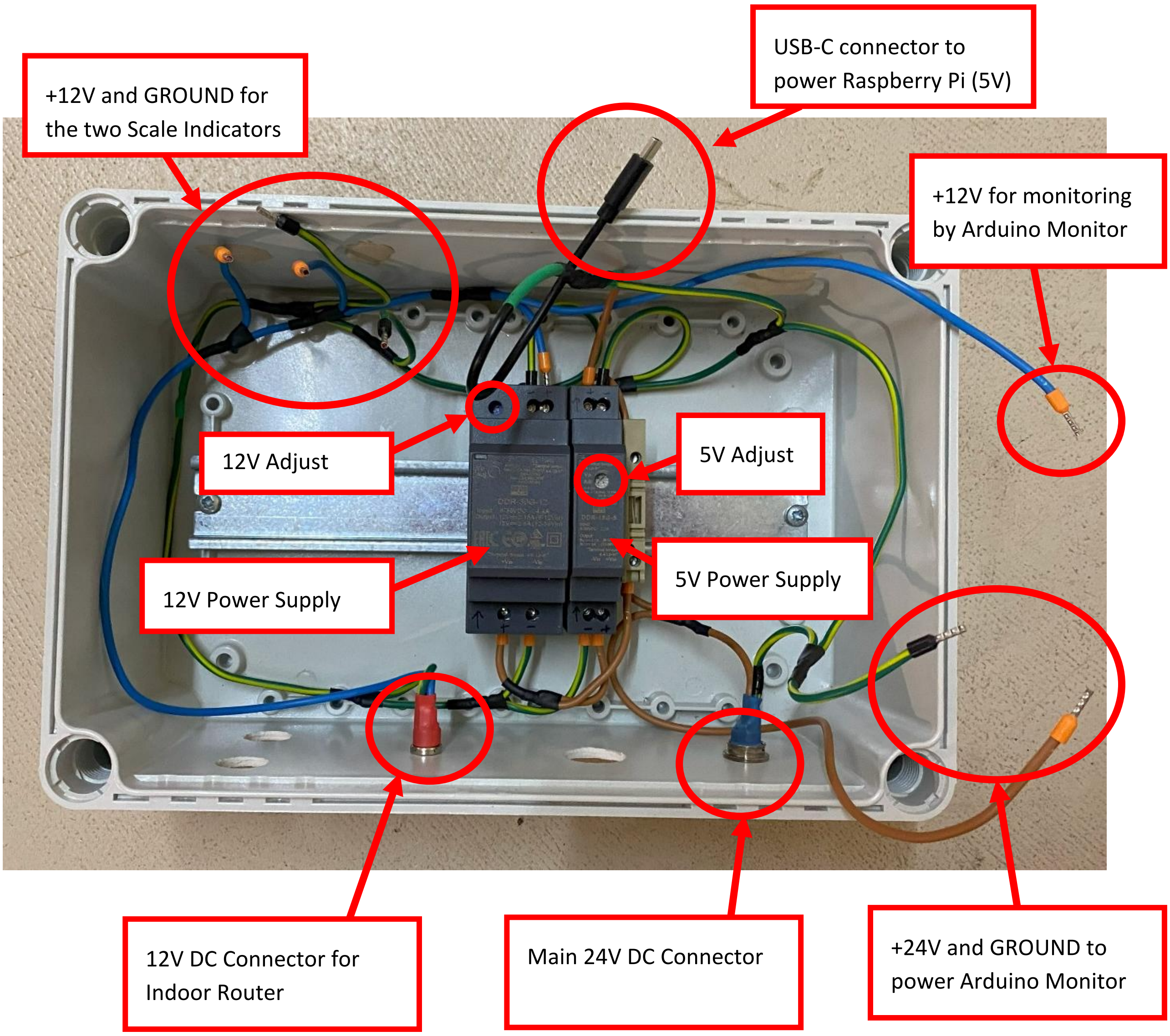


Figure 40 - Power Supplies and Wiring

TESTING THE POWER SUPPLIES

Before connecting any of the components, it is important to check the output of the power supplies.

- Insert a 3A fuse into the fuse holder. The APMS Main Unit uses a 3A fuse (668-5963).
- Connect 24V DC to the main DC connector.
  - To do this, use the DC plug (RS: 705-1519) and make a temporary power cable for testing.
- There should be a red light on the fuse holder and a blue light on each power supply.
- Measure the output of the 5V power supply and set it to 5.1V. The supply can be adjusted by turning the "5V Adjust" potentiometer, as seen in FIGURE 40 - POWER SUPPLIES AND WIRING.
- Measure the output of the 12V power supply and set it to 12.1V. The supply can be adjusted by turning the 12V Adjust potentiometer.

* Please take note of the polarities, as the two power supplies are not the same.

INSTALLING THE SCALE INDICATORS

The Scale Indicators used in the APMS are made by Laumas, model TLB485. The indicators run on 12V and communicate with the Raspberry Pi via RS-485. RS485 USB-to-Serial Converters are used. (RS: 687-7834)

INDICATOR POWER AND DATA

- The Scale Indicator uses an 8-pin screw connector for power and serial communication. The connector, as well as the layout of the two RS485 Serial Converters can be seen in FIGURE 41 - SCALE INDICATOR CONNECTIONS
- Shorten the cables of the RS485 converters to fit the layout shown.

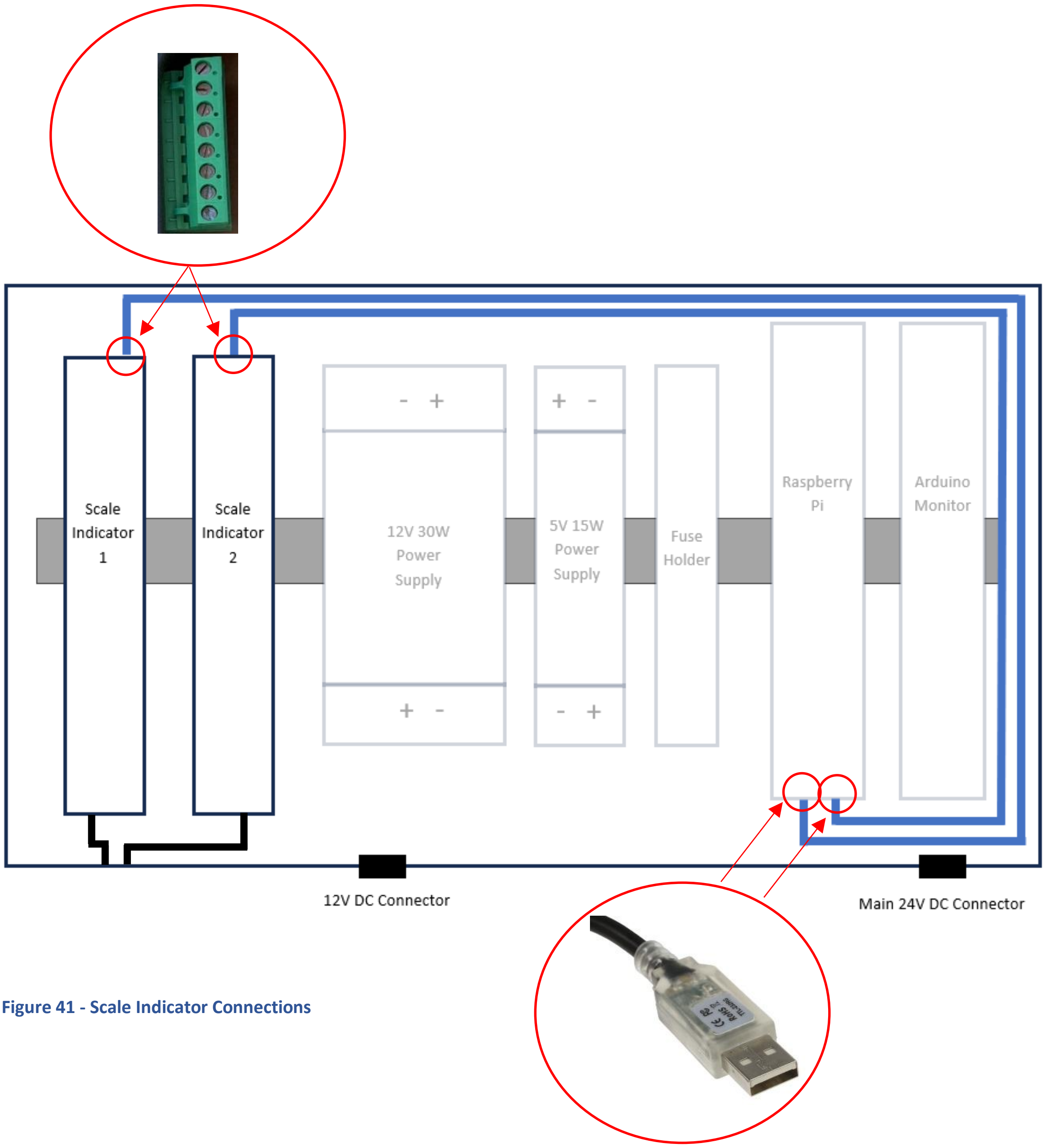


Figure 41 - Scale Indicator Connections

- Only the orange and yellow wires of the RS485 converter is used. Strip and crimp ferrules on the end of the wires and connect as see in FIGURE 42 – USB-TO-RS485 WIRING
- Connect the GROUND and +12V power wires.

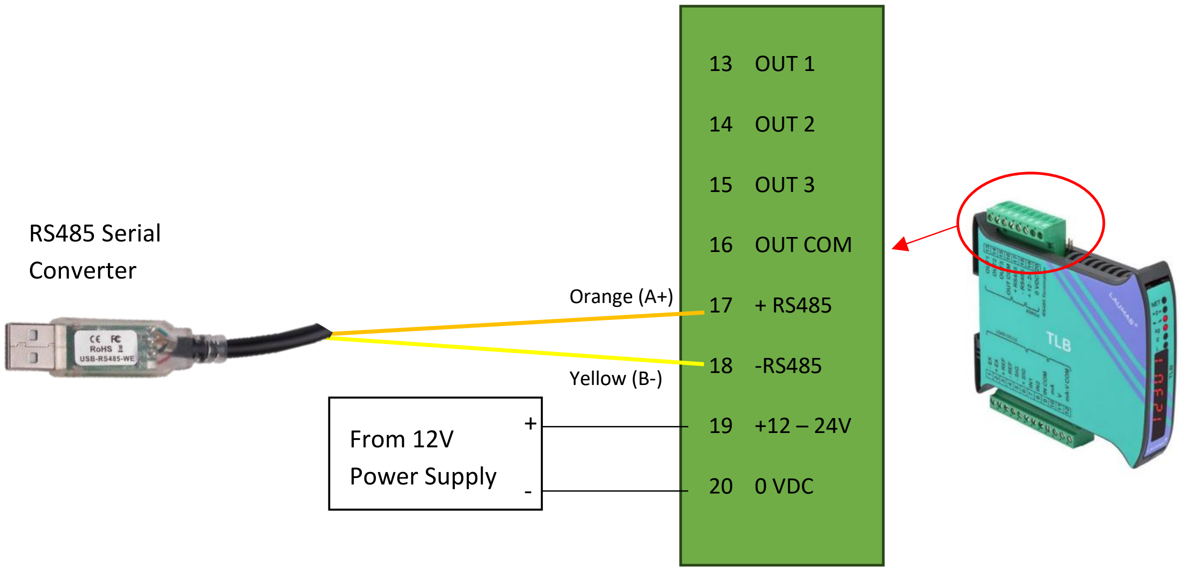


Figure 42 – USB-to-RS485 Wiring

INDICATOR SCALE CONNECTION

- Each scale connects to the APMS Main Unit via a 6-pin socket (111-5764), mounted on the front.
- A short cable runs from the 6-pin socket to the 9-pin connector on the indicator, as shown in FIGURE 43 - LOAD CELL WIRING

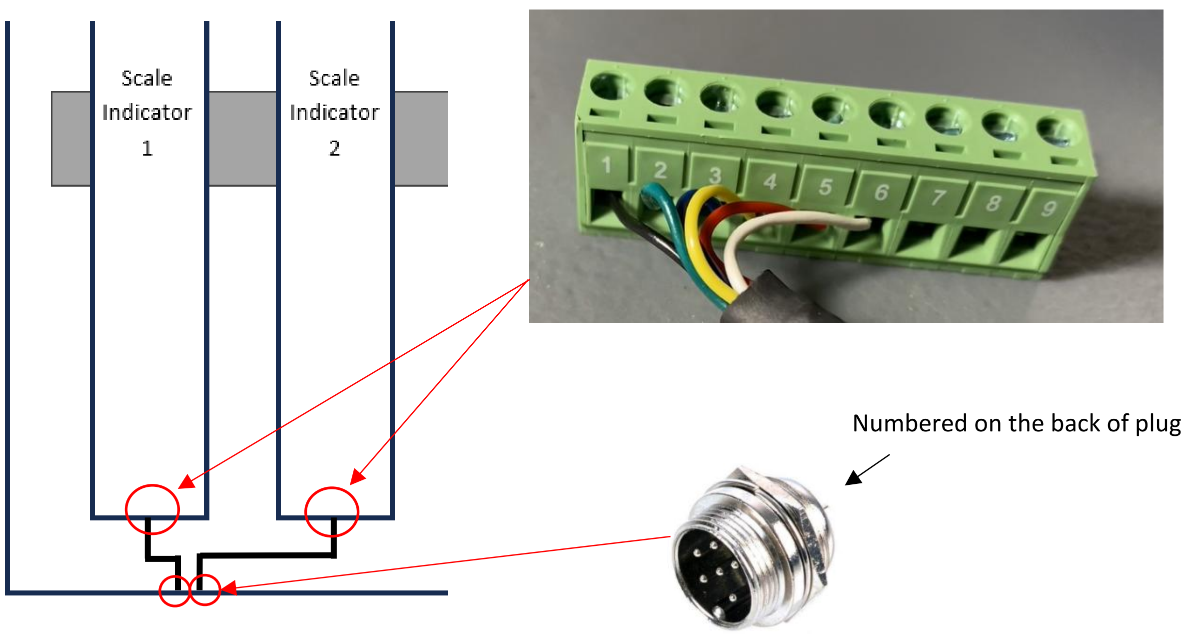


Figure 43 - Load Cell Wiring

- Each pin on the socket (111-5764) is numbered. Match the numbering on the 9-pin connector with that of the socket (Only 1 - 6 used) See FIGURE 44 - INSTALLING 6-PIN SCALE CONNECTOR
- Solder the wires to the sockets and add ferrules before inserting it into the scale connector screw terminals.

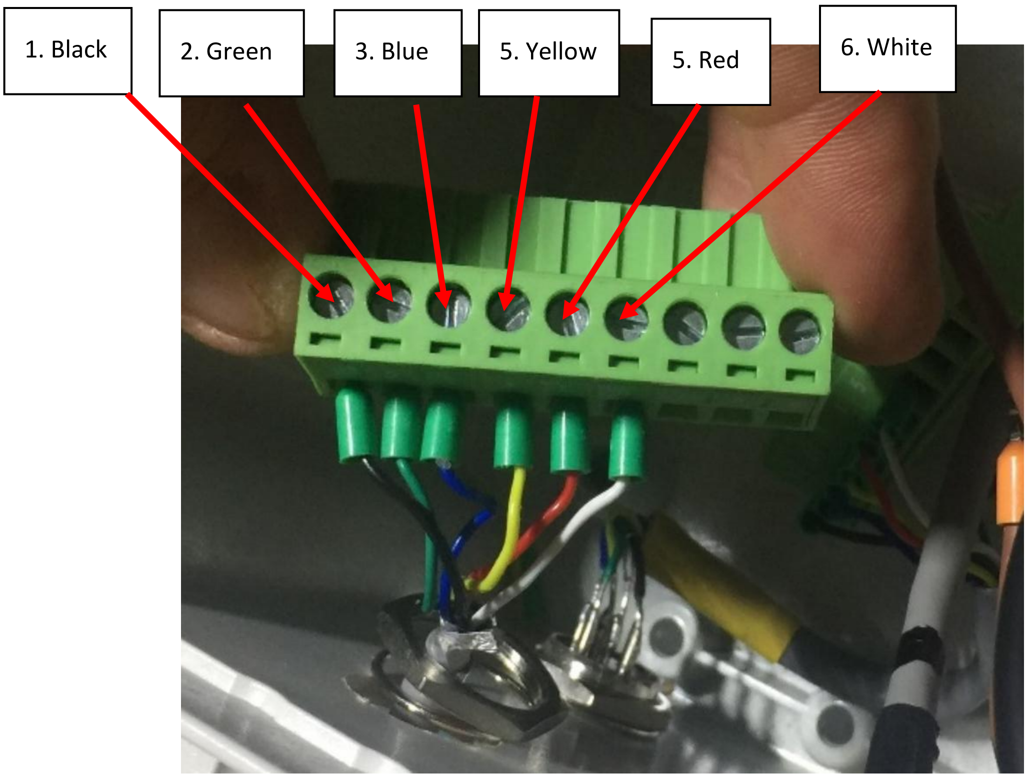


Figure 44 - Installing 6-pin Scale Connector

JUMPER INSTALL

As we are communicating with the indicators at a speed greater than 9600 BAUD, two jumpers need to be installed. Please see the excerpt from the manual below.

If the RS485 network exceeds 100 metres in length or baud-rate over 9600 are used, two terminating resistors are needed at the ends of the network: close the two jumpers indicated in the picture on the furthest instruments. Should there be different instruments or converters, refer to the specific manuals to determine whether it is necessary to connect the above-mentioned resistors.

CONFIGURING THE SCALE INDICATORS

- Before configuring the scale indicators, check that it has been wired correctly and working properly:
  - Plug in both the 8- and 9-pin scale connectors.
  - Plug the scales into their respective scale sockets
  - Apply power to the Main Unit – the indicators should light up and show a weight after start-up.
  - Apply light pressure on Scale 1 and Scale 2 and confirm the correct indicator changes its weight reading.

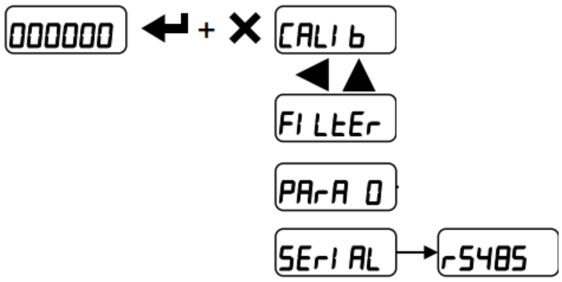
To configure the scale indicators, the front cover is flipped open and the buttons on the device are used.

There are 4 buttons on each indicator. They are:

CANCEL/BACK	✕
ENTER	↵
LEFT	◀
UP	▲

- Set "Filter" to 1

- Configure the RS485 Settings



Name	Value
Mode	Continuous
BAUD rate	115200
Hertz	100
Parity	None
Stop Bit	1
Mode	nOd t

The settings above will be checked when the testing scales script. See EACH PART OF THE SYSTEM CAN NOW BE TESTED INDIVIDUALLY.

TESTING THE SCALES.PY SCRIPT

INSTALLING THE APMS MONITOR

If the APMS Monitor has been tested as explained in TESTING THE MONITOR CIRCUIT BOARD, it can be installed in the Main Unit. The Monitor is mounted on a 3D printed mounting plate, as discussed earlier.

- The Monitor mounting plate clips on to the DIN rail, with the power connector facing the front of the Main Unit, and the 12V monitoring connector facing towards the back. See FIGURE 45 - MONITOR MOUNTED INSIDE MAIN UNIT
- Connect the power and 12V monitoring wires.
- Connect the 40-pin Raspberry Pi ribbon cable.

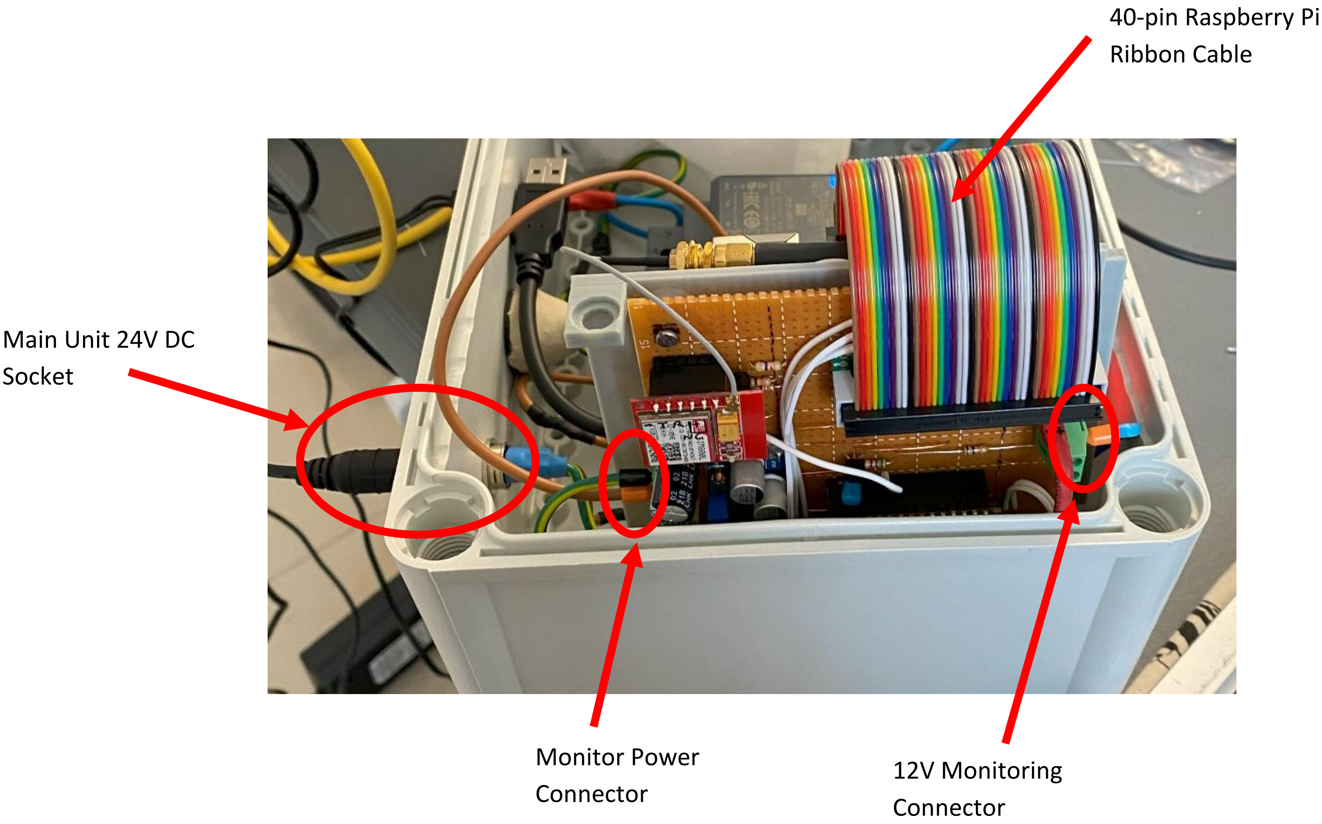


Figure 45 - Monitor mounted inside Main Unit

AWS S3 CLOUD STORAGE

The software and data are stored on the AWS S3 Server.

ADMINISTRATOR DETAILS

To access the software, download a S3 Client, such as S3 Browser. (https://s3browser.com/).

The following information describes the current setup and provides all necessary login details. **Please keep these details secure**.

AWS S3 Cloud Storage Information:

The root username and password for all administrative duties are as follows:

Root username:	*apmsbirdlife@gmail.com*
Password:	*Africanpenguins!@*

The current users and their rights are listed in the table below:

Username	Description	Rights	Access Key
stony_point	Stony Point APMS (Writes data to cloud)	Read/Write	AKIA6DO2JOYFORMYC4KH
bird_island	Bird Island APMS (Writes data to cloud)	Read/Write	AKIA6DO2JOYFJGDUKBPH
dyer_island	Dyer Island APMS (Writes data to cloud)	Read/Write	AKIA6DO2JOYFPHIDWRJG
robben_island	Robben Island APMS (Writes data to cloud)	Read/Write	AKIA6DO2JOYFNAMVVH7F
server	APMS Server (Currently a laptop in the BirdLife office) Reads data from cloud. Uploads website plots to cloud.	Read/Write	AKIA6DO2JOYFNMCEW5WC
pierre_retief	Engineer working on project	Read/Write	AKIA6DO2JOYFAUJR3R4G
ninepoint	Ninepoint downloads short-term and long-term plots, daily, for the website	Read	AKIA6DO2JOYFHOHK5BX7
katta_ludynia	Scientist	Read	AKIA6DO2JOYFLEQE4POG
Eleanor	Scientist, Admin	Read/Write	AKIA6DO2JOYFJJLO6ANR

Table 5 - AWS Users

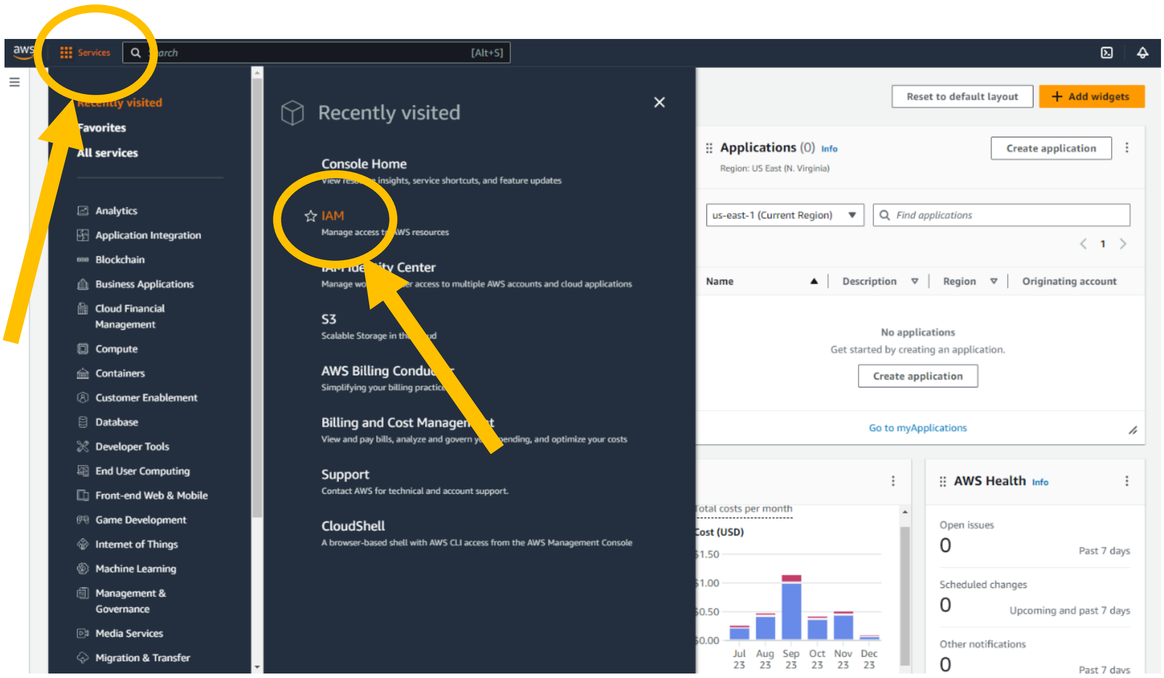
There are currently five main objects:

s3://apmsbirdlife/APMS/data/	This holds all the scale, rfid and system status data.
s3://apmsbirdlife/APMS/software/	This holds all the necessary software to set up a new system.
s3://apmsbirdlife/APMS/documents/	This holds documentation such as the manual, parts lists etc.
s3://apmsbirdlife/APMS/3D_prints/	This holds the design files for the various 3D printed parts
s3://apmsbirdlife/APMS/website/	This holds the daily plots for the website run by Ninepoint

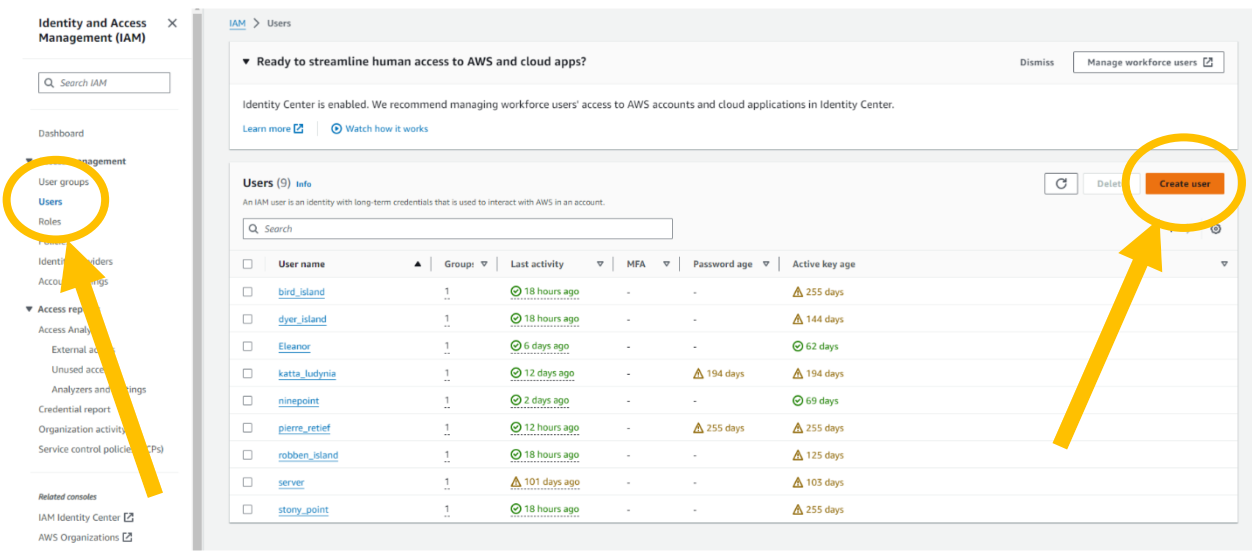
CREATING A USER ON AWS S3

Each Raspberry Pi has an AWS S3 Username, Access Key and Secret Access Key assigned to it. This enables the data and log files to be uploaded to the cloud every day. When installing an APMS at a new site, a user for the site must be created. This manual is accurate as of writing in December 2023, but as websites and services change, slight variations might appear in future. Please update this manual when deemed necessary and update the revision number. Further improvement regarding security, such as using IAM Identity Center instead of IAM, using Multi Factor Authentication or using AWS CLI2, are left up to the discretion of the administrator

- Go to https://aws.amazon.com/s3/ and sign in as the root user, using the login details provided above.
- Click on "Services" and then "IAM" [NOT "IAM Identity Center"]

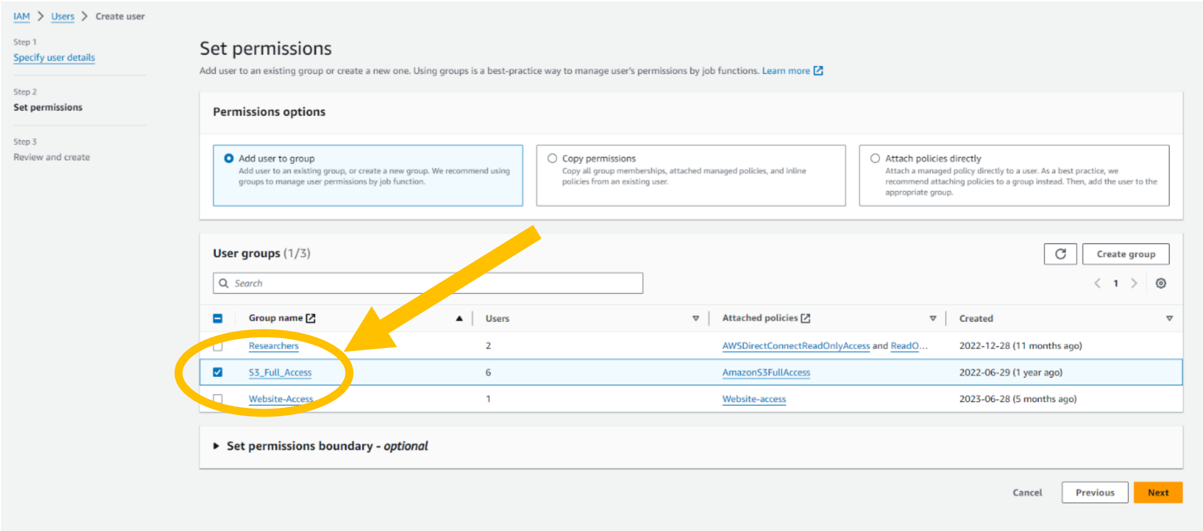


- Click on "Users" and then "Create User"

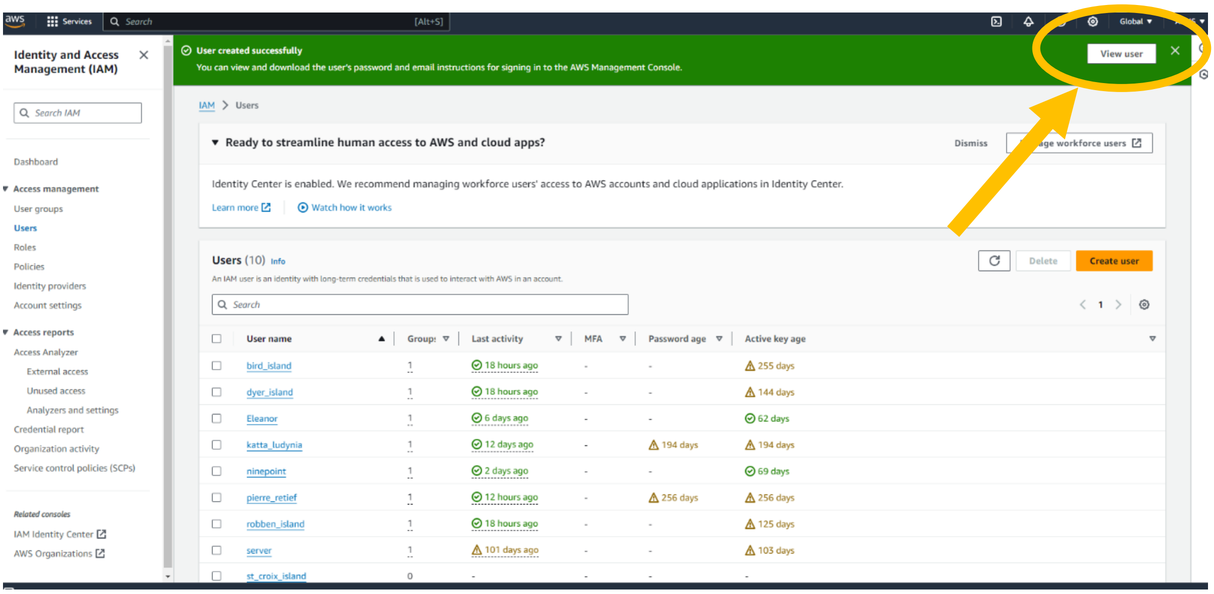


- Under “User name”, enter the site name, all lower case letters. Example: “*bird_island*”.
  - Keep all the defaults as is.
  - Click Next to continue.

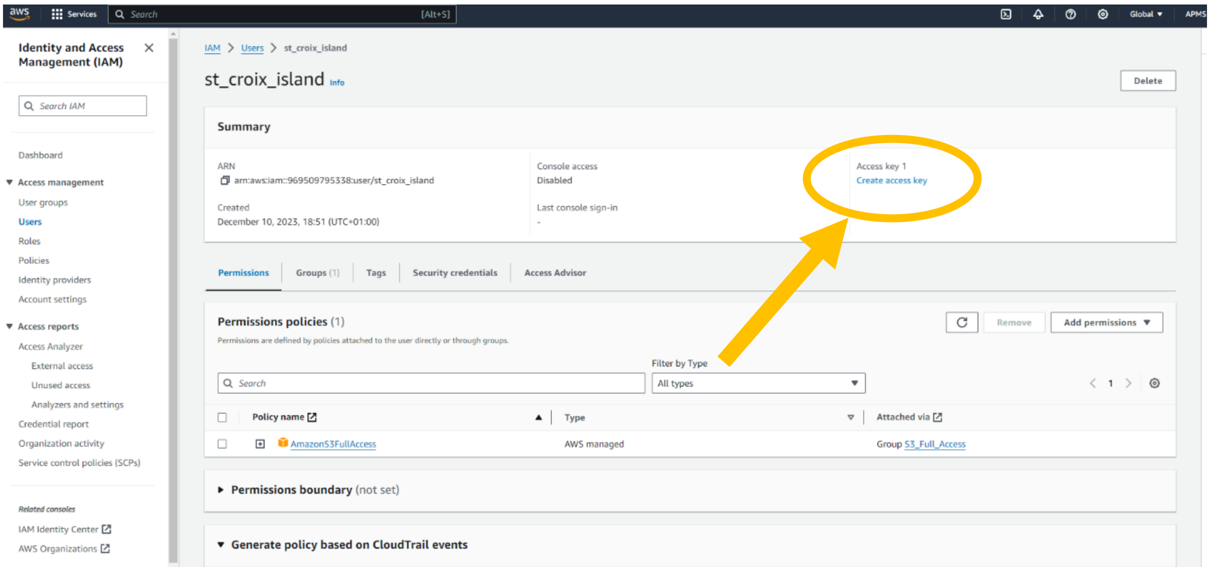
- On the “Set permissions” page, select the S3_Full_Access user group and click “Next”



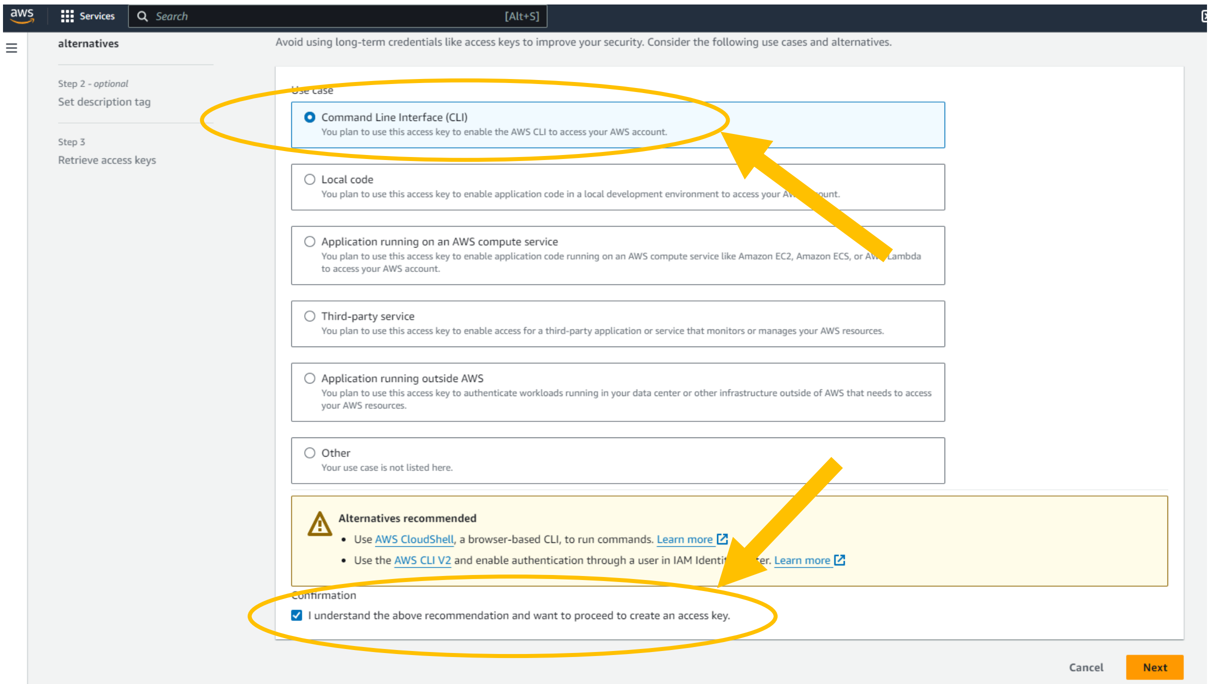
- On the “Review and create” page, click on “Create user”
- Now click on “View user”



- Click on “Create access key”

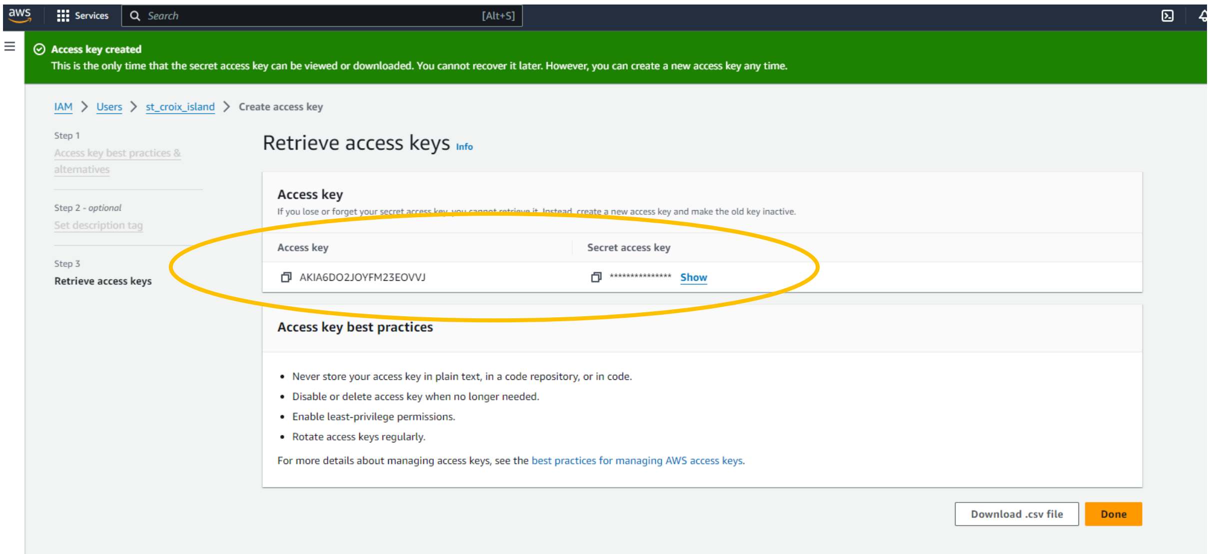


- Click on Command Line Interface (CLI), click the “Confirmation” and click “Next”



- On the “Set description tag” page, enter “APMS upload” and click on “Create access key”

- You have now successfully created a user for the new site and created an access key. Copy down the Access key and Secret access key (Click “Show”). You can also download the .csv file containing this information. Please keep this information very secure as it has full read/write access to the cloud. Also, please note that you will not be able to view this information at a later stage, so make sure it is stored safely. If this information is lost, and is needed in future, create a new user as described in this section.



This access key and secret access key will be needed when the APMS is configured. [See APMS Configuration].

RASPBERRY PI OPERATING SYSTEM

The Raspberry Pi operating system must be installed first, together with all required libraries and applications. A microSD card, SD card reader and a computer running Windows is needed for this step. If the operating system is being installed on a spare Raspberry Pi, a USB-C cable and 5V 3A power supply will be needed to power the device.

PREPARE THE MICRO SD CARD

- Get a new SD card (Currently using a 64GB SanDisk) and insert it into the MicroSD card reader.
- Download the Rpi Imager and Rpi OS from the cloud (APMS/software/RaspberryPi)
    - One way to do this is to download S3 Browser (https://s3browser.com/)
- Install and run Raspberry Pi Imager (Version 1.7.4 for Windows used for this manual)
    - CHOOSE OS → Select "Use custom" → Select the raspberry pi image you downloaded from the cloud.
    - Currently using "2023-02-21-raspios-bullseye-arm64-lite.img".
- CHOOSE STORAGE → Select the SD card.
- Click on the settings cog.
    - Choose a hostname. Example: "apms-dyer" or "apms-stcroix"
    - Select "Enable SSH"
        - "Use password authentication"
    - Select "Set username and password"
        - username: apms --- password: africanpenguins
    - Select "Configure wireless LAN"
        - SSID: apms --- password: africanpenguins --- Wireless LAN country: ZA
    - Select "Set locale settings"
        - Time zone: ZA
    - Leave all other settings unchanged and click on SAVE
- Click on WRITE
- Click YES after confirming you selected the correct storage device
    - Wait for RaspiOS to finish installing and click CONTINUE after installation.
- Eject the uSD card and insert it into the Raspberry Pi. The MicroSD card slot is located on the underside of the Pi.

SSH INTO THE PI (LOCALLY)

- Ensure the Wi-Fi router has been set up. See [INETRNET AND WIFI]
- Power the Raspberry Pi via the USB-C connector. Use a suitable power supply that is capable of deliver 3A of current.
    - The Raspberry Pi will power up when the APMS main unit is powered.
      If not using an APMS main unit, power the Pi with a 5V 3A power supply and USB-C cable
      There will be a solid red light on the Pi when booting is complete.

- On a PC, connect to the Wi-Fi router. This can be done via Wi-Fi or a LAN cable
- Connect to the router's admin page in a browser to find the IP of the new Raspberry pi.
    - The IP address of the router can usually be found on the back of the router or in the router's manual.
    - Alternatively, after connecting to the Wi-Fi, open Powershell and run "***$ ipconfig***".
    - Under "Wireless LAN Adapter Wifi", find the IP address under "Default Gateway".

- To connect to the router, In an internet browser, type the IP address, found in the previous step, into the address bar.
- The default username and password can usually be found on the back of the router or in the router's manual. Commonly, the username is "admin" and the password is either "admin" or left blank. It could also be "africanpenguins" if set up previously.
- After connecting to the router's admin page, look for the IP address of the Raspberry Pi. You would typically find this information in a section called "Connected Devices" or something similar. The Raspberry Pi should appear under the name "apms"

- After obtaining the IP of the Raspberry Pi, SSH into the pi: ***$ ssh [username]@[IP address]***
  - The username will be "apms"
    - Example: *ssh apms@192.168.0.111*
  - The password is "africanpenguins".

- A message similar to the following might appear

PS C:\Users\elwei> ssh apms@192.168.0.103
The authenticity of host '192.168.0.103 (192.168.0.103)' can't be established.
ED25519 key fingerprint is SHA256:/kWcHC3/ADaEtjABjP8hUbrXdEmp09IIDBDLm6Pg8CY.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])?

  - Type yes and if asked for a password, enter "africanpenguins"

- While trying to SSH into the PI from another PC, especially after changes were made, you might encounter the following message:

```
PS C:\Users\elwei> ssh apms@192.168.0.100
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:/kWcHC3/ADaEtjABjP8hUbrXdEmp09IIDBDLm6Pg8CY.
Please contact your system administrator.
Add correct host key in C:\\Users\\elwei/.ssh/known_hosts to get rid of this message.
Offending ED25519 key in C:\\Users\\elwei/.ssh/known_hosts:1
Host key for 192.168.0.100 has changed and you have requested strict checking.
Host key verification failed.
PS C:\Users\elwei> ssh apms@192.168.0.100
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:/kWcHC3/ADaEtjABjP8hUbrXdEmp09IIDBDLm6Pg8CY.
Please contact your system administrator.
Add correct host key in C:\\Users\\elwei/.ssh/known_hosts to get rid of this message.
Offending ED25519 key in C:\\Users\\elwei/.ssh/known_hosts:1
Host key for 192.168.0.100 has changed and you have requested strict checking.
Host key verification failed.
PS C:\Users\elwei> ssh apms@192.168.0.100
The authenticity of host '192.168.0.100 (192.168.0.100)' can't be established.
ED25519 key fingerprint is SHA256:/kWcHC3/ADaEtjABjP8hUbrXdEmp09IIDBDLm6Pg8CY.
This host key is known by the following other names/addresses:
    C:\Users\elwei/.ssh/known_hosts:6: 192.168.0.103
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.100' (ED25519) to the list of known hosts.
apms@192.168.0.100's password:
Linux apms-stcroix 6.1.21-v8+ #1642 SMP PREEMPT Mon Apr  3 17:24:16 BST 2023 aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Dec  6 09:42:46 2023 from 192.168.0.100
apms@apms-stcroix:~ $
```

  - Edit the "known_hosts" file: ***$ sudo nano ~/.ssh/known_hosts***
  - Remove the line containing the IP of the Raspberry Pi:
  - Save the file and exit. (Hold **CTRL** and press **x**. Then press '**y**' to save, then press Enter to exit)

UPDATE THE RASPBERRY PI OPERATING SYSTEM

- Check the time is correct by running: ***$ date***
    - If incorrect, run:
      $ sudo raspi-config → Localisation Options → Timezone → Africa → Johannesburg
      Use the right arrow key to move to FINISHED and press Enter
- Update package list ***$ sudo apt-get update*** Say “yes” to any messages that appear
- Install latest packages ***$ sudo apt-get upgrade*** Say “yes” to any messages that appear

***** Please note *****

Downloading the software via the 4G router could be costly and unnecessary if a local WiFi hotspot is available. To use local Wifi for initial setting up, the following changes to the above steps must be made:

1. When setting up the Raspberry Pi in Rpi Imager, use the Wifi settings of the local WiFi network.
2. To find the IP address of the Raspberry Pi, the easiest option is to plug a monitor and keyboard into the Pi. Power up the Pi and log in using the username and password set up during the installation. In the terminal of the Pi, run **$ ifconfig**
    a. **T**he IP address can be found under “wlan0” -> “inet”
3. To change the WiFi setting on the Raspberry Pi back to the 4G router: Run **$ *sudo raspi-config*** then select “System Options” -> “Wireless LAN“

INSTALL APPLICATIONS AND LIBRARIES

- The following Linux applications are needed for the APMS. The currently used version is listed next to each application. If a future version becomes incompatible, try the listed version instead.

git	[v 2.34.1]	Used to download and update the firmware
ntpdate	[v 4.2.8]	To update the system time via NTP
tmate	[v 2.4.0]	Used to SSH into the Pi, via the internet
python3	[v 3.9.2]	Programming language used to run most of the firmware
minicom	[v 2.8]	Useful program for monitoring the serial port
pip	[v 20.3.4]	Program used to install python packages
vnstat	[v 2.6]	Program to monitor internet usage

  - To install the latest versions of all the applications above:
    - Update the package list: ***$ sudo apt-get update***
    - Install the packages:
      $ sudo apt-get install git ntpdate tmate python3 minicom pip vnstat
    - Answer Yes to any questions that may appear.

  - If installing an older version, add =[version] for example: ***$ sudo apt-get install ntpdate=4.2.8***

- The following python packages are also needed.

pyserial	[v 3.5]	Used for serial communication
vcgencmd	[v 0.1.1]	Used to monitor CPU temperature
psutil	[v 5.9.5]	Used to monitor disk space and CPU usage

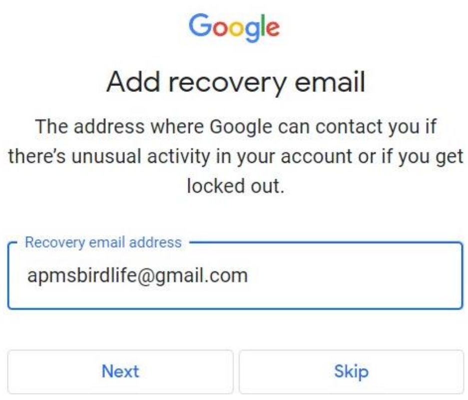
  - To install the python packages, use pip: ***$ pip install pyserial vcgencmd psutil***
    - Answer Yes to any questions that may appear

  - If installing an older version, add =[version] for example: ***$ pip install psutil=5.9.5***

SSH INTO THE PI (VIA TMATE)

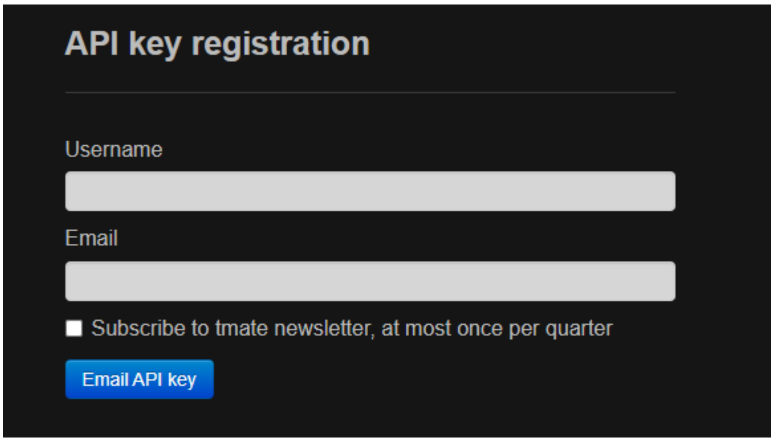
"tmate" is a utility that provides remote access to the Raspberry Pi in the field. It provides a way to SSH into the Pi via the internet and is the main tool for testing, debugging and updating.

- Create a google account for this site.
  - For the username use "apmsXXXXXX" where XXXXXX is the name of the site. Keep it all lowercase letters and one word, with only letters. Example: *"apmsrobbenisland"* or *"apmsdyerisland"*
    For the password use *"africanpenguins".*
    When a recover email is requested, use: apmsbirdlife@gmail.com (see below)



- Check that the gmail account is working by logging in and sending/receiving an email.

- After creating and testing the new gmail account, Go to https://tmate.io/ and scroll down to the section where an API key can be registered. See below.



  - Under "Username", use the same name as for the google account, for example "apmsdyerisland"
  - For "Email", use the newly created gmail address, for example: apmsdyerisland@gmail.com
  - Click on the "Email API Key button" and wait for the key to be sent to the newly created inbox.
  - Check the spam folder if it does not arrive.
    This API key will be used in the next step.

- With the tmate API key in hand, SSH into the Pi as described earlier in this manual.
- Install tmate (If you have not already installed it): ***$ sudo apt-get install tmate***
- To configure tmate, in the home directory, create a text file called "*.tmate.conf*" *(Note the first fullstop)*
  - *To create this file and edit it in a text editor, run:* ***$ sudo nano ~/.tmate.conf***

- Copy the following into the file:

  set tmate-api-key "KEY" This is the API key from the previous step

  set tmate-session-name "USERNAME" "Username" chosen for the API key registration

  EXAMPLE:

  set tmate-api-key "tmk-kdhemOSl2eujXDQrpH7QqmlR5c"

  set tmate-session-name "apmsbirdisland"
- Save the file and exit. (Hold **CTRL** and press **x** -- Then press '**y**' to save -- then press Enter to exit)

- Start tmate. Run: ***$ tmate***
- Press q to detach from session and go back to the terminal.
- To test if tmate is configured properly, try to SSH into the pi via the internet
  - Open a new Git Bash or PowerShell terminal
  - SSH into the Pi by running: ***$ ssh username/session-name@nyc1.tmate.io***
    - "username" and "session-name" = "Username" chosen for the API key registration

    Example: **$ ssh apmsrobbenisland/apmsrobbenisland@nyc1.tmate.io**

    This should log you into the Pi, but now, via the internet.

After successfully connecting to the Pi via the internet, detach from the session:

- Hold **CTRL** and press **b** -- Then release keys – Then press '**d**' to detach

tmate shortcuts

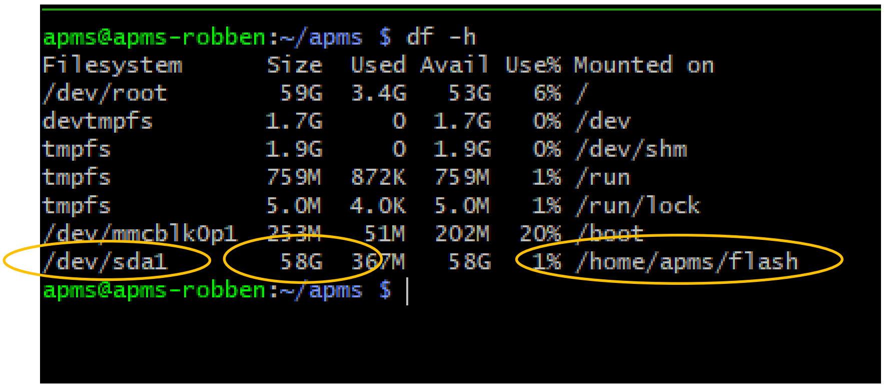
- Hold **CTRL** and press **b** -- Then release keys – Then press ' **d** ' to detach
- Hold **CTRL** and press **b** -- Then release keys – Then press ' **%** ' to split screen vertically
- Hold **CTRL** and press **b** -- Then release keys – Then press ' **"** ' to split screen horizontally
- Hold **CTRL** and press **b** -- Then release keys – Then press ' c ' to create new window
- Hold **CTRL** and press **b** -- Then release keys – Then press ' n ' to switch to next window
- Hold **CTRL** and press **b** -- Then release keys – Then press ' [' to enter "scrolling" mode
  - Press ' q ' to exit scrolling mode

SET UP THE USB FLASH DRIVE

A 64GB USB Flash drive is used as a backup for the APMS data. Every morning, the data gets backed up to the flash drive, before being transferred to the cloud. To configure Flash drive, a NTFS partition must be created.

- SSH into the Pi as described earlier in this manual.
- Insert the USB Flash drive into any of the available USB ports on the Pi.
- Unmount the flash drive (if mounted): ***$ umount /dev/sda1***
- Run fdisk on the drive: ***$ sudo fdisk /dev/sda***
    - Create a new empty DOS partition: press 'o' and enter
    - Create a new partition: press 'n' and enter – use all the default options
    - Change partition type: press 't' and enter – choose 7 - 'HPFS/NTFS/exFAT'
    - Save and exit: press 'w' and enter

- Format disk: ***$ sudo mkntfs -f -L "apms_backup" /dev/sda1***
- Create mounting point ***$ mkdir /home/apms/flash***
- Mount: ***$ sudo mount /dev/sda1 /home/apms/flash***
- To check the USB Flash drive has been properly mounted, run: ***$ df -h***
    - Partition sda1 should be mounted at /home/apms/flash and should show the size of the drive.

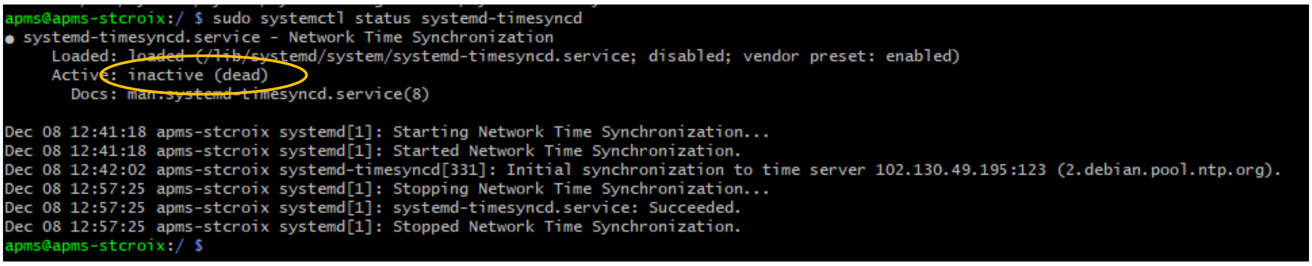


```
apms@apms-robben:~/apms $ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        59G  3.4G   53G   6% /
devtmpfs        1.7G     0  1.7G   0% /dev
tmpfs           1.9G     0  1.9G   0% /dev/shm
tmpfs           759M  872K  759M   1% /run
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
/dev/mmcblk0p1  253M   51M  202M  20% /boot
/dev/sda1        58G  367M   58G   1% /home/apms/flash
apms@apms-robben:~/apms $ |
```

TURN OFF AUTOMATIC TIME UPDATING

To prevent the Raspberry Pi from updating its system clock at an inopportune time, automatic time updating must be disabled. A script called “*update_system_time.py*” will handle the updating of the system time. (See the section: “SETTING UP THE CRONTAB” for more details.

- SSH into the Pi as described earlier in this manual
- To stop time-sync process run: ***$ sudo systemctl stop systemd-timesyncd***
- To prevent the process from starting at boot, run: ***$ sudo systemctl disable systemd-timesyncd***
- To check it is indeed off, run: ***$ sudo systemctl status systemd-timesyncd***



DISABLE SERIAL CONSOLE

The Raspberry Pi communicates with the Monitor via serial, using GPIO14 and GPIO15. The Raspberry Pi Serial Console must be disabled to allow this serial port to be used:

- Run the following: ***$ sudo raspi-config***
- Select “Interfaces”
- Select “Serial Port”
- Select “NO” to login shell
- Select “YES” to enable serial support
- Save and exit

CONFIGURE UART3

UART3 is used to program the Arduino Monitor.

Enable serial port for programming by adding the following:

- Edit /boot/config.txt: ***$ sudo nano /boot/config.txt***
- Add “dtoverlay=uart3” after “enable_uart=1”
- Save the file and exit. (Hold CTRL and press x -- Then press ‘y’ to save -- then press Enter to exit)
- Reboot: ***$ sudo reboot***

CONFIGURE GPIO3 FOR POWER ON/OFF

To enable GPIO3 to be used to power the Raspberry Pi on/off, the config file must be edited:

- Edit the config file. Run ***$ sudo nano /boot/config.txt***
- Add “dtoverlay=gpio-shutdown” to the bottom of the file.
- Save the file and exit. (Hold CTRL and press x -- Then press ‘y’ to save -- then press Enter to exit)
- Reboot: ***$ sudo reboot***

CONFIGURE GIT AND DOWNLOAD FIRMWARE

- SSH into the Pi as described earlier in this manual
- Install git (If you have not installed it already): ***$ sudo apt-get install git***
- Configure git:
  - ***$ git config --global user.name "apms"***
  - ***$ git config --global user.email apmsbirdlife@gmail.com***

> All the firmware is stored on github: https://github.com/apmsbirdlife/apms
>
> The username for the GitHub account is: *apmsbirdlife@gmail.com*
>
> The password for the GitHub account is: *africanpenguins*!

- Generate an SSH key pair: ***$ ssh-keygen -t rsa -b 4096 -C apmsbirdlife@gmail.com***
  - Press Enter for any questions that appear.
- Start SSH agent: ***$ eval $(ssh-agent -s)***
- Add the private key to the agent to manage keys: ***$ ssh-add ~/.ssh/id_rsa***
- Go to www.github.com and log in using the details provided above.
  - Click on the user image (top right of the page) and choose “settings”.
    - “settings” → “SSH and GPG keys” → Click on “New SSH key”
    - Under “Title”, put “APMS [site_name]” Example: “APMS St Croix”
      If replacing a pi, delete the previous one first.
    - In the “Key” field, copy the contents of the following file: “ ~/.ssh/id_rsa.pub ”
      To do this, run the command: ***$ cat ~/.ssh/id_rsa.pub***
      Copy the code using your mouse and paste it into the “Key” field
    - click “Add ssh key ” .
- Go back to the Raspberry Pi terminal and authenticate: ***$ ssh -T git@github.com***
  Type “yes” if asked to continue.
- Go to the home folder and clone the repository:
  - ***$ cd ~***
  - ***$ git clone git@github.com:apmsbirdlife/apms.git***
- All the APMS firmware should now be downloaded to ~/apms/. Go to the directory: ***$ cd ~/apms***
- Make sure you are on the correct branch: ***$ git checkout main***
- To update to the latest firmware, go to ~/apms/ and run: ***$ git pull***

FINDING THE RS485 SERIAL NUMBERS

Each scale is connected to the Raspberry Pi via a RS485 converter. Each converter has a unique serial number and that is how the Pi can determine which scale is which. This is why it is important to label the indicator, scale and RS485 converter as “scale 1” or “scale 2”. These should not be mixed and matched once the system has been set up.

- To identify the serial number of each converter, a simple script was created. The script will be run twice, once for each converter. Run the script and follow the on-screen instructions:

  $ python3 ~/apms/tools/find_rs485_serial_number.py

- Note down the serial number for each converter as it will be used when configuring the APMS. See CONFIGURING THE APMS

CONFIGURING THE APMS

- All the parameters that can and must be set for the system to work properly can be found in the config file: ~/apms/config.json
- A new system needs a config file to be created, an existing system will already have the file.
- To check if the config file already exists, list the contents of the apms directory and see if the file is present: ***$ ls ~/apms/***
- If a config file already exists, it can be edited with: ***$ sudo nano ~/apms/config.json***
  Save the file and exit. (Hold **CTRL** and press **x** -- Then press ‘**y**’ to save -- then press Enter to exit)
- For a new system, the file “config_template.json” can be used.
  - Make a copy of the template: ***$ cp ~/apms/config_template.json ~/apms/config.json***
  - To edit the config file, run: ***$ sudo nano ~/apms/config.json***
  - Refer to the table below when editing the config file. The values given in bold are defaults and do not need to be changed.

The table below shows the parameters and their descriptions.

Parameter	Description
SITE_NAME	Unique site name. Example: “stony_point” All lowercase, no spaces
SITE_CODE	Unique 3 Letter site code. Example “STO” for Stony. All capital letters, must be 3 letters
SCALE_1_SERIAL_NUMBER SCALE_2_SERIAL_NUMBER	Unique serial number for the RS485 converter. See: “Finding the rs485 serial numbers”
AWS_S3_DATA_PATH	This should be “s3://apmsbirdlife/APMS/data/[site name]/” Example: *“s3://apmsbirdlife/APMS/data/stony_point/”*
AWS_ACCESS_KEY	The access key you obtained when the user was created. See “ Creating a user on aws s3”
AWS_SECRET_ACCESS_KEY	The secret access key you obtained when the user was created. See “

	Creating a user on aws s3"
AWS_BUCKET	Name of AWS S3 Bucket: **"apmsbirdlife"**
APMS_ABS_PATH	Absolute path where APMS is installed: **"/home/apms/"**
DATA_PATH	Path where raw data is stored: **"/home/apms/data/"**
LOGS_PATH	Path where log files are stored: **"/home/apms/logs/"**
ERROR_PATH	Path where error files are stored **"/home/apms/error/"**
USB_PATH	Path when USB Flash drive is mounted **"/home/apms/flash/"**
USB_PARTITION	USB Flash drive partition **"/dev/sda1"**
EMAIL_ADDRESS	APMS main email address: **"apmsbirdlife@gmail.com"**
EMAIL_PASSWORD	This is the password to the main email address: **"dayammxlcvvwmdza"**
EMAIL_ADDRESS_ADMIN	Email address of APMS administrator: **"el.weideman@gmail.com"**
SCALE_SAMPLING_FREQ	Sampling frequency of scales (samples per second) **100**
SCALE_BAUD_RATE	RS485 Serial Converter BAUD rate **115200**
MIN_PENGUIN_WEIGHT	This is the minimum penguin weight (in kg) expected to be encountered in the field: **1.9**
MAX_PENGUIN_WEIGHT	This is the maximum penguin weight (in kg) expected to be encountered in the field: **4.3**
MAX_DEAD_WEIGHT	Maximum weight allowed (in kg) for sand/debris on the scales before warning: **0.5**
SCALE_EMPTY_TIME	Duration (in seconds) the scales must be empty before the event stops. **3**
RFID_SERIAL_NUMBER	This is the serial number of the serial converter inside the BioMark RFID reader: **"0001"**

RFID_BAUD_RATE	The is the BAUD rate of the BioMark serial converter **115200**
SERIAL_PORT_INTERNAL	This is the name of the serial port used for communication with the Monitor: **"/dev/ttyS0"**
INTERNAL_BAUD_RATE	The is the BAUD rate for communication between the Pi and the Monitor **115200**
MAX_SCALE_ERROR_COUNT	This is the number of times a scale can fail before it gets disabled. **3**
CPU_TEMP_WARNING	CPU temperature (in °C) at which to send out warning message **60**
CPU_TEMP_DANGER	CPU temperature (in °C) at which Raspberry Pi will start throttling the CPU: **70**
UPLOAD	This variable is set to **1** to allow daily data uploads to the cloud (0 to disable) **1**

- As an example, the configuration file for Stony Point can be seen below.


```
{
        "SITE_NAME" : "stony_point",
        "SITE_CODE" : "STO",
        "SCALE_1_SERIAL_NUMBER" : "FT613MNU",
        "SCALE_2_SERIAL_NUMBER" : "FT613Q9L",
        "AWS_S3_DATA_PATH" : "s3://apmsbirdlife/APMS/data/stony_point/",
        "AWS_USERNAME" : "apmsbirdlife@gmail.com",
        "AWS_ACCESS_KEY" : "AKIA6DO2JOYFORMYC4KH",
        "AWS_SECRET_ACCESS_KEY" : "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",
        "AWS_BUCKET" : "apmsbirdlife",
        "APMS_ABS_PATH" : "/home/apms/",
        "DATA_PATH" : "/home/apms/data/",
        "LOGS_PATH" : "/home/apms/logs/",
        "ERROR_PATH" : "/home/apms/error/",
        "USB_PATH" : "/home/apms/flash/",
        "USB_PARTITION" : "/dev/sda1",
        "EMAIL_ADDRESS" : "apmsbirdlife@gmail.com",
        "EMAIL_ADDRESS_ADMIN" : "pierreretief@gmail.com",
        "EMAIL_PASSWORD" : "dayammxlcvvwmdza",
        "SCALE_SAMPLING_FREQ" : 100,
        "SCALE_BAUD_RATE" : 115200,
        "MIN_PENGUIN_WEIGHT" : 1.9,
        "MAX_PENGUIN_WEIGHT" : 4.3,
        "MAX_DEAD_WEIGHT" : 0.5,
        "SCALE_EMPTY_TIME" : 3,
        "RFID_SERIAL_NUMBER" : "0001",
        "RFID_BAUD_RATE" : 115200,
        "SERIAL_PORT_INTERNAL" : "/dev/ttyS0",
        "INTERNAL_BAUD_RATE" : 115200,
        "MAX_SCALE_ERROR_COUNT" : 3,
        "CPU_TEMP_WARNING" : 60,
        "CPU_TEMP_DANGER" : 70,
        "UPLOAD" : 1
}
```

- Note the format regarding colons, commas etc. Do not add unnecessary spaces.
- After editing, Save the file and exit.
    - Hold **CTRL** and press **x**. Then press '**y**' to save, then press Enter to exit
- The configuration file has now been created.

PROGRAMMING THE ARDUINO MONITOR VIA THE PI

As discussed in UPDI PROGRAMMING, the Arduino is programmed via UPDI.

INITIAL SETUP

- Before soldering the link and connecting the UPDI wire, compile and upload the test sketch as discussed in PROGRAMMING VIA THE ARDUINO SOFTWARE.
- Ensure the Monitor Circuit Board has been fully tested before plugging it into the Pi. See TESTING THE MONITOR CIRCUIT BOARD.
- Ensure the Serial Console has been disabled as discussed in DISABLE SERIAL CONSOLE.
- Ensure UART3 has been configured as discussed in CONFIGURE UART3.
- Ensure GPIO3 has been configured to power the Raspberry Pi on/off as discussed in CONFIGURE GPIO3 FOR POWER ON/OFF.
- Provide 24V DC to the Main Unit, providing power for the Raspberry Pi and Arduino to boot up.
- SSH into the Pi.
- Ensure the latest firmware has been downloaded: ***$ cd ~/apms*** and then ***$ git clone***

- To compile the firmware, arduino-cli must be installed. Run***:***

$ curl -fsSL https://raw.githubusercontent.com/arduino/arduino-cli/master/install.sh | sh

Add to path:

- Edit bashrc: ***$ nano ~/.bashrc***
- Add the following to the bottom: "export PATH=$PATH:/home/apms/bin"
- Save the file and exit. (Hold **CTRL** and press **x** -- Then press '**y**' to save -- then press Enter to exit)
- Type: ***$ source ~/.bashrc***

To install the AVR core, run: ***$ arduino-cli core install arduino:megaavr***

To set up the config file, run ***$ arduino-cli config init***

The Software Serial port buffer must be increased:

- Determine the version number. Run: ***$ ls ~/.arduino15/packages/arduino/hardware/megaavr/***

- The run the following:

$ sudo nano ~/.arduino15/packages/arduino/hardware/megaavr/[VERSION NUMBER]/libraries/SoftwareSerial/src/SoftwareSerial.h

- Look for the line "#define _SS_MAX_RX_BUFF 64 // RX buffer size" and change 64 to 100
- Save the file and exit. (Hold **CTRL** and press **x** -- Then press '**y**' to save -- then press Enter to exit)

COMPILING AND UPLOADING

- SSH into the Pi
- Go to the Monitor firmware directory: ***$ cd ~/apms/system_monitor***
- Compile the firmware: ***$ ./compile.sh***

```
apms@apms-robben:~/apms/system_monitor $ ./compile.sh
Sketch uses 17189 bytes (34%) of program storage space. Maximum is 49152 bytes.
Global variables use 1159 bytes (18%) of dynamic memory, leaving 4985 bytes for local variables. Maximum is 6144 bytes.

Used library   Version Path
EEPROM         2.0     /home/apms/.arduino15/packages/arduino/hardware/megaavr/1.8.8/libraries/EEPROM
SoftwareSerial 1.0     /home/apms/.arduino15/packages/arduino/hardware/megaavr/1.8.8/libraries/SoftwareSerial

Used platform    Version Path
arduino:megaavr 1.8.8    /home/apms/.arduino15/packages/arduino/hardware/megaavr/1.8.8
```

- Upload the firmware to the monitor: ***$ ./upload.sh***
    - The output can be seen in the below screenshot.
    - If the upload fails, check the UPDI wire and link, Schottky diode and the wiring of the level shifter.
    - The Monitor should reboot and send an SMS to the operator.

```
apms@apms-robben:~/apms/system_monitor $ ./upload.sh
Connecting to SerialUPDI
Pinging device...
Ping response: 1E9651
Erasing device before writing from hex file...
Writing from hex file...
Writing flash...
Done.
```

BIOMARK READER

The BioMark RFID Reader is device used to record penguins that have been tagged. The device outputs the tag number via a serial-to-USB converter, accessible via a mini-USB port on the device.

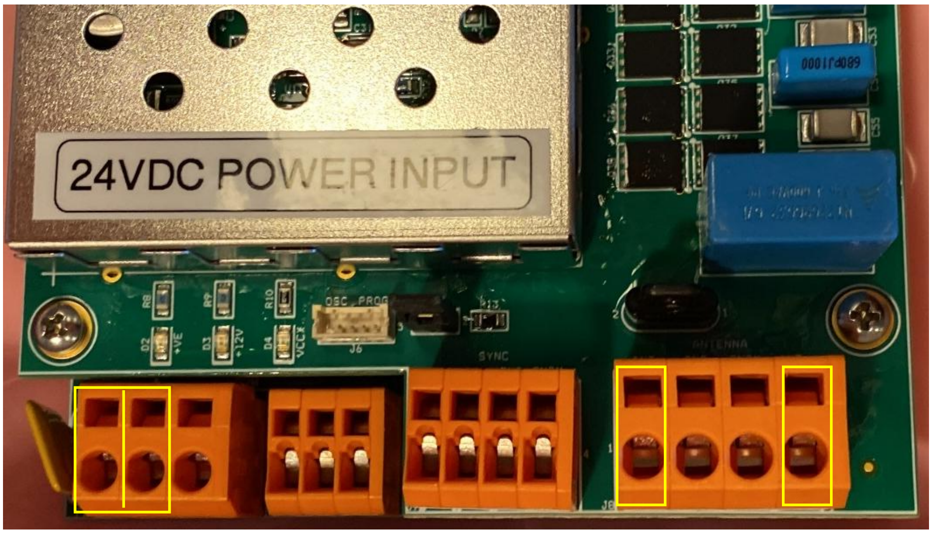
The reader has many different configurations and the latest BioMark manual should always be followed.

OVERVIEW

Below is some information of the currently used model, for reference. The Raspberry Pi reads the tags via a USB cable, plugged into the front of the Main Unit.

For the reader to operate, it needs 24V power and an antenna connected. For the pinout, see FIGURE 46 - BIOMARK PINOUT.

To make an antenna, see MAKING A BIOMARK ANTENNA.



+ -

24V

+ -

ANTENNA

Figure 46 - BioMark Pinout

The BioMark reader can be configured either via the BioMark Software (Device Manager) or via the Raspberry Pi terminal. Both methods described here use a USB-A to mini-USB cable to connect to the Reader.

It is also possible to connect to reader via a RS232 or Bluetooth connection. Refer to the BioMark documentation for more information if this is required.

Application Firmware Version v 1.7.1

CONNECT TO BIOMARK VIA DEVICE MANAGER

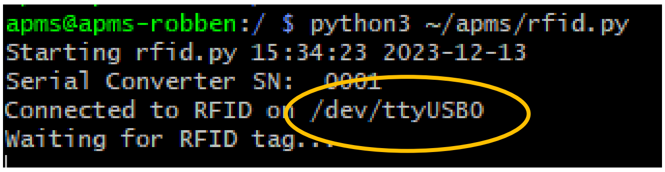
- BioMark Device Manager, can be installed on a Windows PC to configure the system. Currently version 1.2.15.1 is being used and can be found at: **s3://apmsbirdlife/APMS/software/BioMark/DeviceManagerStandalone_1.2.15.1.zip**

- The Windows 10 and Windows 11 driver for the serial converter, used by the BioMark Reader, can be found at: **s3://apmsbirdlife/APMS/software/BioMark/CP210x_Universal_Windows_Driver_11.3.0.zip**

- Please refer to the IS1001 Reader User Manual and the Device Manager Software Manual for instructions on how to connect and configure the reader. Both can be found at:
    - **s3://apmsbirdlife /APMS/Documents/IS1001_Reader_User_Manual_Rev_11.pdf**
    - **s3://apmsbirdlife/APMS /Documents/Device-Manager-Software-Manual-v1.1.pdf**
- Refer to the section CONFIGURING THE BIOMARK READER for the required settings.

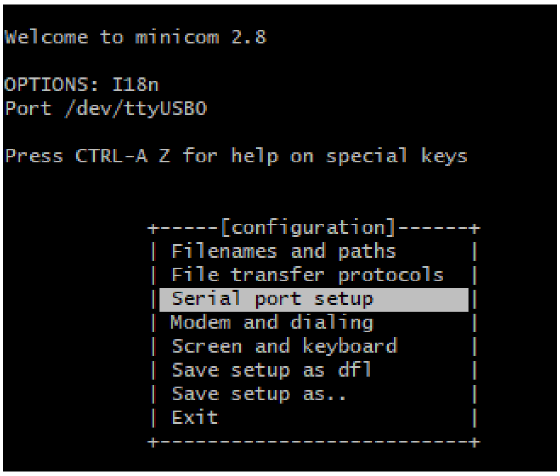
CONNECT TO BIOMARK VIA RASPBERRY PI TERMINAL

- To connect to the BioMark Reader from the Raspberry Pi terminal, ensure that both the BioMark Reader and Raspberry Pi are powered up.
- SSH into the Pi
- Check that the rfid service is not running in the background and stop the service if it is:
    - Check with: $ ***systemctl status rfid***
    - If active, stop the service: ***$ sudo systemctl stop rfid***
- Connect the USB cable to the mini-USB port on the BioMark reader and to any of the USB ports of the Raspberry Pi. If an APMS Main Unit has been built, the front USB port can be used.
- After the USB cable is plugged into the Pi, the BioMark serial converter will be assigned to a device file. This will either be "ttyUSB0", "ttyUSB1" or "ttyUSB2", depending on what other USB devices are plugged in. To determine which device file to use:
    - Run the rfid script: ***$ python3 ~/apms/rfid.py***
    - Find the device file name as indicated in the picture below.

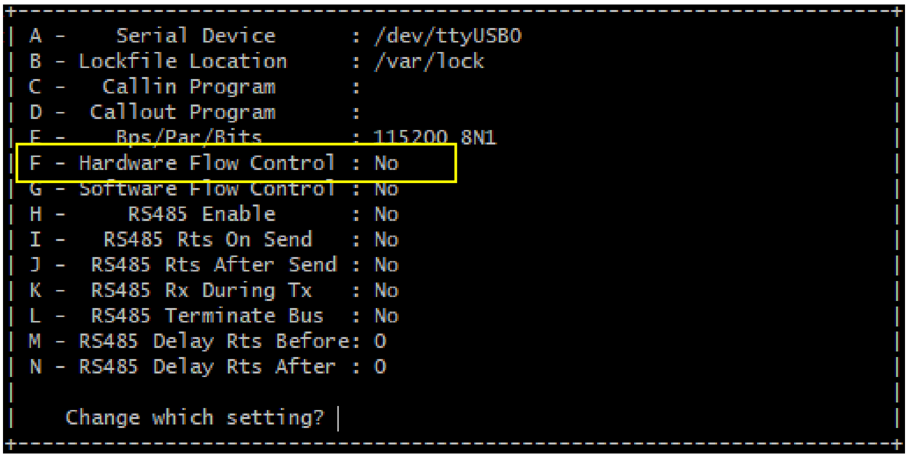


    - Close the script: Hold **CTRL** and press **c**

- To connect to the BioMark reader from the Pi, a program called Mincom will be used. This was installed when the Raspberry Pi was set up. See INSTALL APPLICATIONS AND LIBRARIES.
  - Run: **$ minicom -D /dev/[device file]**
    - Example: *$ minicom -D /dev/ttyUSB0*
  - Once inside minicom, disable “Hardware Flow Control”:
  - Hold **CTRL** and press **a** -- Then press ‘**o**’ to bring up the options menu, go to “**Serial port setup**” and press Enter. See screenshot below



  - Ensure “Hardware Flow Control” is off (It must have “no” next to it) – see picture below.
    - If “Hardware Flow Control” is on, turn it off by pressing ‘ **f** ’.



  - Press **Enter** to save and return to the Options menu.
  - Press **Esc** to exit the Options menu

- Type “**rfs**” and press **Enter**. The current configuration should be printed to the screen.
- You will not see the letters “rfs” until you press **Enter**.
- An Example for Robben Island can be seen below.

```
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.8 | VT102 | O
rfs
INF: Start Of Full Status Report
 Reader:
  ID:                                              01
  Model:                                       IS1001
  S/N:                                      1625.1566
  Date:                                    12/13/2023
  Time:                                      16:22:31
  Application Firmware Version:                 1.7.1
  Operation Mode:                                Scan
  Network Mode:                     IS1001 Standalone
  Exciter Sync. Mode:                      Standalone
  Date/Time Sync.:                           Disabled
  Beeper:                                    Disabled
  Tag Display Format:                             DEC
  Initiation Delay:                          Disabled
  Auto Standby Voltages:                   16 V, 18 V
  Periodic Standby Start Time:                   0:00
  Periodic Standby Duration:                 Disabled
  Idling Time:                               Disabled
 Alarms:
  Antenna Current Low Alarm:                    3.5 A
  Noise High Alarm:                               50%
  Tuning Capacit. High Alarm:                     970
  Tuning Capacit. Low Alarm:                       10
  Alarms Unique Delay:                       3600 sec
 Antenna/Tuning:
  Exciter Voltage Level:                            2
  Dynamic Tuning:                             Enabled
  Tuning Target Phase:                            387
  Tuning Target Phase Deviation Threshold:          4
 Communication:
  Local Port Speed:                            115200
  Tags To Local Port:                         Enabled
  Alarms To Local Port:                       Enabled
  Messages To Local Port:                     Enabled
  Remote Port Protocol:                         ASCII
  Remote Port Transfer Rate:                     Full
 Detection:
  HDX Tag Detection:                          Enabled
  Fastag Detection:                           Enabled
  FDXB Tag Signal Level Detection:            Enabled
  BioTherm Tag Temperature Detection:        Disabled
  Detection Counter Enabled:                      Yes
  Unique Mode:                                  Delay
  Unique Delay:                              1200 sec
  FDXB Detection Scan Time:                    120 ms
  VTT Level:                                      128
  Auto VTT Delay:                             360 min
 Measurements:
  Antenna Current Gain:                           120
  Antenna Current Offset:                         110
 Memory:
  Tags Memory Size:                             78583
  Status Reports Memory Size:                    1023
  Store Tags To Memory:                       Enabled
  Store VTT To Memory:                        Enabled
  Store Stat. Reports To Memory:              Enabled
 Reports:
  Auto Noise Report Delay:                    360 min
  Auto Status Report Delay:                  1440 min
 Diagnostics:
  Detection Counter:                              678
  Tags In Memory:                         64729 (82%)
  Status Reports In Memory:               1023 (100%)
  Input Voltage:                               25.9 V
  Exciter Voltage:                             14.0 V
  Antenna Tuning:                               Tuned
  Antenna Current:                              4.3 A
  Tuning Capacitors:                              227
  Tuning Phase:                                   387
  Tuning Relative Phase:                            0
  FDXB Signal Level:                     214 mV (23%)
  Temperature:                                 44.2 C
  Sync. Input Present:                            N/A
  Sec. Master Active:                             N/A
 Active Alarms:
  Reports Memory Full
INF: End Of Full Status Report
```

- If you see the configuration page as above, it means you have successfully connected to the BioMark Reader and can now send/receive commands in the console.

CONFIGURING THE BIOMARK READER

After successfully connecting to the BioMark Reader as discussed previously, the system can be configured.

- If using the "Device Manager" software, please refer the BioMark documentation and use the settings provided below.
- If connecting via a Raspberry Pi terminal, use the commands below to set the BioMark parameters.
- **Before executing any commands, it is always a good idea to run "rfs" and check that a status report is received. This confirms the reader is working properly and clears the serial buffer of any garbage characters that may be present.**

The settings given below are used for a BioMark Reader connected to an APMS.

- If connecting using the Device Manager software, refer the BioMark documentation.
- If connecting to the BioMark Reader via the Raspberry Pi terminal (minicom), the commands given in the table below can be used to configure the system. Keep the default for all other settings. Compare the values in the table with the values obtained from executing "rfs", and change the settings as needed.

Setting	Value	Command
Operation Mode	"Scan"	**ROM1**
Network Mode	IS1001 Standalone	**RNMS**
Exciter Sync. Mode	Standalone	**RSM0**
Beeper	Disabled	**RBS0**
Tag Display Format	DEC	**RDFD**
Exciter Voltage Level	3	**AVE3**
Dynamic Tuning	Enabled	**ADT1**
Tags To Local Port	Enabled	**CTL1**
Remote Port Protocol	ASCII	**CPRA**
Fastag Detection	Enabled	**DFH1**
Detection Counter Enabled	Yes	**DCS1**
Unique Mode	Delay	**DUMD**
Unique Delay	1200 sec	**DUD1200**
Auto VTT Delay	360 min	**DTD360**

- After changing all the necessary parameters, run "rfs" again to confirm all the settings are correct.
- Take a test tag and move it close to the antenna. The tag should be picked up and displayed in minicom. If so, you have successfully configured the BioMark reader.
- Exit minicom
  - Hold **CTRL** and press **a** -- Then release keys – Then press ' **q** ' and then **Enter** to exit.

MAKING A BIOMARK ANTENNA

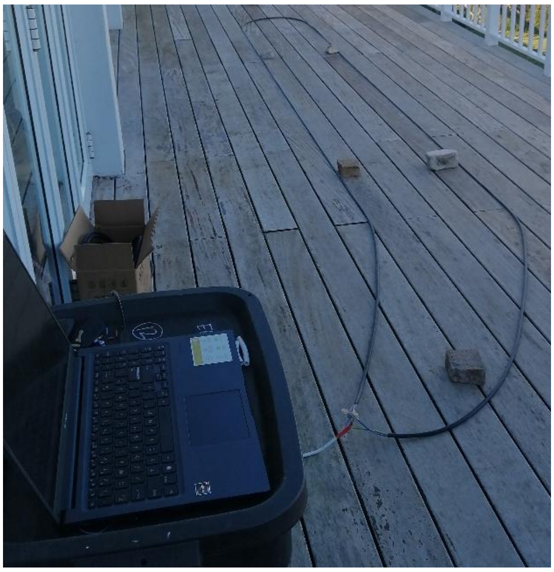
An experimental antenna has been made with some success. To test a new antenna, first ensure a connection to the BioMark Reader has been made, as discussed previously.

As the wire being used might be slightly different, the basic method will be described below. The antennas at Dyer Island and Robben Island were made using the same 3-core wire as used to run power from the solar supply to the APMS. The main idea behind making an antenna is to make multiple loops and cutting it down to the correct length to achieve resonance.

- Start with a length of wire, L, strip both ends and connect the wires as shown in FIGURE 47 - CABLE LAYOUT WHILE MAKING ANTENNA.

Figure 47 - Cable Layout while making Antenna

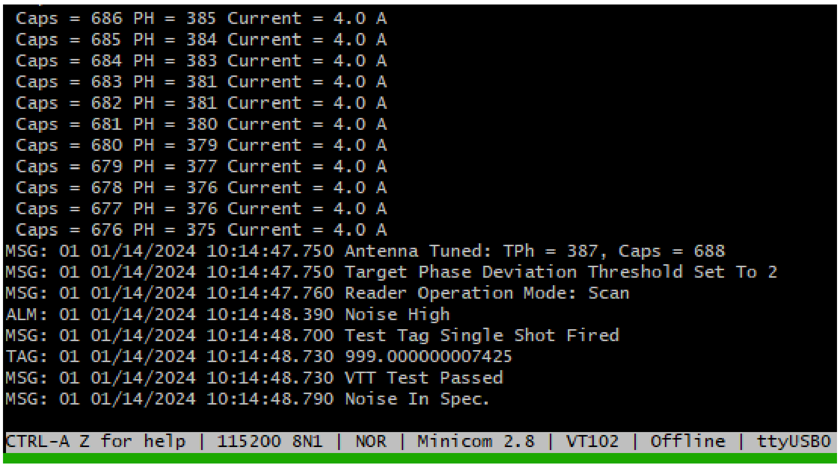
- The length to start at is a bit of a guess. The three core wire used for the latest antennas ended up being approximately 15m. Thicker wires or more loops will result in a shorter length.
- Add a “fly-lead” of approximately 1m to join the antenna and the BioMark reader.
- During the test, lay the wires parallel, 60cm apart, and connect the antenna to the BioMark reader. See below.



- Once the antenna is connected, and a connection has been made with the BioMark, the antenna can be tested. In the BioMark console, send the command “aft”. This is “Antenna Full Tune”. The reader will try to tune the antenna by adding capacitance and give a final “caps” value between 0 and 1024. As the antenna cable is too long at the start, the number should be low.
- Cut a piece of the cable off, connect in the same fashion and test again. The number might remain the same until the antenna starts getting close to resonance.

- Repeat this process until a caps value of approximately 512 is reached.
    - Be careful not to cut too much off, as the number starts growing rapidly once it gets close to resonance.
- At this stage, the antenna current should also have increase. (>3A)
- Test the range of the new antenna with a test tag. Use the same test procedure when placing an antenna in the field.
- Solder the wires and insulate using heat shrink to prevent moisture from corroding the joints

Note, when testing in the field, the caps number will vary wildly as the antenna shape is changed. The important thing is that the antenna tunes, and a tag is picked up from a decent distance. A screenshot of an antenna tested in the field (caps = 688) can be seen in FIGURE 48 - ANTENNA TUNE EXAMPLE


```
 Caps = 686 PH = 385 Current = 4.0 A
 Caps = 685 PH = 384 Current = 4.0 A
 Caps = 684 PH = 383 Current = 4.0 A
 Caps = 683 PH = 381 Current = 4.0 A
 Caps = 682 PH = 381 Current = 4.0 A
 Caps = 681 PH = 380 Current = 4.0 A
 Caps = 680 PH = 379 Current = 4.0 A
 Caps = 679 PH = 377 Current = 4.0 A
 Caps = 678 PH = 376 Current = 4.0 A
 Caps = 677 PH = 376 Current = 4.0 A
 Caps = 676 PH = 375 Current = 4.0 A
MSG: 01 01/14/2024 10:14:47.750 Antenna Tuned: TPh = 387, Caps = 688
MSG: 01 01/14/2024 10:14:47.750 Target Phase Deviation Threshold Set To 2
MSG: 01 01/14/2024 10:14:47.760 Reader Operation Mode: Scan
ALM: 01 01/14/2024 10:14:48.390 Noise High
MSG: 01 01/14/2024 10:14:48.700 Test Tag Single Shot Fired
TAG: 01 01/14/2024 10:14:48.730 999.000000007425
MSG: 01 01/14/2024 10:14:48.730 VTT Test Passed
MSG: 01 01/14/2024 10:14:48.790 Noise In Spec.

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.8 | VT102 | Offline | ttyUSB0
```

Figure 48 - Antenna Tune Example

TESTING EACH SCRIPT

Each part of the system can now be tested individually.

TESTING THE SCALES.PY SCRIPT

Before testing, ensure the APMS is configured - See CONFIGURING THE APMS.

- Plug both scales into their respective sockets.
- Power up the APMS Main Unit
- SSH into the Pi.
- Check if the scales service is running in the background and stop the service if it is:
  - Check with: $ ***systemctl status scales***
  - If active, stop the service: ***$ sudo systemctl stop scales***
- Ensure the indicators show weight readings of close to 0 kg (It should be within 1g after calibration)
- Check the scale frequencies: ***$ python3 ~/apms/test_scale_frequency.py***
  - Currently the scales are set to 100 Hz.
  - Check that the scale frequencies are within 1% (between 99 Hz and 101 Hz)
- Run the scales script: **$ python3 ~/apms/scales.py**
  - You should see the following on the screen:

```
apms@apms-robben:~/apms $ python3 scales.py
scales.py running...
Identifying Serial Ports...
Serial port for Scale 1 is: /dev/ttyUSB2
Serial port for Scale 2 is: /dev/ttyUSB1
Weighbridge running...
```

  - Put a <u>1kg</u> weight on <u>scale 1</u>.
  - As it is less than the weight of a penguin, it should detect a high dead weight.
  - Check that it is indeed <u>scale 1</u> showing the high dead weight.
  - Remove the <u>1kg</u> weight. A message should state the dead weight has reduced
  - Repeat for scale 2
  - Please see FIGURE 49 - 1KG TEST.

```
Now plug the serial converter into any of the USB ports and press ENTER

Serial number detected: B000A3RA
Thank you. Goodbye
apms@apms-stcroix:~ $ sudo nano ~/apms/config.json
apms@apms-stcroix:~ $ systemctl status scales
Unit scales.service could not be found.
apms@apms-stcroix:~ $ python3 ~/apms/scales.py
scales.py running...
Identifying Serial Ports...
Serial port for Scale 1 is: /dev/ttyUSB1
Serial port for Scale 2 is: /dev/ttyUSB0
Weighbridge running...
Check dead weight on scale_1: 1.002 kg
Dead weight on scale_1 reduced to acceptable level
Check dead weight on scale_2: 1.026 kg
```

Figure 49 - 1kg Test

- Place a 5kg weight on scale 1.
    - As this weight is greater than the minimum penguin weight, an event should be started. As scale 1 was triggered first, it should indicate a penguin heading out to the ocean.
    - Wait a few seconds and remove the 5kg weight. After removing the weight, there will be a slight pause and then the event should finish, and the data stored to a file. This pause is the time period set in the config file for "SCALE_EMPTY_TIME".
    - A screenshot is shown below:

```
11/12/2023 20:41:34 - Scale 1 has been triggered. Heading out to ocean.
Max serial 1 buffer: 8/128
Max serial 2 buffer: 8/128
11/12/2023 20:42:14 - Scale event finished.
Saving to /home/apms/data/2023/20231211/scale/S20231211204134STC.csv
```

    - Repeat for scale 2. This time, the system should detect scale 2 being triggered first, displaying a message that a penguin is coming back from the ocean.
      Remove the weight and wait for the event to finish.

- The testing of the scales is complete. The files generated in the above tests will be checked for accuracy after uploading them to the cloud. See TESTING THE DATA_TRANSFER.PY SCRIPT

TESTING THE RFID.PY SCRIPT

The APMS picks up RFID tags via an external RFID reader made by BioMark (Model IS1001).

- Before testing the rfid script, please ensure the BioMark Reader has been fully configured and the BioMark antenna installed. See CONFIGURING THE BIOMARK READER.
- Connect a cable between the mini-USB port on the BioMark Reader and the front USB port of the APMS main unit.
    - If an APMS main unit is not being used, the Raspberry Pi will have to be powered by a 5V power supply and any USB port on the Pi can be used to connect to the BioMark Reader.

- Power up the BioMark Reader and APMS main unit.
- SSH into the Pi.
- Check that the rfid service is not running in the background and stop the service if it is:
    - Check with: ***$ systemctl status rfid***
    - If active, stop the service: ***$ sudo systemctl stop rfid***

- Run the rfid script: ***$ python3 ~/apms/rfid.py***
- You should see the following on the screen.
    - Note that the RFID reader could be assigned to "*ttyUSB0*", "*ttyUSB1*" or "*ttyUSB2*".

```
apms@apms-robben:/ $ python3 ~/apms/rfid.py
Starting rfid.py 15:34:23 2023-12-13
Serial Converter SN:  0001
Connected to RFID on /dev/ttyUSB0
Waiting for RFID tag...
```

- Take a test tag and move it close to the antenna. The tag should be registered and saved to a file. This file will also be checked after uploading to the cloud.

```
python3[4056]: Starting rfid.py 10:47:22 2023-08-01
python3[4056]: Serial Converter SN: 0001
python3[4056]: Serial port for RFID is /dev/ttyUSB1
python3[4056]: Connected to RFID on /dev/ttyUSB1
python3[4056]: Waiting for RFID tag...
python3[4056]: Tag Number: TAG: 01 08/01/2023 11:35:09.180 999.000000007425
python3[4056]: Saving data to /home/apms/data/testdata/2023/20230801/rfid/R20230801113509ROB.csv
python3[4056]: Data saved succesfully
```

TESTING THE SYSTEM_STATUS_SHECK.PY SCRIPT

The system_status_check.py script automatically runs every hour, at 10 past the hour. The script is used to monitor the health of the entire APMS. See FIGURE 50 - OUTPUT OF SYSTEM_STATUS_CHECK.PY SCRIPT for a screenshot after running the script manually. The main features of the script are:

- Send an SMS to the operator if the Raspberry Pi has rebooted.
- Check Last time data was uploaded and backfill data if needed.
  - The script will check when last the data was uploaded to the cloud and if more than a day, the data transfer script will be executed to upload all outstanding data.
- Check CPU temperature
  - If the CPU temperature exceeds the WARNING or DANGER thresholds, an email will be sent to the operator. Thresholds are set in the configuration file. See CONFIGURING THE APMS and adjust CPU_TEMP_WARNING and CPU_TEMP_DANGER variables.
- Checks CPU load and memory usage
- Checks USB Flash drive is mounted and storage available
- Tests the internet connection
- Check if a BASH command has been received from Monitor and executes it (still in development)
- Receives status report from Arduino Monitor (Used for Dashboard)
- Sends a report to the Arduino Monitor
  - Used when requesting a status report from the Monitor, via SMS. See USING THE APMS MONITOR.

All the information is logged to a daily system status file. This file will normally have 24 entries, one for every hour. On occasion, the request for a status report from the Monitor might fail. The operator should receive an email alert. If this happens, wait one hour and check if the next status report is received (at 10 past the hour). If not investigate.

The daily status file can be viewed with ***cat*** and is located at:

~/data/ [YEAR]/[YYYYMMDD]/[YYYYMMDD]_STATUS.txt

For example: ***$ cat ~/data/2023/20231230/20231230_STATUS.txt***

This file is uploaded to the cloud daily and most of the info on the Dashboard is obtained from this file.

To test the system_status_check.py script, do the following:

- Run the script: ***$ python3 ~/apms/system_status_check.py***

A typical output is shown in FIGURE 50 - OUTPUT OF SYSTEM_STATUS_CHECK.PY SCRIPT.

```
############################################################
############          APMS Status Report         ###########
############ Date: 20240113       Time: 13:25:50 ###########
############################################################

Opened serial port: /dev/ttyS0
Rpi Last Boot:  2023-11-08 14:00:25
Last Upload:  2024-01-13 01:00:29
The current CPU temperature is 55.5 deg C
CPU temperature OK
CPU max load is 8.3%
Memory Used: 6.9%
uSD storage used: 19.5%
Checking USB Flash Drive...
USB drive partition /dev/sda1 mounted to /home/apms/flash/
USB drive storage used: 26.5%
Testing internet connection...
ping Google at 8.8.8.8
Device is connected to the internet
Checking command request from Monitor
No command request
Requesting Monitor status report...
-------------------------------------------------------
           Start of Arduino Status Report
-------------------------------------------------------
Arduino boot time:       2023-12-07 11:49:03
Arduino system time:     2024-01-13 13:25:56
Stored Mobile Number:    +27728391080
Ambient temperature:     47.41 deg C       Temperature OK
4V SIM800L Supply:       4.09 V            Voltage OK
5V (Raspberry Pi):       5.06 V            Voltage OK
12V (Router, scales):    12.05 V           Voltage OK
24V Solar Supply:        27.62 V           Voltage OK
-------------------------------------------------------
          End of Arduino Status Report
-------------------------------------------------------

Requesting to send Raspberry Pi System Status Report.
Response was: ACK$S
Sending Raspberry Pi Status Report
STO
BT:20231108 14:00
CH:20240113 13:26
S:Y
R:Y
T:Y
I:Y
CPU:55.5
MEM:6.9
SD:19.5
USB:26.5
--------------------------------------------
Raspberry Pi System Status Report sent


Saving data to:  /home/apms/data/2024/20240113/20240113_STATUS.txt
---No alerts were triggered---


############################################################
############       End of APMS Status Report     ###########
############ Date: 20240113       Time: 13:26:11 ###########
############################################################
```

Report received from Monitor

Raspberry Pi Status Report sent to Monitor

Figure 50 - Output of system_status_check.py script

TESTING THE DATA_TRANSFER.PY SCRIPT

The APMS system stores data in three places. Initially, the raw files are saved to the internal micro-SD card of the Raspberry Pi. Then, every morning at 1AM the new data is transferred to the USB Flash drive and to the cloud. The APMS uses AWS S3 for cloud storage. It is the responsibility of the administrator to manage the storage space available on the USB Flash drive and the internal micro-SD card. Both these parameters can be viewed on the dashboard and an alert is set for when either one reaches 75% capacity.

- Ensure there are some data files to transfer and that the APMS has been configured. If you have already completed the testing of the scales and rfid scripts, there should be some test files to transfer. See TESTING THE SCALES.PY SCRIPT and TESTING THE RFID.PY SCRIPT.
- Run the data transfer script: ***$ python3 ~/apms/transfer_data***
    - You should see something similar to the screenshot below

```
##################################################################
Data Transfer Started            14-12-2023 - 20:46:46
##################################################################
Checking USB Flash Drive...
USB drive partition /dev/sda1 mounted to /home/apms/flash/
Backing up to USB flash drive...
Data succesfully copied to the USB flash drive

Testing internet connection...
ping Google at 8.8.8.8
Device is connected to the internet
Uploading log file...
Uploaded log file to the cloud succesfully
Uploading data for the following days:
20231214
Sending data remotely for: 20231214...
local path:  /home/apms/data/2023/20231214
remote path:  s3://apmsbirdlife/APMS/data/robben_island/2023/20231214/
Remote upload to s3://apmsbirdlife/APMS/data/robben_island/2023/20231214/ successful
Uploaded data to the cloud succesfully
Data for robben_island succesfully transfered to the cloud

Email Message:
 APMS Data Transfer Report for robben_island:
Data succesfully copied to the USB flash drive
Log file for robben_island succesfully transfered to the cloud
Data for robben_island succesfully transfered to the cloud

Sending email
email sent
##################################################################
Data Transfer Finished           14-12-2023 - 20:47:00
##################################################################
```

- Every time data is successfully uploaded, the file *~/logs/last_upload.txt* is updated.
    - Verify today's date and time was added to the bottom of the file after the transfer:
      $ cat ~/logs/last_upload.txt

- To check if the data transferred to the USB Flash drive:
    - Ensure it is mounted. See SET UP THE USB FLASH DRIVE.
    - List the files in the scale and rfid data directory. For example, to list the files for today, the 15th of December 2023:
        - ***$ ls ~/flash/2023/20231215/scale/***
        - ***$ ls ~/flash/2023/20231215/rfid/***

    - Adapt the lines above for the day the test is being conducted.
- To check if the data transferred to the cloud:
    - Open an S3 Client, such as S3 Browser.
    - Check the scale and rfid data directory. For example, the scale and rfid directories for Stony Point, for the 15th of December 2023, can be found at:
        - *s3://apmsbirdlife/APMS/data/stony_point/2023/20231215/scale/*
        - *s3://apmsbirdlife/APMS/data/stony_point/2023/20231215/rfid/*

TESTING THE APMS_EMAIL.PY SCRIPT

The apms_email.py script is a useful script that can be run from the command line to send an email. The script is also capable of sending a file as an attachment. It is important to check this script is working, as it will be used to email the operator in case of a fault.

The syntax is as follows: apms_email "body of email" [path to file]

example: ***$ python3 ~/apms/apms_email.py "this is the body of the email" ~/attachment.zip***

If the script was not successful, examine the email information in the APMS configuration file. See CONFIGURING THE APMS

TESTING THE LOG.PY SCRIPT

The log.py script is crucial and should be used whenever a calibration is done, something unusual is detected with the system or when any work is carried out. The log script will make a backup of the existing log file whenever a log entry is added. The backed-up log files can be found at: **~/logs/backup**

To use the log script:

- SSH into the Pi.
- Run the log script: ***$ python3 ~/apms/log.py***
- Follow the onscreen commands to log an entry

If this is the first log entry, a new log file will be created (~/logs/log.txt)

- View the log file to check the entry was added ***$ cat ~/logs/log.txt***
- Edit the log file if a change needs to be made ***$ sudo nano ~/logs/log.txt***

The log entries are displayed on the Dashboard. See

APMS DASHBOARD.

TESTING THE ON/OFF BUTTON

To test the ON/OFF button, simply press it. The Raspberry Pi should start shutting down if it was on. The lights will flash and eventually a solid red light will be seen. If the Raspberry Pi was off, it should boot up. If it does not work, check the wiring of the button on the Monitor, as well as the configuration of GPIO on the Pi. See BUZZER AND BUTTON and CONFIGURE GPIO3 FOR POWER ON/OFF for more info. Another way to isolate the fault is to power down the Pi via the Monitor command: “CMD:P”. If this works, the Pi has been configured properly.

CONFIGURE THE CRONTAB

The crontab is a task scheduler and is responsible for the following:

- Check the system every hour.
- Transfer APMS data to the cloud every day.
- Update the system clock once a week and at boot.

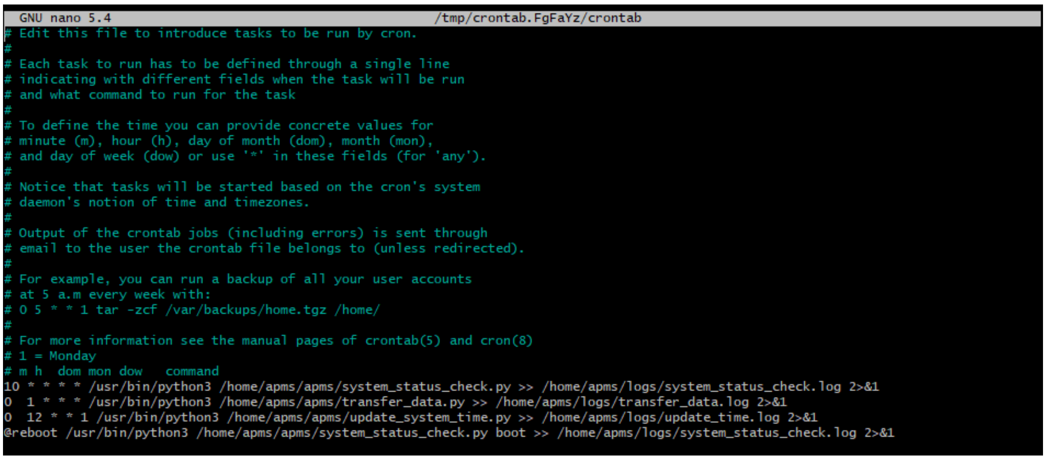
The crontab sends the output of each script to a log file. See ~/logs/

SET UP THE CRONTAB

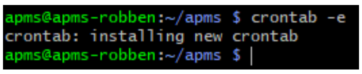
To configure the crontab:

- Run: $ **crontab -e**
- If a message comes up asking to choose a text editor, choose “nano”
- Copy and paste the following into the text editor, at the bottom of the file. See *** for an example.

```
10 * * * * /usr/bin/python3 /home/apms/apms/system_status_check.py >> /home/apms/logs/system_status_check.log 2>&1
0  1 * * * /usr/bin/python3 /home/apms/apms/transfer_data.py >> /home/apms/logs/transfer_data.log 2>&1
0  12 * * 1 /usr/bin/python3 /home/apms/apms/update_system_time.py >> /home/apms/logs/update_time.log 2>&1
@reboot /usr/bin/python3 /home/apms/apms/system_status_check.py boot >> /home/apms/logs/system_status_check.log 2>&1
```



- Save and exit. (Hold **CTRL** and press **x** -- Then press ‘**y**’ to save -- then press Enter to exit)
- A message should state that the crontab has been successfully updated.



TESTING THE CRONTAB AND TIME UPDATE SCRIPT

To test whether the crontab is working correctly, view the log files of the scripts:

$ cat ~/logs/system_status_check.log
$ cat ~/logs/transfer_data.log
$ cat ~/logs/update_time.log

COMMISIONING A NEW SYSTEM IN THE FIELD

After each component of the APMS have been built and tested, as outlined in this manual, the system can be installed in the field. It has also been assumed that a 24V Solar Power Supply has been built and power is available at the location of the APMS.

Please take note of the following:

APMS LOCATION

When choosing a location to install the APMS, ensure the following:

- A high rate of penguins crossing, going to, <u>and</u> coming back from the ocean
- The distance between the scales and the APMS Main Unit should not be excessive.
- The distance between solar supply and the Main Unit should be taken into consideration.
- The cell signal strength is strong enough at the exact location where the 4G Router and APMS Monitor will be operating. Use *www.speedtest.net*
- Try to disturb the area as little as possible as the penguins are very sensitive to changes in their environment. This could lead to them using a different path to avoid the weigh bridge. This is also why every inch of the Scales Platform is covered in dark green canvas.

APMS ENCLOSURE

- At Stony Point, the APMS is housed in a large steel box, together with the BioMark RFID Reader. The RFID reader itself is in a pelican case.
- At Dyer Island and Robben Island a rugged 125l box, made by Skitch is used. This allows enough space for both the APMS Main Unit, BioMark circuit board and the 4G Indoor Router, when used. Drill holes for the scale cables, RFID antenna and for power to enter the Skitch box. Use grommets or feed irrigation piping directly into the box. Use silicone to seal around the areas where the pipe enters to avoid water getting in. Where possible, angle the pipes so that they enter the Skitch box in an upward angle, to prevent water running down the pipe and into the box. See FIGURE 51 - SKITCH BOX HOUSING APMS and FIGURE 52 - CLOSED SKITCH BOX WITH IRRIGATION PIPING
- At Bird Island a smaller Skitch box is used, with the BioMark in its own enclosure.
- It’s a good idea to paint the enclosure white, as it will help reduce the temperature inside. Decide whether it is necessary to bolt the Skitch box down. At Bird Island it was bolted down as the location was very much exposed to the wind.

Figure 51 - Skitch Box Housing APMS

Figure 52 - Closed Skitch Box with Irrigation Piping

WIRING THE SKITCH BOX

After finding a suitable location for the APMS, drilling the holes and laying the irrigation pipes, the wiring inside the Skitch box can start.

- Feed the 24V DC power wires from the solar supply, through the irrigation piping and into the box. From here, it must split to provide power to both the APMS Main Unit, as well as the BioMark Reader.
- Solder a DC Plug (RS: 705-1519) to the 24V power coming into the box, to power the Main Unit.
- The BioMark reader should be equipped with a circuit breaker, which can also act as a power switch when needed.
- If an indoor router is used, plug the router into the 12V DC OUT socket.
- Feed the BioMark antenna into the box and connect to the BioMark reader as in the photo above.
    - See PLACING THE BIOMARK ANTENNA for information regarding the placement of the antenna
- Plug the one end of the USB Cabe (RS: 822-3226) into the mini-USB port on the BioMark reader, and the other end into the USB-A connector on the front of the Main Unit.

INSTALL THE SCALES PLATFORM

- It is important to get the Scale Platform as level as possible. Use a spirit level and check both sides of the platform. Ensure that the scale pans are level before leaving the site.

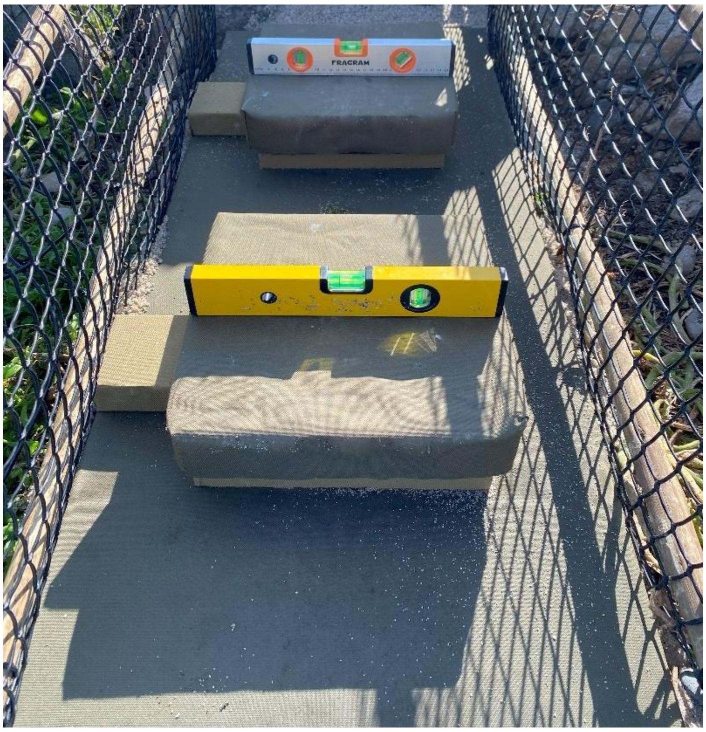


Figure 53 - Check Scales using a Spirit Level

- Use sand and rocks to provide a level surface for the platform to rest on.
- Feed the scale cables through irrigation piping into the Skitch box. This is to protect the cable, especially if the cable has been lengthened and there is a joint.
- At all times, the canvas covered pan must rest on the frame and not be touching anything else. If twigs and branches are touching the pans, the weight reading will be affected.

Figure 54 - Robben Island Scales Platform

PLACING THE BIOMARK ANTENNA

Choose a suitable location for the BioMark Antenna. Avoid running the antenna close to other cables and do not let the antenna overlap itself. Change the shape of the antenna and do a full tune. When the antenna is tuned, test with a test tag to check for detection at an adequate distance.

PUTTING UP A WARNING SIGN

Putting up a warning sign is important in order to notify anyone in the vicinity that they should keep clear. The sign can, however, alert troublemakers of the weighbridge's existence. The decision to put up a sign thus lies with the installer. The sign is made by [insert info here] and the file is stored at [insert info here].

FINAL CHECKS

The system can now be started, but before doing so, it is important to ensure everything is in their final positions. The following is a description of the checks

- The Solar Power supply is fully operational
- All enclosures are fully closed and locked: BioMark Reader, APMS Main Unit, Skitch Box, Router, etc
- The BioMark system has been securely connected, been fully configured and is in its final position.
- The BioMark Antenna has been laid out, cable tied to the RFID mount, tuned and tested.
- The Arduino Monitor has been tested and the SIM800L antenna is in its final position.
- The Main Unit is properly closed and in its final position.
- The router is plugged in, working and in its final position. Check outdoor router is properly closed.
- All tools and metal objects must be removed from the area
- If a warning sign is to be used, it must be put up and secure.

STARTING A NEW SYSTEM

For a new system, the rfid and scales scripts need to be configured to run in the background, as processes. To do this, do the following:

- Copy the service files
- Start the service
- Check the service is running
- Enable the service

- SSH into the system.
    - Run the following:
        - ***$ sudo systemctl daemon-reload***
        - ***$ systemctl start tmate***
        - ***$ systemctl enable tmate***
    - You can check that the tmate service is active by running: ***$ systemctl status tmate***
- Reboot the Pi: ***$ sudo reboot***
- *Wait a minute and try to SSH in remotely (via the internet). If this works, you have successfully configured tmate.*

APMS DATA PROCESSING

The data processing section consists of the data processing script, the code necessary to display the data on the Dashboard and a script to generate graphs for a website. Instructions below are given for Linux, but the data_processing.py script also works on Windows.

OVERVIEW

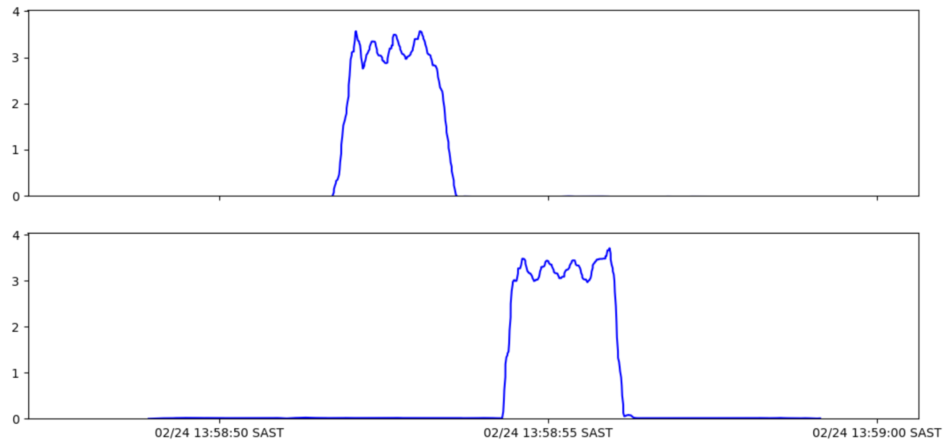
The data processing script has the task of taking in raw scale and RFID data and producing a csv file containing a list of penguins. The script can be used to process multiple days over any given date range. For the example below, only one day's data from one site is discussed. This process is repeated for all the days of all the sites requested.

A basic overview of how the script processes the penguin data for the previous day.

1. Download the scale and RFID data for the previous day.
    a. This is stored in *~/apms_data_processing/data/downloaded_raw/*
2. Copy the raw files to *~/apms_data_processing/data/processing* directory for further processing
3. Each scale file contains data for both scales and can be seen as one event.
4. For each scale, the event is broken up into subevents.
5. The penguins in each sub-event are added to the list of penguins for that scale
6. The lists from both scales are compared and the direction and weight of each penguin are determined.
7. The penguins for each event are added to the list of penguins for the day.
8. The list of penguins for the day is compared to a list of RFID tags for the day.
9. Penguin weights are linked to RFID tags where appropriate.
10. A CSV file is generated with a list of penguins and their weights. The list contains the following:
    a. Time of crossing
    b. RFID tag (if assigned),
    c. Penguin weight
    d. weight discrepancy between scale 1 and scale 2
    e. Direction (out to ocean or back from ocean)

DETERMINING THE WEIGHT AND DIRECTION OF A PENGUIN

The first step in determining the weight of a penguin is to read in a scale file and break it up into subevents. An example of a subevent is shown below. The weights are sampled at 100 times per second.



f. A sub-event is only successfully processed if:
    i. The weights on both scales are reasonable penguin weights.
    ii. All the penguins walking over the one scale is also picked up on the other scale
    iii. The weight difference between the two scale readings is small (set in config file)

g. The events that can not be processed

To process and display the penguin data, download the code from github

- See the section CONFIGURE GIT AND DOWNLOAD FIRMWARE to authenticate.
- Create a directory in the home drive, called “apms_data_processing”.
- Navigate to the apms_data_processing directory and clone the directory from github:

  $ git clone git@github.com:apmsbirdlife/apms_data_processing.git

- Once the data has been downloaded, the first step is to edit the config file

EDITING THE CONFIG FILE

All the main parameters for data processing and the Dashboard must be set in the config file. An example file is included, called “config_example.py”. Edit this file and change the parameters as required. Many of the settings can be left with their default values.

```
import os


HOME_PATH = os.path.expanduser('~')

INSTALL_PATH = os.path.join(HOME_PATH, 'apms_data_processing')

DATA_PATH = os.path.join(INSTALL_PATH, 'data')

LOG_PATH = os.path.join(INSTALL_PATH, 'logs')

################################################################################


################################################################################

# AWS S3

################################################################################

s3_data_path = 'APMS/data/'

bucket_name = 'apmsbirdlife'

aws_access_key='AKIA6DO2JOYFAUJR3R4G'

aws_secret_access_key='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'

################################################################################



################################################################################

# SCALE PARAMETERS

################################################################################

SAMPLING_FREQ = 100

SCALE_OVERLOAD = 15

SCALES = ['scale_1','scale_2']

MAX_SCALES_DIFFERENCE_ALLOWED = 0.3

# Allowed 1% error in average sampling period

# MAX_AVG_SAMPLING_PERIOD = 0.0101 # 10.1 ms

# MIN_AVG_SAMPLING_PERIOD = 0.0099 # 9.9 ms

MAX_AVG_SAMPLING_PERIOD = 0.011 # 10.1 ms

MIN_AVG_SAMPLING_PERIOD = 0.009 # 9.9 ms

MAX_DEADWEIGHT_DIFF = 0.05
```

```
MAX_DEAD_WEIGHT = 0.5
SCALE_1_OFFSET = 0
SCALE_2_OFFSET = 0
################################################################################


################################################################################
# PENGUIN PARAMETERS
################################################################################
PENGUIN_WALKING_PERIOD_MIN = 0.18 # 5.6 Hz
PENGUIN_WALKING_PERIOD_MAX = 0.4  # 2.5 Hz
MIN_PENGUIN_WEIGHT = 1.9
MAX_PENGUIN_WEIGHT = 4.7
MAX_OVERSHOOT_PERCENTAGE = 50
CLIMB_DURATION_MIN = 0.05

# For the box system, often there is a longer climb on/ climb off
CLIMB_DURATION_MAX = 2


# CLIMB_DURATION_MAX = 1.0
TRANSITION_WINDOW_INCREMENT = 0.05
################################################################################


################################################################################
# DATA PROCESSING PARAMETERS
################################################################################
# MIN_GAP_BETWEEN_TRANSITIONS = 0.3
MIN_TIME_BETWEEN_SCALES = 0
MAX_TIME_BETWEEN_SCALES = 1
MIN_TIME_BETWEEN_SUB_EVENTS = 0.3
# MIN_DURATION_VALID_WEIGHT = 0.21
MIN_DURATION_VALID_WEIGHT = 0.35
```

```
# PENGUIN_ISOLAION_INTERVAL = 180
# EVENT_FILE_ISOLATION_INTERVAL = 180
PENGUIN_ISOLAION_INTERVAL = 60
EVENT_FILE_ISOLATION_INTERVAL = 60

STD_DEV_STATIC = 0.015
# STD_DEV_DYNAMIC = 0.6
STD_DEV_DYNAMIC = 0.5

STD_DEV_MAX = 0.8
##########################################################################################



##########################################################################################
# RFID PARAMETERS
##########################################################################################
TEST_TAGS = ['999.000000007425', '3E7.0000001D01', '999.000000007426']
RFID_IGNORE_INTERVAL = 1200      # seconds window in which the same tag will be ignored (1200 = 20 min)
RFID_ISOLATION_INTERVAL = 600   # window of seconds in which no other tag must be registered (300 = 5 min)
RFID_TO_PENGUIN_INTERVAL = 60 # time between penguin crossing scales and rfid trigger
##########################################################################################



##########################################################################################
# DASHBOARD PARAMETERS
##########################################################################################
HOST_IP = '192.168.1.32'

# Data Process
DOWNLOAD_ALL = True

# Dashboard
DOWNLOAD_STATUS_FILES = True
```

```
DOWNLOAD_LOG_FILE = True

LOG_TO_INFLUX = True

STATUS_TO_INFLUX = True

PENGUINS_TO_INLFUX = True

#######################################################################################



#######################################################################################

# SITE PARAMETERS

#######################################################################################

STONY_POINT   = {'name':'stony_point','rfid_location':'middle_only', 'start_date':'20230603'}

BIRD_ISLAND   = {'name':'bird_island','rfid_location':'outer_only'  , 'start_date':'20230512'}

DYER_ISLAND   = {'name':'dyer_island','rfid_location':'middle_only' , 'start_date':'20230718'}

ROBBEN_ISLAND = {'name':'robben_island','rfid_location':'middle_only', 'start_date':'20230801'}


SITES = [STONY_POINT, BIRD_ISLAND, DYER_ISLAND, ROBBEN_ISLAND]
```

DATA PROCESSING SCRIPT

The APMS data processing script takes in raw scale and RFID data and analyses it to produce a CSV file, with a list of penguins. The list contains the time of crossing, penguin weight, direction, weight error and RIFD tag (if applicable). The process_data.py script can be used from the terminal to process date for a given site and date range. To download and use the script, do the following:

The APMS Dashboard shows the health of each site and is run using InfluxDB and Grafana server. Install a fresh copy of Ubuntu on a PC to act as a server.

To process data using the script, use the following syntax:

$ python3 process_data.py *[SITE] [DATE RANGE]* *where:*

- [SITE] can be left blank to process data for all sites, or a site name can be given.
    - The following site names can currently be used:
        - "stony_point"
        - "bird_island"
        - "dyer_island"
        - "robben_island"

[DATE RANGE] can be either be left blank, a date range such as "20230601-20231101" or number of days back, like "5". If left blank, it will process only yesterday's data.

Some examples:

(Process the last 7 days' worth of data for Stony Point)
$ python3 process_data.py stony_point 7

(Process yesterday's data for Bird Island)
$ python3 process_data.py bird_island

(Process yesterday's data for all the sites)
$ python3 process_data.py

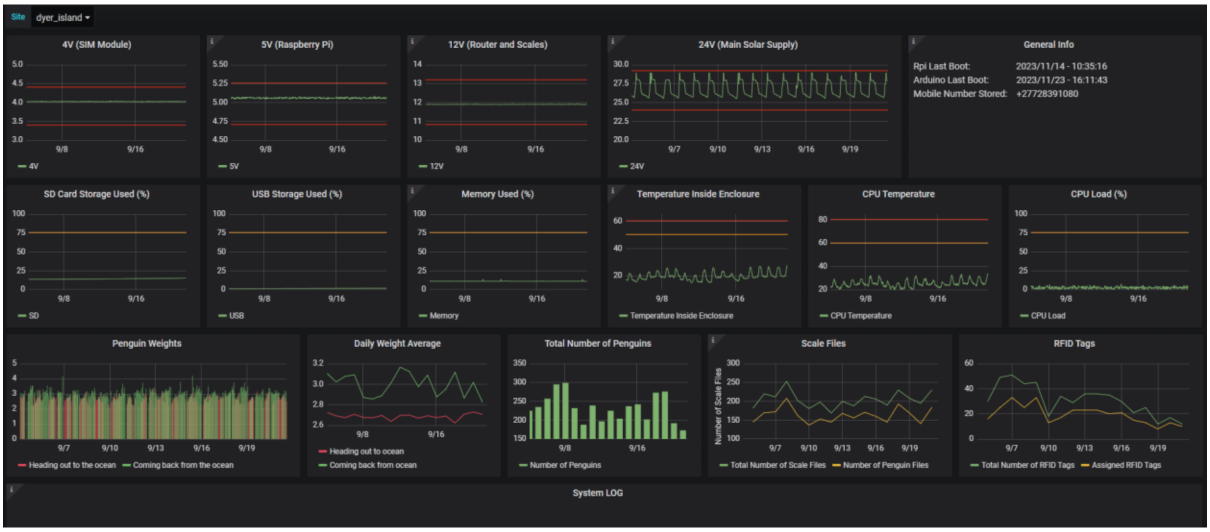
(Process data from the first of September 2023 till the first of November 2023, for Bird Island)
$ python3 process_data.py dyer_island 20230901-20231101

This will create a CSV file for each day. The CSV file will be stored in the "processed" directory. An example of a CSV file and where it is stored, is shown belowwith the following path:

~/apms_data_processing/data/processed/stony_point/2023/20231227/stony_point_20231227_processed_data.csv

APMS DASHBOARD

The APMS Dashboard allows the operator to view the health of the system at each site. A screenshot of the Dashboard for Dyer Island can be seen below.



INSTALL INFLUXDB AND GRAFANA SERVER

THE DASHBOARD USES THE PROCESSED DATA FILES TO DISPLAY PENGUIN DATA. THE DATA PROCESSING SCRIPT MUST THUS BE FULLY FUNCTIONING BEFORE SETTING UP THE DASHBOARD. SEE TESTING A NEW SYSTEM

APMS DATA PROCESSING.

Access the terminal and install the InfluxDB database, which will be used to hold our Dashboard data, as well as the Grafana server, which will be used to display the data.

- Update the system: ***$ sudo apt update***
- Install InfluxDB: ***$ sudo apt install influxdb***
  - Start the influx service: ***$ sudo systemctl start influxdb***
  - Enable to start at boot: ***$ sudo systemctl enable influxdb***
  - Install the client: ***$ sudo apt install influxdb-client***

ROUTINE MAINTENANCE

When visiting a site, check the following:

- Tighten stay ropes
- Check outdoor router antennas and masts are secure
- Check for water ingress into the Skitch Box
- Check RFID antenna is still in good condition

CLEANING AND CHECKING THE SCALES

- Clean the scales whenever visiting the site. As long as very little pressure is applied to the scales themselves, the system does not need to be turned off.
- Check around the scales and ensure no branches or debris are touching the scale pans. Anything touching the pans on the sides will alter the weight reading. Sand on top of the scales (if not too excessive) is not a problem.
- Check that the canvas is still in good condition and not torn/peeling off. Replace as necessary.

ROUTINE CALIBRATION OF SCALES

R&D Weighing recommends the scales be calibrated every 3 to 4 months to ensure accuracy. To check if the system needs to be calibrated, place a 1kg, 5kg and 6kg (5+1) weight on each scale and observe the readings on the scale indicators. If the reading is out by more than 5g, recalibrate. The wind can have a huge influence on the scales and thus <u>only calibrate on good weather days</u>.

There are 4 buttons on each indicator. They are:

CANCEL/BACK	✖
ENTER	↵
LEFT	◀
UP	▲

- SSH into the Pi
- Stop the scales process: ***$ sudo systemctl stop scales***
- Log the start of the calibration event: ***$ python3 ~/apms/log.py***
    - Log the weight discrepancy for each scale using the 5kg weight.
    - You can run **$ systemctl status scales** and **$ systemctl status rfid** to check that the scripts have indeed stopped
- Remove all weights from scales.

- Tare the scale:
    - Hold in the CANCEL button until it displays: 2ErO Then press ENTER
    - All the Lights should be flashing. Press ENTER again to zero the scale.
    - The weight should read 0. Press CANCEL until you are back at the main weight display.

- Start with Scale 1 (colony side) and then do Scale 2 (ocean side). Start by placing both the 1kg and 5kg weights in the centre of the scale. Then proceed to the indicator, and flip open the cover to reveal the buttons as described above. [You will need a small flat screwdriver to lift the cover]. It can be difficult to see the LED display without looking through the cover, so flip it down to confirm what is displayed if uncertain.

- While pressing and holding the ENTER button, also press the CANCEL button. This should display CALIb

- Press ENTER. Now use ◀ ▲ to select UEIGHt and press ENTER

- A flashing weight should be displayed. All the lights should be OFF.

- Use ◀ ▲ to adjust the weight until it reads exactly 6.000 kg

  ◀ to change value

  ▲ to change decimal place holder

- Press ENTER. The new weight should be displayed, and all the lights should be flashing.

- Press ENTER one more time. This should show UEIGHt again. Press CANCEL until you see the weight displayed.
- Check with 1kg, 5kg and 6kg weights again. It should be within 1g. If not, recalibrate.
- Start the APMS processes. See **Error! Reference source not found.**.
- Log the end of the calibration event: ***$ python3 ~/apms/log.py***

Testing and Calibration of System

To test the accuracy of the APMS weighbridge, the weight estimates are compared to that obtained by an independent scale. The scale currently used is the *** and is accurate to 5g. Video footage is captured of the two main scales, as well as the independent scale. The independent scale has an LCD display, and this should be visible to the camera. It does have a night light and should thus be hidden as much as possible from view, as the penguins are very aware of changes to their surroundings and get easily frazzled. The video clips showing penguins stationary on the independent scale will be used to find the "true" weight, while the footage of the main two scales will be used to characterise the data. The penguin should be stationary on the independent scale, long enough to confidently read a stable reading from the LCD display.

<u>You will need the following:</u>

1. A laptop with Wi-Fi. (Take a LAN cable with just in case)
2. Certified 1kg and 5kg weights
3. Independent scale with fresh batteries and canvas cover.
4. Camera to capture footage. Currently using the Reolink Go Plus. Ensure it is charged and that you have the solar panel + zipties
5. In addition to this, always take a basic tool kit and power supply with, as you never know what you might find at the site.

<u>To perform the testing/calibration:</u>

1. Choose good weather and observe when the best time is to set up. (Usually in the afternoon between morning and evening rush).
2. SSH into the system and log the start of the testing/calibration event in the log file ~/data/log.txt
3. The independent scale must be set up as described in the section "Setup Procedure for Independent Scale".
4. Calibrate the scales as described in the section "Routine Calibration of Scales"
5. Ensure the camera has a clear view of the 2 main scales, the independent scale and the LCD display
6. After at least one day of capturing footage, SSH back into the system and log the end of the event in the log file.
7. Remove the camera and independent scale, leaving the site as undisturbed as possible.
8. The data can then be manually processed or the user can wait for the daily file which is automatically generated by the algorithm at 2AM every morning.
9. Compare the weights obtained from the footage with the weights generated by the algorithm
10. If the penguin weight is out by more than 50g, adjust the following parameters and run the algorithm manually to check again.

a. MAX_PENGUIN_WEIGHT - The maximum weight of a penguin expected to walk over the scales.
b. MIN_PENGUIN_WEIGHT - The minimum weight of a penguin expected to walk over the scales
c. MAX_SCALES_DIFFERENCE_ALLOWED – Difference allowed between weight measured on Scale 1 and Scale 2.
d. CLIMB_ON_DURATION_MIN – Minimum amount of time it takes penguin to climb on

   ….

Setup Procedure for Independent Scale

a. Ensure scale has fresh batteries
b. Ensure the scale has the canvas cover on
c. Put the scale on a flat surface and ensure the scale is level.
d. Test with 1kg, 5kg and 6kg weight. It should be spot on.
    i. Calibrate according to the user manual, if required
        ii. ADD INSTRUCTIONS HERE… (Please take a photo and send it my way…, a pic of the scale will be nice too)
e. At the site, also ensure the scale is level, and recheck with weights.
f. Place the display as far away and out of sight from the penguins as possible.

TROUBLESHOOTING

There are too many permutations of how things can go wrong to cover it all in this manual. However, some common things to look out for, as well as how to go about getting more information, will be discussed below.

The Monitor is your friend when diagnosing what's wrong.

USING THE APMS MONITOR

If the Monitor has power, it should operate and be able to give information about the system, even if the Main Unit is completely dead. Ensure the SIM card has airtime to be able to reply via SMS.

- The Monitor will send a SMS to the operator in the event of the following:
  - If the 4V of the SIM800L is too low/high
  - If the 5V of the Raspberry P is too low/high
  - If the 12V of the scales/router is too low/high
  - If the 24V solar supply is too low/high
  - If the ambient temperature is too high. (Warning: 50°C - Danger at 60°C)
  - If the Raspberry Pi or Monitor reboot

  If any of these messages are received, the operator should investigate.

MONITOR COMMANDS

A simple set of commands can be used by the administrator to diagnose system faults via SMS.
To send a command to the Monitor, send a SMS starting with **"CMD:"**
For example, to request a status report, send: "**CMD:S**"
The monitor should reply, via SMS, with a brief status report. The status report is discussed in ***
Note the commands are not case sensitive.
The table below is a list of valid commands.

Command Name	**What to send**	**Description**
Request Status Report	CMD:S	The Monitor should reply by sending a status report via SMS
Reboot	CMD:R	This power cycles the Raspberry Pi
Power On/Off	CMD:P	This powers the Raspberry Pi on if it was off, and off if it was on
Update Time	CMD:U	This updates the Monitor system time.
Change Phone Number of Operator	CMD:M+27*********	This changes the phone number of the operator, saved in the non-volatile memory of the Monitor.
Turn Off SMS Notifications	CMD:X	To turn off SMS notifications from the Monitor.
Turn On SMS Notifications	CMD:Y	To turn on SMS notifications from the Monitor
Send BASH command	CMD:C*****	IN DEVELOPMENT Send a BASH Command for the Pi to execute next time it checks in.

STATUS REPORT (CMD:S)

An example of a status report as received from the Stony Point Monitor can be seen below. The example will be used to explain the different parameters.

```
Boot:2023-12-07 11:49:03
Temp:46.9
24V:27.3
12V:12.0
5V:5.1
4V:4.1
Rpi: STO
BT:20231108 14:00
CH:20240105 12:10
S:Y
R:Y
T:Y
I:Y
CPU:55.0
MEM:7.0
SD:19.5
USB:26.4
11:25
```

GENERAL

To monitor internet usage, use "vnstat", which should already be installed. Run ***vnstat -w*** or ***vnstat -m*** for weekly/monthly report

Allow power down via monitor:

Then reboot: $ sudo reboot

CONCLUSION

Lastly, the system was built as a platform for future expansions and experiments. There are GPIO ports available on the Raspberry Pi, the CPU has the capacity to do more work and the Monitor can be programmed with more features.

To add – prevent sd from filling up

Add suod apt-get install awscli

Add “aws configure”

Region “af-south-1”